Integration measures and labour market success of refugees and beneficiaries of subsidiary protection in Austria

FIMAS Project Research Report

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INTEGRATION MEASURES AND LABOUR MARKET SUCCESS OF REFUGEES AND BENEFICIARIES OF SUBSIDIARY PROTECTION IN AUSTRIA

FIMAS Project Final Report

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Executive summary

Analysing the development of the labour market status of refugees in Austria represents a key field of research for integration policy. The **survey** on **integration** of refugees and beneficiaries of subsidiary protection in Austria, which was conducted as part of the FIMAS project,[1] provides a **socio-scientific dataset** for the analysis of the **status quo of labour market integration** and **integration processes of members of the target group in Austria.**

The project surveyed close to **1,200 refugees in five Austrian federal provinces**. The survey was conducted through personal interviews between August 2016 and May 2017 in Vienna, Upper Austria, Salzburg, Styria and Tyrol. Target groups for the interviews were persons of working age (15-60) from **Syria, Afghanistan, Iraq** and the **Russian Federation (mainly Chechnya)**, who had been granted refugee status or subsidiary protection in Austria in the preceding ten years.

The findings of the survey pointed out **major differences in educational qualifications** across the different analytic groups. The share of persons who were illiterate was highest among Afghans, as was the share of persons with low educational qualifications (two thirds). Half of Chechens/Russians and Iraqis were highly educated, and Syrians had the highest share of persons with intermediate level qualifications compared to the three other groups. Female respondents often had lower levels of education, though were more likely to hold university degrees.

The findings pointed to a mismatch or **devaluation of educational qualifications.** Seemingly, degrees earned abroad are not directly transferable to the Austrian labour market. Thus, it was primarily those with higher qualifications who had particular difficulties in finding a job in Austria. Only the **formal recognition of foreign higher education degrees**, although only a very small number of respondents had sought to do this, or the **obtainment of Austrian degrees** increased the chances for labour market integration.

Employment experience prior to migrating to Austria, particularly professional training, contributed positively to respondents' labour market integration in Austria. However, it was primarily former agricultural workers and elementary workers who had a higher chance of finding gainful employment. Even though Austria is facing a shortage of skilled workers, former engineers, in particular, had very little chance of finding a job. The **mismatch** between qualifications and employment was high – more than three quarters of respondents worked in an industry not (or only partially) related to their educational or training background.[2] This is a result of the high concentration of respondents who were employed in Austria but working in **elementary occupations**, particularly in the federal provinces. Only in Vienna there was a wider distribution of jobs across different occupational fields, such as social work or technical occupations. The chance of having gainful employment was highest and the risk of unemployment lowest in Salzburg and Tyrol.

The strong concentration of low-skilled jobs, high rates of part-time employment, temporary employment and overall low salaries of employed persons could be summarised as a situation characterised by **occupational segregation** and **labour market segmentation**. Put simply, the Austrian labour market offers only limited opportunities for recognised refugees and beneficiaries of subsidiary protection. These opportunities regularly involve low-status jobs that non-migrants typically avoid (such as kitchen help, cleaners, warehouse or security workers). Respondents were therefore particularly likely to work on **atypical employment contracts**, and were significantly more likely than other migrants or Austrians to do so, especially when they took up their first occupation in Austria.

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[1] "Integration measures and labour market success of refugees and beneficiaries of subsidiary protection in Austria" (FIMAS)

[2] It should be noted that respondents often were employed in industries that did not correspond to their qualifications even before migrating to Austria.

Women were significantly less likely to participate in the labour market, and therefore had a **lower rate of employment**. On average, they found their first paid job one-and-a-half years **later than their male counterparts.** This is primarily a result of limited capacity for job searching and taking up a job as such due to household work and childcare. When employed, women were more likely than men to work in academic professions and were less likely to work as elementary workers. **Women** and **highly qualified workers** had the highest chances of having jobs that somewhat matched their educational backgrounds.

Of all the respondents, 30% had participated in an official **values course** – training on fundamental values and rules governing social life in Austria – and 25% in a **competence check** conducted by the Public Employment Service. Multivariate models showed that participation in competence checks and values courses had no direct and immediate impact on current labour market integration. Notwithstanding this, potential positive effects of these recent measures could be expected to become evident in the medium term. The participation in **German language courses** impacted labour market integration at least indirectly by improving language skills.

The groups analysed as part of this study differed structurally by the length of time they had been in Austria, as well as by age, gender and educational level. When controlling for these effects, **Afghans,** in particular, were more likely to be employed than other recognised refugees or beneficiaries of subsidiary protection. Despite **Afghans'** educational orientation tending to be weaker and **often having lower levels of education**, labour market integration did not take any longer, nor were Afghan' chances of labour market participation any worse, than for other groups. Compared to other groups, however, Afghans tended to work in low-skilled occupations, and were rarely employed in highly-paid jobs.

Syrians represent the group who had been in Austria for the shortest amount of time, and as such were often still looking for jobs, undertaking training or education, or learning German. However, among the labour force, they stood out as the group that found **a first job the fastest.**

Respondents from the **Russian Federation** and from **Iraq** were also covered in the sample, but due to small subsamples and potential biases, **results for these two subgroups may not be robust, and may not be representative of the respective refugee population.**

As yet, little is known about the social situation of **Chechens** in Austria. When compared to other groups, their **high rates of secondary and tertiary degrees** on the one hand, and their **slow path to the labour market** on the other, are striking. The length of time it took to start their first job is much longer than it was for other groups (eight years). Higher rates of labour force participation would be expected for this group due to the longer amount of time they have been in Austria.

All told, **Iraqis** were not able to bring their **high levels of education** (40% are university graduates) to bear in terms of overall employment, higher incomes or better qualified occupations. With 51% of Iraqi respondents reporting they completed post-secondary education, this made them the group with the highest proportion of highly educated respondents in the study.

Longer **periods of residence in Austria** correlated to both significantly increased labour force participation and employment rates among refugees. After **six to nine years** of duration of stay in Austria, **labour force participation reached 75% – equivalent to that of the overall population.** Among recognised refugees and beneficiaries of subsidiary protection, **roughly 55%** found **employment** after **nine or ten years.** Due to under-coverage of employed persons in the sample this rate might actually be higher and more or less match the overall Austrian employment rate.

Regarding the respondents' **experiences** in Austria and their **hopes** for the future, the survey revealed a high desire to find work and to pursue an education. Family and friends were above all the most important support factors for starting a new life in Austria, but respondents also frequently pointed to the Austrian state, rights and freedoms, the welcoming culture and friendliness of Austrian society, and support from NGOs in helping them get settled. The greatest **barrier** for integration was the issue of **language**, with six out of ten respondents noting this. Unemployment/job search and the overall living

situation took second and third place, respectively. In terms of their hopes for the future, respondents most often mentioned **finding a job**, followed by learning the **language**, with pursuing **training or education** taking third place. Perceptions towards respondents' experiences and hopes for the future were asked via open-ended questions and allowed for multiple answers. The intentions about returning to refugees' home countries varied by group. Chechens were the only group where more than half of respondents noted their wish to return once their country of origin was safe again.

Based on the individual information provided on barriers and hopes for the future, it became clear that nearly every group of respondents had **strong intentions to focus on education and the labour market**. These were limited by the need to acquire language skills and the right educational qualifications, whilst women faced additional barriers of childcare and household work. Combined with **group-specific differences** relating to educational and skill levels, the discussion of the **integration of refugees in the labour market** can therefore **not be generalised**. Instead, group-specific differences based on country of origin as well as gender, education and training/profession must be duly taken into account when designing appropriate policy measures.

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List of acronyms used

AMS	Public Employment Service Austria
EUROSTAT	Statistical Office of the European Union
Exp(B)	Odds ratio
FIMAS	Project abbreviation for the "Integration measures and labour market success of refugees and beneficiaries of subsidiary protection in Austria" project
ICMPD	International Centre for Migration Policy Development
ILO	International Labour Organization
ISCED	International Standard Classification of Education
n	Sample size
N	Population size
Ö-ISCO 08	2008 Austrian Classification of Occupations (based on the ISCO, or International Standard Classification of Occupations)
UNHCR	United Nations High Commissioner for Refugees
WG	Shared accommodation
ZSI	Centre for Social Innovation

1 Baseline situation

Austria is one of the main receiving countries in Europe in terms of recognised refugees, especially when compared to the size of its population. Based on a per capita analysis, only Germany surpassed Austria in granting asylum in 2016, with 322 positive decisions per 100,000 inhabitants (Austria: 294 per 100,000 inhabitants, Statistics Austria, 2016, 2017g). According to the relevant literature, recognised refugees or beneficiaries of subsidiary protection are among the societal groups most affected by various forms of exclusion in Europe and the problems that go along with it. As stated in the Austrian National Action Plan for Integration, integrating refugees represents a **particular challenge**. Thus, both in the European and Austrian context empirical research on the social status of this group is largely missing. Until recently, only a few case studies have been conducted on specific refugee groups. In this regard, the 2016 Austrian integration report stated that on the basis of the existing data it would be very difficult to conclusively answer the question of the prospects of refugees or beneficiaries of subsidiary protection for sustaining their lives in Austria (Expert Council for Integration, 2016, p. 7). The majority of related studies have focused on refugees' labour market participation in terms of its macroeconomic impacts. The individual labour market situation and labour market trajectories of different groups among the overall population of recognised refugees and beneficiaries of subsidiary protection are still under-researched areas, as are a number of closely associated issues, including education, professional qualifications, recognition of degrees, health, accommodation and social networks.

The aim of the "Integration measures and labour market success of refugees and beneficiaries of subsidiary protection in Austria" (FIMAS) research project was to bridge these gaps and gain greater understanding of the processes involved in labour market integration and the barriers and factors for success related to this, using a quantitative survey. The main **research questions** were: Which factors impact successful labour market participation of members of the target group? What are their qualifications, what characteristics differentiate them, and to which extent do qualifications influence their labour market success? What are refugees' employment histories, what were their professions abroad and what role does this professional experience play in Austria? How does employment status differ between different refugee cohorts?

Thus, the FIMAS survey tried to improve the knowledge base regarding a number of questions drawn from and priorities defined by the Austrian **National Action Plan for Integration – NAP.I** (2010).[3] The NAP.I points out the specific challenges inherent in integrating recognised refugees and beneficiaries of subsidiary protection. The 50 Point Action Plan regarding the integration of recognised refugees and beneficiaries of subsidiary protection building upon the NAP.I (Austrian Federal Ministry for Europe, Integration and Foreign Affairs, 2015) explicitly points to the necessity of assessing qualifications (paragraph 11) and accompanying scholarly research (paragraph 50). For interlinking existing statistical data (paragraph 49) – such as data on minimum income and residence permits – survey data also opens up a number of possibilities, such as through cooperation and exchange between research institutions that conduct research on refugees. Tailor-made measures, such as expanding job-specific language courses (paragraph 14), could build upon research findings, including upon those of the FIMAS survey.

FIMAS, in essence, covers all the topics included in the National Action Plan for Integration, particularly with respect to the **fields of language and education** as well as **work and employment.** The regional dimension of integration is also covered by the survey being conducted in the participating federal provinces and cities. Moreover, the focus on the target group of recently arrived refugees and the

[3] The Austrian NAP.I intends to bring together all integration policies, programmes and measures of the federal level, provincial governments, local authorities, cities, and social partners. The NAP.I was the result of a comprehensive work process between a wide number of public stakeholders, interest groups and NGOs. In parallel, international experts, migrant associations and Austrian citizens brought additional input to the NAP.I. The NAP.I is supposed to provide a platform for nation-wide cooperation of all stakeholders for successful integration measures and optimise their implementation.

envisaged longitudinal perspective with regard to this target group provide for the possibility to empirically study integration from the start.

The literature on educational credentials of immigrants shows that the qualifications of recent immigrants are sometimes higher than those of the resident population (Buber-Ennser et al., 2016). Contrary to commonly held beliefs that migrants are poorly educated or not educated at all, at least in terms of formal qualifications, one cannot assume lower levels of education. However, the relevant literature also shows that it is difficult to make any generalisations; the distribution of educational qualifications differs from one nationality to the other and between genders, and formal qualifications earned abroad cannot be directly compared to corresponding Austrian degrees.

Research also suggests that immigrants, refugees and beneficiaries of subsidiary protection often fall under the category of a "labour reserve", namely the share of the workforce that more than any other group is dependent on the ups and downs of the economic cycle. In times of crisis, members of this group tend to be among the first to be affected by cuts and layoffs; they are overrepresented in precarious employment arrangements and are – statistically speaking – at a higher risk of unemployment. Conversely, when the economy picks up again, they re-enter the labour market at disproportionally higher levels; however, often only in part-time jobs, temporary employment contracts or other forms of atypical employment. Furthermore, they might face multiple disadvantageous and exclusionary factors. Citizenship, residence status, ethnicity, language skills, gender, age, health and disability, prejudice and discrimination might come into play individually or in combination. The resulting multiple disadvantage in employment thus manifests in significantly poorer labour market positioning or even complete exclusion.

For refugees, the granting of refugee status or subsidiary protection in Austria opens the formal **access to the labour market**. As a consequence, they enjoy rights similar to Austrians – with the exception of occupations limited to Austrian citizens and EU citizens residing in Austria. Their legal rights, however, do not translate into a comparable labour force participation rate to Austrian or EU citizens. There is a long and cumbersome path refugees must follow from formal equal access to the labour market to participation and positioning comparable to other migrant groups or the non-migrant population.

1.1 Immigration's effects on the labour market

A broad number of studies have addressed the question of what macroeconomic impacts immigration has on receiving countries; only a smaller number has dealt with the question of the macroeconomic impacts and labour market participation of recognised refugees or persons with subsidiary protection. The vast majority of studies on the economic impact of immigration conclude that, generally speaking, migration contributes positively to economic growth and development in countries of destination, although these positive effects are comparatively small in their overall impact. This applies to refugees as well, since, like other migrant groups, they on average tend to be younger than the overall population, contribute more to welfare and tax-paid services than they receive, and contribute to growth through consumption from the beginning, and through employment in the medium to long-term. They **increase labour supply;** however, initially, they **also increase the number of unemployed**.

Recent model calculations assumed an additional increase in the Austrian unemployment rate of 0.4% by 2030, with this increase **primarily attributable to the admission and residence of refugees** (EcoAustria 2017). This is not expected to impact the labour market position of the resident population – Austrian citizens, EU citizens and other third country nationals who are integrated in the Austrian labour market. Expected displacement effects among the resident population are expected to be close to zero (0.01%); only low-skilled residents are expected to be affected by increased displacement at a still very moderate rate of 0.2% by 2030. This group regularly comprises third country national immigrants rather than Austrian or EU citizens.

The modelling of labour market effects of admission of recognised refugees and beneficiaries of subsidiary protection is based on a number of assumptions, such as age and qualification levels. Recent modelling for Austria (Berger et al., 2016; Holler and Schuster, 2016) uses, among others, research findings on the labour market integration of refugees in Sweden to assess labour market and employment participation. Thanks to recent studies (Buber-Ennser et al., 2016; Kohlbacher et al., 2017) and surveys on the subject (AMS, 2016c; Federal Province of Salzburg, 2017) the availability of data is continuously improving.

At present, it is not precisely clear whether, how well, and how quickly labour market integration of refugees and beneficiaries of subsidiary protection is taking place, since members of this target group are not identifiable as such in the datasets created for the purpose of labour market monitoring (UNHCR, 2013, p. 46). Neither are attributes that would allow for easy identification of members of the target group included in the samples of regular population surveys like the Labour Force Survey (LFS, part of the Austrian microcensus) or EU-SILC, or in register data from the Austrian Central Civil Register, social insurance data or payroll tax data. One notable exception is the data held by the Public Employment Service Austria, which includes the category of “refugee status” (asylum seeker, granted asylum or granted subsidiary protection) for the individuals recorded in its database. This information is regularly included at the beginning of the initial registration. Whether the information is complete and whether subsequent changes in status are recorded is not fully certain. The LFS ad-hoc modules included in 2008 and 2014 are another example, which allowed for identifying refugees among the sample on the basis of questions asked on reason and purpose of immigration to Austria.

Of particular note is the study conducted by Joanneum Research (Prettenthaler et al., 2017) on the economic effects of the admission of refugees to Austria. It analysed data from the Austrian Labour Market Database between 2000 and 2015, comprising data from the AMS (Austrian Public Employment Service) and social insurance agencies, which includes the attribute of refugee status. All persons insured during this period – i.e. those not active in the labour market as well as unemployed and employed persons – were included. In addition to **clear, gender-specific differences** in the employment rate, the study made evident how labour market integration processes develop over time, suggesting an increasing curve during the first four years, a peak after five years and a stable to stagnant development subsequent to this peak. Compared to the total population, the **employment rate** of recognised refugees and beneficiaries of subsidiary protection was **below average**, even nine years after the time of entering into Austria. One key recommendation made by the authors of the study was to integrate refugees in the labour market earlier, for longer periods and more continuously than is the case now. After all, the economic effects of refugee immigration largely depend on the success of these integration processes.

2 Labour market situation and social status of refugees in Austria

The labour market situation of migrants and persons born in Austria differ in a number of aspects: migrants are twice as likely to be **unemployed**, twice as likely to be **working as labourers** and twice as likely to be employed in **elementary occupations**(Statistics Austria, 2015). **Over-qualification** is nearly three times as common among migrants as it is among those born in Austria.

Recognised refugees and beneficiaries of subsidiary protection are particularly likely to be overqualified for their jobs on the Austrian labour market, and – if at all – manage to reach their full labour market potential only several years after arriving in the country. Based on microcensus data, including the additional module in the 2008 micro census, Huber demonstrated that there are **lower labour force participation rates** (75.9%), **employment rates** (70.4%) and a distinct **qualification mismatch** among refugees (41.8% for men, 49.6% for women) in Austria compared to other groups (Huber, 2010). An Austrian Chamber of Labour study (Riesenfelder et al., 2011, p. 126) similarly found that refugees, as a group, tend to be overqualified for their jobs (43% in total), among the overall immigrant population. Certain types of atypical employment, specifically **shift work** and **night work**, were particularly common among refugees, even when compared to other migrant groups. There was no data available on refugees with respect to temporary employment.

A study conducted by Bock-Schappelwein and Huber (2016), which focused specifically on the labour market integration of refugees based on data from the 2014 micro census and the additional module, also indicated lower labour force participation and employment rates, as well as a higher rate of unemployment among refugees compared to other migrant groups. The **labour force participation rate among refugees** was **71.8%** in 2014 (compared to 73.1% among all migrants); the **employment rate** was **60.4%** (compared to 64.9% for all migrants); and the **unemployment rate** was **15.9%** (11.3% among all migrants).

Within the group of recognised refugees and beneficiaries of subsidiary protection, the labour force participation rate for persons who had arrived between 2005 and 2014 (primarily Chechens and Afghans) was particularly low (35.2%). By contrast, labour force participation among cohorts that had arrived earlier was nearly twice as high. At 65.5%, the labour force participation rate for women was ten percentage points lower than the rate of men (75.8%). Thus, OECD reports suggest that refugees in EU Member States in general integrate in labour markets only gradually and over longer time periods. Data aggregated across 25 states suggest that they achieve the employment rate of the general population only after a period of residence of 15 to 19 years (OECD, 2016a, p. 21; OECD, 2016b; OECD/UNHCR, 2016).

The analysis of demographic data of respondents participating in the Bock-Schappelwein and Huber study indicated that they tended to be overwhelmingly **young (85% were aged between 16 and 40 when they entered the country)** and **male (61%)** (Bock-Schappelwein and Huber, 2016). This also implies that refugees were younger than labour migrants and family migrants, but older than educational migrants. The rate at which formal qualifications were recognised (16%) and not recognised (74%) was roughly equivalent to that among economic and family migrants. Furthermore, respondents who came to Austria for asylum **rarely completed** their **education or training in Austria** (10%), had **poorer German language skills** among those of working age, and had a **higher than average proportion of persons with intermediate qualifications** (ISCED 3 or 4).

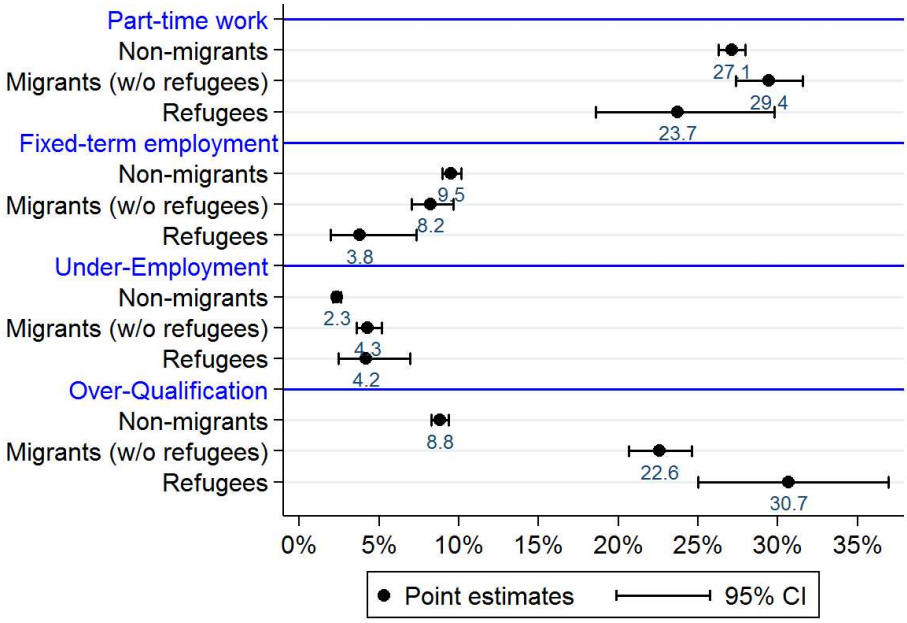
In addition, **qualification mismatch**, or a formal over-qualification or under-qualification of employees in their jobs, was **more common** among refugees than in other groups. **Over-qualification** was especially significant among refugees: whilst 20% of all migrants were formally overqualified for their jobs, among refugees this rate was at 27%. This implies that refugees succeed less often than other

migrant groups in utilising their formal qualifications on the labour market – for example, due to language barriers.

The legal status of asylum seekers in Austria must be taken into account since they are only granted full access to the labour market once they have been granted asylum. Comparisons of labour market success across different migrant groups, and even within immigrant cohorts, are therefore difficult to make. The case of refugees who came to Austria from the former Yugoslavia in the early 1990s showed that their labour market and employment participation was even higher than among labour and family migrants from the same period.

The authors' own analyses of the **2014 microcensus**, including the additional module on the labour market situation of migrants, indicate that certain types of atypical employment appeared less often among refugees than other migrants, even though the differences are not statistically significant. Among refugees in paid employment, 24% were employed on a part-time basis. Among other migrants, this rate was 29%, and among non-migrants, it was 27% (Graph 1). Fixed-term employment contracts were also less common among refugees, particularly in comparison to non-migrants. Underemployment was twice as common among migrants (4%) as it was for non-migrants (2%), but refugees (4%) did not differ significantly from other migrants. The most **striking differences could be found in the area of over-qualification**. Only 9% of non-migrants, but 23% of migrants and 31% of refugees stated that they held higher qualifications than would be required for their job.

Graph 1. Atypical employment among refugees (n=472), other migrants (n=3,657) and non-migrants (n=19,240) in Austria, with confidence intervals



Source: Statistics Austria, 2014; data from the 2014 microcensus and ad-hoc module, authors' own calculations, weighted data, not including civil servants and members of the armed forces

As regards asylum seekers' educational and qualification levels, ICMPD carried out two research projects in 2015 and 2016. The two projects collected socio-structural data from asylum seekers in Austria using a similar research design. As part of the EQUAS pilot study (on behalf of the Austrian Federal Ministry of the Interior), quantitative interviews were conducted with asylum seekers in 2015 (ICMPD, 2015). A total of 315 personal interviews were conducted with respondents from Syria, Afghanistan, Kosovo and the Russian Federation in order to gather information on education, training, qualifications, language and labour market experience.

Building on this, the EQUAS+ research project was implemented between 2015 and 2016 (with funding from the Asylum, Migration and Integration Fund (AMIF) and co-financed by the Austrian Federal Ministry of the Interior (ICMPD, 2016). For this cross-sectional study, 1,025 asylum seekers from

Afghanistan, Iraq, Syria, Iran and other countries were interviewed about their education, training, qualifications, language skills, labour market experience, migration routes and choice of destination country. Both studies confirmed the high degree of heterogeneity within the whole population of asylum seekers in terms of education, qualifications and professional experience. Higher education qualifications were particularly common among Russians/Chechens (EQUAS), Iranians and Iraqis (EQUAS+), whilst respondents from Afghanistan only had basic qualifications in the majority of cases (both studies).

The DiPAS study conducted in late 2015 by the Wittgenstein Centre aimed to report on the labour market potential of asylum seekers who had arrived in the summer of 2015 (Buber-Ennser et al., 2016). In it, 514 asylum seekers in Vienna and eastern Austria were asked about **education** and **employment histories, values and attitudes**, as well as about socio-demographic attributes. The focus of this survey was on persons with Syrian (36%), Iraqi (38%) and Afghani (16%) citizenship; citizens of other countries accounted for 10% of respondents.

The findings illustrate **widely varying levels of education by group:** more than two thirds (70%) of respondents had completed at least the compulsory levels of education in each country; 47% had at least upper secondary qualification; and 26% had post-secondary qualifications. The proportion of respondents from Iraq (31%) and Syria (27%) with post-secondary qualifications was comparable to that among Austrians (28%). Although the proportion of Afghan respondents with post-secondary qualifications was significantly lower at 10%, relative to the Afghan population, they held relatively high educational qualifications. What is also interesting is that gender-specific differences in educational qualifications were marginal – with the exception of the group without formal training, which was 10% for women but only 4% for men. The results indicate a certain degree of **education-related self-selection** among the asylum seekers who had arrived in Austria in 2015. Comparing the DiPAS data to the census data in each country allows leads to the conclusion that asylum seekers from Syria and Afghanistan were more highly educated than the general population in their countries of origin.

Potential and willingness to integrate in the labour market became evident as well. Of the respondents, 72% had labour market experience in their countries of origin and 67% wanted to find work in Austria as quickly as possible. Younger respondents, in particular, wanted to continue their education (71% of 15 to 19-year-olds and 46% of 20 to 24-year-olds). Furthermore, 75% of respondents expressed the desire to remain in Austria in the long term, even if the situation in their country of origin were to stabilise.

The "competence check for professional integration of refugees" project (*"Kompetenzcheck zur beruflichen Integration von Asylberechtigten"*) conducted by the AMS Vienna between August and December 2015 also indicated that, as a whole, the persons seeking asylum in Austria in the summer of 2015 were **more highly qualified than expected** (AMS, 2016c). Nevertheless, based on the responses of the 898 persons who participated in competence checks, it can be concluded that there are **major differences between countries of origin.** Asylum seekers from Iraq, Syria and Iran often held secondary and post-secondary educational qualifications, whilst Afghans often only held primary educational qualifications, and sometimes no qualifications whatsoever. Due to the non-random and sometimes self-selective sample of participants, the AMS also noted that the data merely represented the analysis of project data, not the findings of a scientific study, and therefore did not claim its results to be representative.

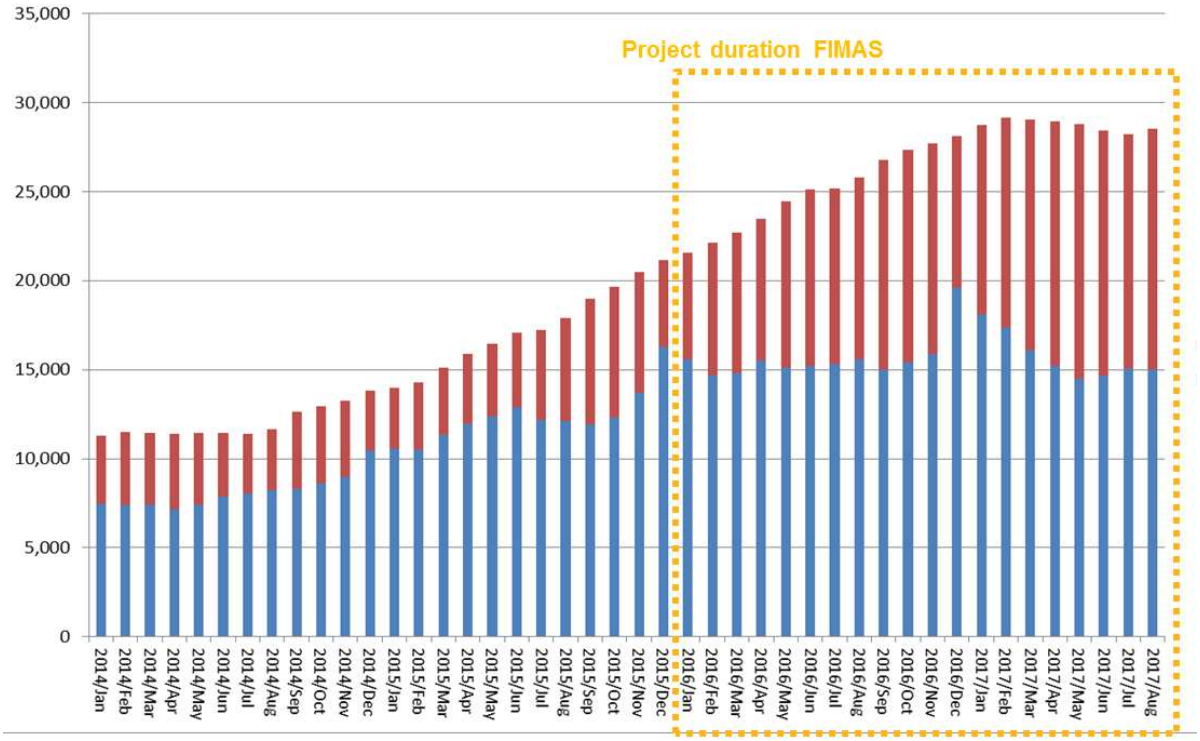
2.1 Unemployed persons and participants in training measures

The basic structural attributes of the group comprising unemployed recognised refugees and beneficiaries of subsidiary protection in Austria are derived from AMS data. Current analyses show that the size of this group has continuously grown between January 2014 and February 2017 (Graph 2), followed by a slight decline. The proportion of persons in training has significantly increased since 2015 compared to those registered as unemployed. As of August 2017, 28,552 recognised refugees or beneficiaries of subsidiary protection in Austria were registered with the AMS, of which 14,993 were

unemployed (52.3%) and 13,559 (47.5%) were in training (AMS, 2017). According to residence status, the group comprised 23,591 recognised refugees and 4,961 beneficiaries of subsidiary protection (17.4%). The percentage of persons in training was higher among beneficiaries of subsidiary protection (53.0%) than among refugees (46.3%).

By far the larger part of this group were men (20,823, or 72.9%) and just over a quarter were women (7,729, or 27.1%).

Graph 2. Changes to the number of refugees and beneficiaries of subsidiary protection registered with the AMS as unemployed or in training, January 2014 to August 2017



red = in training, blue = unemployed
Source: AMS, 2017

Broken down by citizenship, persons from Syria formed the largest group; nearly half (45.0%) of all refugees registered with AMS. In second place came persons from Afghanistan (19.2%), followed by persons from the Russian Federation (11.2%), Iraq (6.5%), Iran (5.1%) and stateless persons (3.8%) (Table 1). The research design of the FIMAS study covers the four largest citizenship groups among recognised refugees and beneficiaries of subsidiary protection.

Nearly two thirds of those registered with the AMS were in Vienna (62.3%). Upper Austria took second place (9.9%), followed by Lower Austria (9.5%) and Styria (6.8%). There were fewer registered refugees and beneficiaries of subsidiary protection in Salzburg (2.9%), Carinthia (2.7%), Tyrol (2.7%), Vorarlberg (2.6%) and Burgenland (0.8%). When compared to the same month of the previous year, the number of registered persons increased more rapidly outside the capital; in particular, in Upper Austria it rose by 38%, whilst the rate of increase was only 4% in Vienna (AMS, 2017). Five out of the seven federal provinces with the largest number of persons registered are included in this study.

Table 1. Refugees or beneficiaries of subsidiary protection registered with AMS, August 2017

Citizenship	Number of persons	%
Syria	12,836	45.0%
Afghanistan	5,469	19.2%
Russian Federation	3,193	11.2%
Iraq	1,850	6.5%
Iran	1,457	5.1%
Stateless/unclear	1,073	3.8%
Somalia	1,039	3.6%
Other	1,635	5.7%
Total	**28,552**	**100.0%**

Federal province	Number of persons	%
Vienna	17,776	62.3%
Upper Austria	2,826	9.9%
Lower Austria	2,706	9.5%
Styria	1,932	6.8%
Salzburg	821	2.9%
Carinthia	780	2.7%
Tyrol	757	2.7%
Vorarlberg	730	2.6%
Burgenland	224	0.8%
Total	**28,552**	**100.0%**

Source: AMS, 2017

The AMS carefully monitored how the labour market situation developed for these groups. An analysis conducted in July 2016 showed that, of those who had been granted residency in Austria in 2015 and later registered with the AMS, 10% were in employment, 73% were still registered and 17% were out of the labour market (AMS, 2016a).

3 Research design

The standardised survey was designed as a cross-sectional study with the option of continuation in the form of a longitudinal survey. The particular characteristics of this research design were: 1) that the survey covered multiple federal states in Austria and aimed for the best possible comparability; 2) that the focus was on the four citizenship groups of Syria, Afghanistan, the Russian Federation and Iraq; 3) that the survey was locally conducted by way of personal interviews (face to face) and by interviewers recruited and trained by ICMPD; and 4) that the questionnaire was designed in a way that allowed for the use of the simplest possible language and was translated into the five interview languages: Arabic, Dari, Russian, Chechen and English.

At the time of the survey, there were no empirical data publicly available on the **full population** of all recognised refugees and beneficiaries of subsidiary protection residing in Austria. The **population register** maintained by Statistics Austria does include data on the population by citizenship and country of birth but it does not information on residence status. Country of birth can be used to determine the total population of immigrants resident in Austria (including by federal province and age group); however, these data include asylum seekers, persons that have been granted asylum or subsidiary protection, persons residing on humanitarian grounds, persons who have been denied asylum as well as persons with another residence status, e.g. students, family members of Austrians or third-country nationals, or skilled migrants. Consequently, the population register could not be used directly for a specific analysis of the group of recognised refugees and beneficiaries of subsidiary protection.

Asylum statistics have their limitations as well for assessing the size and structure of the total population of recognised refugees and beneficiaries of subsidiary protection. Although the statistics on asylum applications and decisions published by the Austrian Federal Ministry of the Interior on a monthly and annual basis include a breakdown by nationality, the statistics on decisions only relate to the date of the decision. They are therefore not suited for drawing conclusions regarding the residency of the groups of interest in Austria, even though one can assume low rates of return or onward migration of the target group.[4] Internal migration and mortality among the target group are also unknown once their status has been established.[5]

EUROSTAT also provides national asylum statistics of the EU Member States. The EU's statistical office has been publishing data broken down by nationality, type of procedural decision, gender and age group since 2008. The UN High Commissioner for Refugees (UNHCR) also publishes data on applications, decisions and the number of pending cases, etc. **UNHCR** publishes an estimate of current figures for resident refugees based on decision statistics for the previous ten years. However, this information is not broken down by any additional attributes like gender or age. According to the latest UNHCR estimates, **approximately 93,000 recognised refugees (including beneficiaries of subsidiary protection or those granted residency on humanitarian grounds)** and **approximately 76,000 asylum seekers** were living in Austria **at the end of 2016.** Including around 900 stateless persons, around **171,000 persons**, who fall under the UNHCR definition of "persons of concern" (UNHCR, 2018), were living in Austria. Table 2 provides an overview of some of the estimates on which the FIMAS study is based.

Of the total of approximately 93,000 recognised refugees, 76% (or approximately 71,000) belonged to the target group of the FIMAS study: around 31,000 Syrians (33%), around 20,000 Afghans (22%), around 14,000 Russians (15%) and around 6,000 Iraqis (6%).

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[4] For example, 78% of Afghans who arrived in Austria between 2007 and 2011 were still in the country in 2016, the highest rate among all nationality groups (Statistics Austria, 2017g).

[5] Linking the Integrated Foreigners' Administration Database (*Integrierte Fremdenadministration* [IFA]) to the population registry and social insurance data has already been suggested by the Expert Council for Integration to facilitate the processing of current residence and income statistics for refugees (Expert Council for Integration, 2016, p. 76f).

According to both UNHCR estimates and EUROSTAT data, the four nationality groups included in FIMAS represent the largest groups of recognised refugees in Austria, followed by Somalis (4,163), Iranians (3,598), stateless persons (2,398) and persons whose nationality is unknown (1,933) (UNHCR, 2018).

The UNHCR estimates are higher than the data provided by EUROSTAT, particularly for persons from Afghanistan and the Russian Federation. This is likely due to the fact that the EUROSTAT data is limited to first instance decisions, which do not take into account positive appeal and second instance decisions, rather than due to the age restriction and slightly different observation periods of the EUROSTAT data, which appear to not be relevant to the result.

Comparing the UNHCR data with Austrian population statistics shows that citizenship is not per se an ideal indicator for detailing the refugee population. At the start of 2017, 41,672 Syrians were living in Austria – around 11,000 more persons than the approximately 31,000 Syrian refugees estimated by the UNHCR figures (Table 2). The differences in group sizes for Afghan citizens (45,259 persons in total, of which approximately 20,200 are refugees), the Russian citizens (32,018 persons in total, of which approximately 13,800 are refugees) and Iraq citizens (14,802 persons in total, of which approximately 5,600 are refugees) are even more pronounced. Up to two thirds of the persons in these groups in the Austrian population statistics are thus no refugees.

Table 2. Positive asylum decisions, estimates of resident refugees and population statistics in Austria

	Positive first instance decisions, 2008-2016 (EUROSTAT)	Estimate of resident refugees, end of 2016 (UNHCR)	Population statistics as of 1 Jan. 2017 (Statistics Austria)			
	by nationality	by nationality	by citizenship		by country of birth	
	age 0-64	all age groups	all age groups	age 15-59	all age groups	age 15-59
Syria	31,700	30,958	41,672	28,062	41,588	29,032
Afghanistan	13,900	20,220	45,259	34,196	44,684	35,925
Russ. Fed.	7,585	13,773	32,018	21,804	33,851	27,054
Iraq	5,600	5,555	14,802	10,728	16,197	12,326

Sources: EUROSTAT, 2017a; UNHCR, 2018; Statistics Austria, 2017a, b

In summary, the estimates by UNHCR seem to be a plausible approximation of the number of refugees and beneficiaries of subsidiary protection in Austria. However, it remains unclear exactly how many persons have applied for or received which residence permits initially. To put it another way, the number of Syrians residing in Austria who fled their homeland and then applied for asylum in Austria (or elsewhere) cannot be determined precisely on the basis of the available data.

The only nationwide database currently containing updated information and, above all, information that allows for distinguishing recognised refugees or beneficiaries of subsidiary protection from other migrant categories is the **labour market database (AMDB)**, which combines AMS data and data of the Federation of Austrian Social Insurance Entities (*Hauptverband der österreichischen Sozialversicherungsträger*). When a client contacts the AMS for the first time, the attribute convention refugee or beneficiary of subsidiary protection is recorded.

However, it is not possible, based on these data, to answer the question of the number of persons who have found irregular work or are not insured for other reasons. Given the difficult social and economic situation faced by refugees, especially immediately upon their arrival in Austria, and their lack of

networks to help them find informal work or support them financially, it can be assumed that the proportion of these groups among the total refugee population in Austria will be relatively low.

3.1 Target groups

As a consequence of the above, the research design of the survey was based on AMS data; on the size and structure of the population of **recognised refugees** and **beneficiaries of subsidiary protection who had registered** with the AMS at the start of the project. Distributions according to nationality and residence by federal province (Table 3) were known for January 2016. The share of men (74%) and women (26%), and the distribution by age group was known as of September 2015: young persons under the age of 25 accounted for 28% of the target group, adults aged between 25 and 44 made up 56%, and adults aged 45 and older made up 16% (AMS, 2015).

Table 3. Refugees and beneficiaries of subsidiary protection registered with AMS, Jan 2016

Citizenship	Number of persons	%
Syria	6,810	31.6%
Afghanistan	4,789	22.2%
Russian Federation	3,378	15.7%
Stateless/unclear	1,868	8.7%
Iraq	1,125	5.2%
Iran	1,085	5.0%
Somalia	808	3.7%
Other	1,712	7.9%
Total	**21,575**	**100.0%**

Federal province	Number of persons	%
Vienna	14,354	66.5%
Upper Austria	2,007	9.3%
Lower Austria	1,789	8.3%
Styria	1,248	5.8%
Tyrol	637	3.0%
Salzburg	580	2.7%
Carinthia	440	2.0%
Vorarlberg	404	1.9%
Burgenland	116	0.5%
Total	**21,575**	**100.0%**

Source: Salzburg24.at, AMS data

The selection of the **federal provinces** was based on the distribution of members of the target group among them as well as on the distribution of regional sponsors of the study. Since sampling had to be limited to the federal provinces of Vienna, Upper Austria, Styria, Tyrol and Salzburg, it is by definition not possible to draw any generally applicable conclusions for the overall Austrian situation. Notwithstanding this, the survey covered five of the six federal provinces hosting the largest number of recognised refugees and beneficiaries of subsidiary protection. In early 2016, 87% of refugees were registered with the AMS in one of the five federal provinces selected.

The selection of **cities** in which the survey would be conducted was, among other reasons, based on the desire to cover both larger and smaller cities, as well as surrounding areas of cities to capture differences in the distribution of structural attributes and therefore possible differences in integration processes. Ultimately, Innsbruck and its environs, Imst, Wörgl and Kufstein were chosen in Tyrol, the city of Salzburg and its environs in Salzburg, Linz and its environs, plus Wels and its environs in Upper Austria, Graz and its environs in Styria, and the capital Vienna and its environs.

With respect to **citizenship**, the four most prevalent groups among those registered with the AMS were chosen: recognised refugees or beneficiaries of subsidiary protection from Syria, Afghanistan, the Russian Federation and Iraq. The group of persons registered as stateless was not included in the sample due to its unknown and potentially heterogeneous structure, which includes differences in language.

Another consideration included the intention to carry out group comparisons between **working and non-working populations.** This criterion was therefore also taken into account in the stratification of sampling.

Because the aim was to draw conclusions regarding the working-age population as well as to investigate integration processes over time, bearing in mind that such processes may no longer seem relevant to those approaching the retirement age, the **age of respondents was limited to between 15 and 59 years**.

A further limitation related to **the cohorts of immigrants, recognised refugees and beneficiaries of subsidiary protection**: the focus was on comparisons between the more recent cohorts. The decision was taken, therefore, to only sample persons who were first recorded in the social insurance data from 2006 onwards, and those who were at least 15 years of age when their data was initially recorded.

3.2 Sampling frame, drawing, gross sample

Thanks to the support of the AMS main office, it was possible to draw an **address sample** from the **AMS database**, which formed the basis for the first phase of the survey. The AMS data also formed the basis for the **sampling frame**.

Table 4. FIMAS gross sample, random sampling (AMS data)

Federal province	Cities	Status	Russ. Fed.	Syria	Iraq	Afgha-nistan	Total (citiz.)	Total (cities)	Total (federal province)
Vienna	Vienna	A	175	238	175	238	826	1,652	1,652
		I	175	238	175	238	826		
Salzburg	Salzburg	A		158		101	259	845	845
		I		202		384	586		
Upper Austria	Linz	A		125	74	125	324	717	1,012
		I		125	143	125	393		
	Wels	A		75		46	121	295	
		I		72		102	174		
Tyrol	Innsbruck/Imst	A		130		115	245	729	866
		I		197		287	484		
	Wörgl/Kufstein	A		38		32	70	137	
		I		26		41	67		
Styria	Graz	A		365		295	660	1,124	1,124
		I		117		347	464		
Total		**A**	**175**	**1,129**	**249**	**952**	**2,505**	**5,499**	
		I	**175**	**977**	**318**	**1,524**	**2,994**		

A = active; registered as unemployed, I = inactive; previously registered as unemployed
Source: FIMAS gross sample

The attributes federal province, city, citizenship and registered status (currently or previously registered with the AMS) were taken into account as **stratification attributes** for sampling.

- The planned net sample per federal province was at approx. 200 to 240 persons. Due to the much higher number of refugees in Vienna and the greater ease of reaching them, the planned net sample for Vienna was at approx. 320 persons.
- The breakdown for Linz and Wels and the four cities in Tyrol was based on a query of their distributions in the AMS database.
- Of the 1,200 total planned interviews, around 530 were included in the sampling plan for each of the two largest groups – Syria and Afghanistan – as well as around 70 each for the other two groups – the Russian Federation and Iraq. These four groups made up 75% of all recognised refugees and beneficiaries of subsidiary protection registered with the AMS at the start of 2016.

- The project aimed at a balanced sample (50:50) between unemployment (status “A” for active) and gainful employment (status “I”, inactive), which means that the sampling was stratified accordingly, where possible.

Based on the estimated return, a **gross sample** of 5,500 persons was drawn. These included1,652 addresses in Vienna, 845 in the city and area surrounding Salzburg, 1,012 each in Linz and Wels and their environs, 886 in Innsbruck/Imst and their environs as well as Wörgl/Kufstein and their environs, plus 1,124 in Graz and its environs.

For reasons of survey logistics, only two groups were surveyed in every federal state – Syrians and Afghans. Surveys of persons from the Russian Federation were limited to Vienna, and those of persons from Iraq only took place in Vienna and Linz.

Sampling was implemented through an AMS service provider in late July 2016. The selection was random within the categories resulting from stratification combinations.

In addition to the stratification attributes, the resulting gross address sample included the postcode, town, citizenship and registration status (Active/Inactive as well as unemployed/in training, etc.), name and gender, as well as the last date of registration for persons not currently registered with the AMS.

3.3 Questionnaire and translations

The questionnaire was developed in close cooperation with project partners and stakeholders. The survey prioritised comprehensibility of the questionnaire. The low-threshold approach of the questionnaire was deemed necessary because a considerable number of respondents was expected to be illiterate or have a low level of education. To avoid a bias towards better educated persons, the questions and answer categories were formulated to be as simple as possible.

By July 2016, the **source questionnaire in German** was written up over several steps, shortened and the first pre-tests were carried out. The questionnaire was then translated into **Arabic, Dari, Russian, Chechen** and **English.**

Once the translations were available, a **review meeting was held** with the translators and the research team to discuss issues of linguistic features, deviations and the functional equivalence of the different versions of the questionnaire. The meeting also provided an opportunity for clarifying issues related to understanding the concepts included in the questionnaire and to provide instructions to the interviewers. For example, the terms *recognised refugees* and *beneficiaries of subsidiary protection* used in the invitation letters are not commonly used in Arabic. The Arabic version was therefore reformulated accordingly. The terms *occupation, job* and *work* had already been discussed when drafting the source questionnaire, and it was ultimately the term *work* that was most used throughout (conceptually, however, this primarily referred to paid work in certain occupations or employment relationships). In Arabic, the word *work* is usually associated with the word *labourer*, i.e. manual work, whilst *occupation* does not have the same connotations. As such, both terms were included in the first mention of the concept in the Arabic questionnaire for clarification.

The translators were also able to convert all dates stated to the Persian calendar system in the Dari version of the questionnaire. Dates were converted back to the Gregorian calendar during data entry or data cleaning.

The aim was for each interview to last an average of 30 to 40 minutes. Because the focus was on surveying employment and educational histories, questions about attitudes were left out, and blocks relating to possible factors influencing labour integration (such as health) were shortened. This meant the focus lay on the blocks of questions A to H, which covered questions about educational and training qualifications completed abroad and in Austria, recognition of foreign qualifications and participation in competence checks and values courses, as well as on blocks I to M on employment history abroad and

in Austria, labour market availability and volunteer work. This overall document was divided into a main questionnaire with supplemental pages to record information on the highest level of education or training achieved prior to immigrating to Austria. The questionnaire was followed by an optional contact sheet. In addition, an interviewer questionnaire was used for quality assurance purposes:

1) **Personal questionnaire plus supplemental pages with educational/training levels specific to each country of origin**
   - Education and training – BLOCKS A TO H
   - Work experience abroad – BLOCKS I, J, K
   - Work experience in Austria – BLOCKS L, N, O, P
   - Starting work – BLOCK M
   - German – BLOCKS Q, R
   - English and other languages – BLOCK S
   - Social networks – BLOCK T
   - Accommodation – BLOCK U
   - Health – BLOCK W
   - Social demographics – BLOCK X
   - Experiences and aspirations – BLOCK Y

2) **Contact sheet**
   - Consent to be re-interviewed
   - Personal contact information
   - Information on contact details for potential additional interview partners

3) **Interviewer questionnaire – BLOCK Z**
   - Presence of third party
   - Willingness to cooperate
   - Understanding of questions, language barriers

To make it easier for respondents to answer questions about educational levels completed and not completed, it was decided that questions would be asked about **educational levels specific to each country of origin**. To that end, a filter question (A) was asked at the beginning of the questionnaire to determine the country in which the respondent received their education. Depending on the information provided, the interviewers were then instructed to use the appropriate supplemental page. These supplemental pages were provided for Afghanistan, Syria, Iraq, the Russian Federation and Iran because it was known that some of the Afghans had grown up in Iran. In addition, in the event that a respondent had completed their primary education in another country, a general supplemental sheet was provided in German, which included descriptions of the internationally standardised educational levels (ISCED).

The supplemental pages specific to each country of education were developed based on information on country-specific educational systems in order to reflect the educational degrees respondents were familiar with in as much detail as possible. The country-specific overviews on the information portal for foreign vocational education systems compiled on behalf of the German Federal Ministry for Economic Affairs and Energy (Cologne Institute for Economic Research, 2017) provided a certain level of orientation. Country-specific ISCED mappings created by the UNESCO Institute for Statistics (UIS, 2017) were also used.

3.4 Interviewers

The project was able to recruit a number of highly qualified interpreters and social workers to serve as interviewers, primarily through the Diakonie refugee service. Interviewers had to be bilingual, with German and one of the interview languages as their working languages. Nearly every interviewer had Arabic, Farsi/Dari or Russian and Chechen as their first language.

Half-day training sessions were organised for interviewers in August and September 2016 in Vienna and Salzburg. The sessions presented the background of the project and information about asylum statistics, the competence checks and related studies, and went through the project's research design. The training focused on the role of the interviewers, communication in theory and practice, guidelines for different interview situations, features of standardised interviews and ethical standards when surveying refugees. The research team and interviewers kept in regular contact during the field work phase to quickly clarify any outstanding issues related to the questionnaire.

A total of 34 interviewers were used for the survey in all federal provinces, including:

- 19 for Arabic
- 13 for Dari
- 2 for Russian/Chechen

- 8 in Vienna
- 4 in Graz
- 4 in Salzburg
- 11 in Tyrol and
- 7 in Upper Austria

At least four interviewers per federal province were needed to cover, by means of double staffing, the week-long survey and particularly busy interview days during the interview weeks. The high number of interviewers, especially in Tyrol and Upper Austria, was due to the regional distribution within the federal provinces on the one hand, and interviewers occasionally dropping out (this also applied to Vienna) on the other.

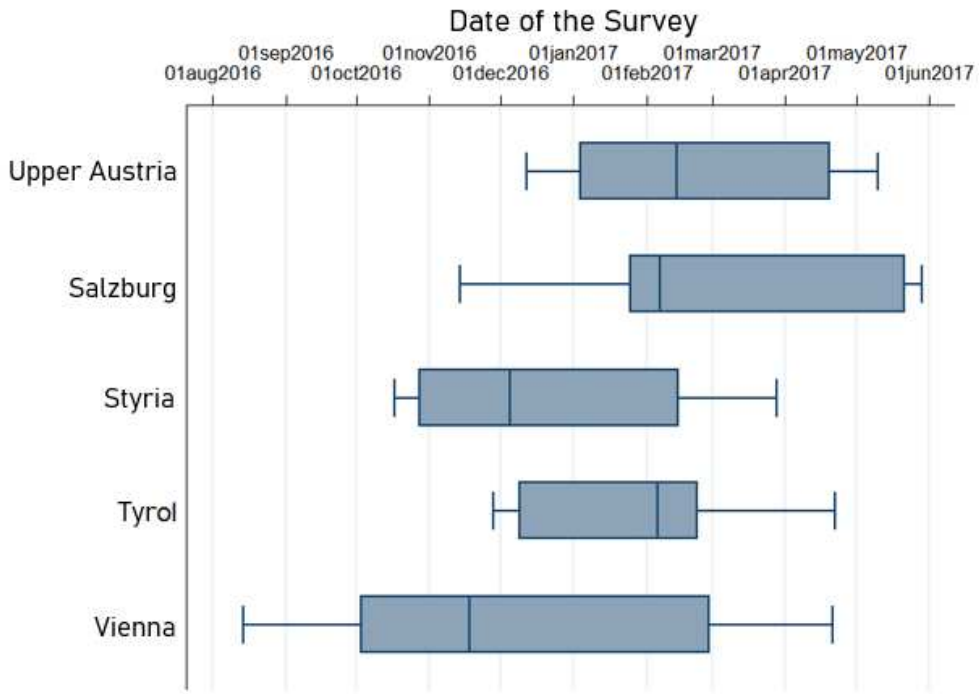
3.5 Data collection, net sample

In order to ensure the quality of the data and to minimise difficulties in communication, personal interviews (face to face) were chosen as the sole survey mode. Based on the address sample, information letters were sent to the target group, with an invitation to attend an interview in their respective city. Depending on the federal province/city and target group, bilingual invitation letters were formulated (in the respective first language and in German), which provided information on the study, the organisations conducting it (ICMPD, Diakonie Flüchtlingsdienst, ZSI and AMS), the voluntary nature and anonymity of the survey, as well as the respective interview location and possible interview times. Additionally, it was noted that the interview could be conducted in person in the respondent's first language using interpreters. A mobile telephone number was also listed for persons to call should they have questions about the study.

Mostly surveys were offered over a period of one or two weeks from Monday to Friday between 8 a.m. and 3 p.m. Most of the interview locations were Diakonie facilities in Vienna, Salzburg and Innsbruck, AMS regional offices in Graz, Linz, Traun, Innsbruck, Imst, Kufstein and Vienna, as well as rooms provided by the Vocational Training Institute (bfi) or the Austrian Chamber of Labour.

The first field phase began in Vienna in August 2016 (Graph 3). Field work began in Graz in October, in Tyrol and Salzburg in November, and in Upper Austria in December 2016.

Graph 3. Field phases by federal province



Source: FIMAS data

Based on a random gross sample of 5,497 postal addresses, letters were sent out to several hundred respondents in phases, depending on the federal state. Non-response due to errors in the address data were hardly relevant; almost all neutral non-response was accounted for by persons who had moved (354) and returns marked "unknown" (367) (Table 5), which could either indicate the person had moved or incorrect address information was provided. After deducting the returns, the adjusted gross sample was 4,758 addresses.

Table 5. FIMAS gross sample and response rate

Phase 1: Probability Sampling	n	Shares
Gross sample	**5,497**	**100%**
Neutral non-response		
wrong address	6	0.1%
no indication	7	0.1%
not accepted/deposit not used	5	0.1%
moved	354	6.4%
unknown	367	6.7%
total	739	13.4%
Adjusted gross sample	**4,758**	**100%**
Non-neutral non-response		
interview abortions	1	0.0%
total	1	0.0%
Realised interviews/response rate	514	10.8%

Source: authors' calculations

The first field phase, using probability sampling, yielded 514 complete interviews, making for a 10.8% response rate, which amounted to around half of the responses that had been expected. Presumed reasons for this included a) the lack of incentives – the persons invited were not offered any financial compensation for the expense incurred by taking the time to participate; b) the excessive distance from

home to the survey location in some cases, especially for those living in outer conurbation areas; c) survey times being limited to weekdays and times of day that often conflicted with the working hours of persons working full time; and (d) reservations or unfamiliarity with the concept of a scientific survey in general.

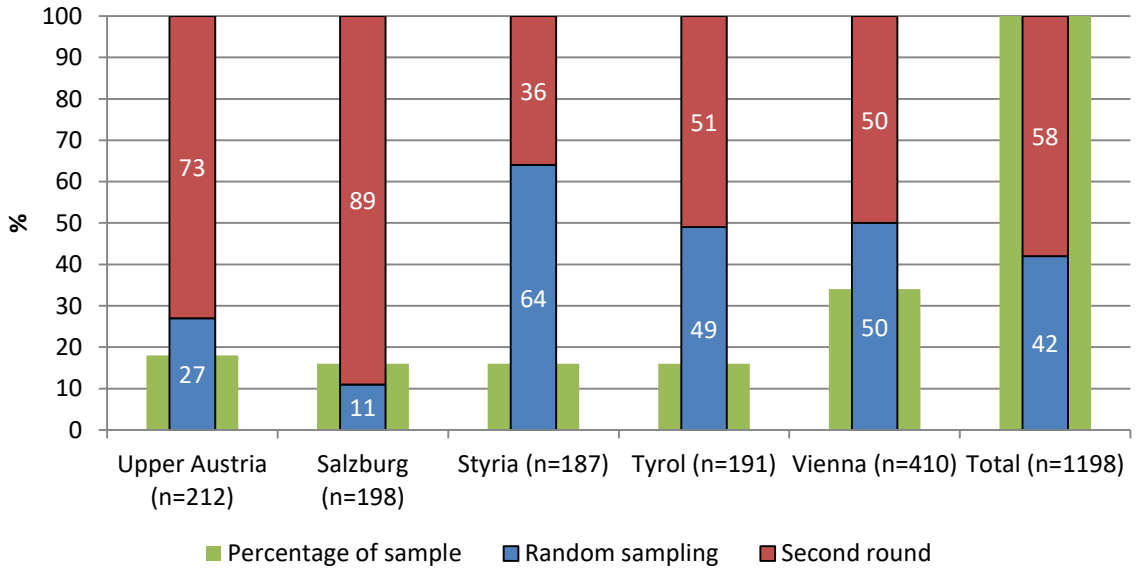
Towards the end of this first field phase, it became apparent that the goal of 1,200 interviews could not be reached with this survey strategy. Thus, several additional strategies were developed to reach the total number of planned interviews as well as the sub-samples by federal province. The primary focus was on institutions in contact with the target groups that had agreed to lend their support to the survey.

The highest proportion of interviews achieved through random sampling was in Styria, where a total of 64% of respondents were interviewed after receiving an invitation through a personal letter. Salzburg and Upper Austria had the lowest number of interviews during the first survey phase. Most of the participants in these two federal provinces were found through contacts during the second wave of the survey.

In Vienna, during the second phase, the interviews were held at Interface Vienna, at the Youth College run by Start Vienna, at the Zinnergasse base of the Diakonie refugee service, and, especially for interviews with Chechens, through the interviewers' personal contacts and networks.

In Upper Austria, interviews were conducted at bfi Linz and at the "die Berater" training facility in Linz. In the city of Salzburg, interviews during both phases were held at the "Integrationshaus" run by the Diakonie refugee service. During the first phase, respondents were contacted by postal invitation to do the interview at the Integrationshaus, whilst interviews were conducted with Integrationshaus residents and clients during the second phase. Caritas provided support for the study in Graz by conducting interviews with clients. In Tyrol, as in Salzburg, clients of the Diakonie refugee service were interviewed. In addition, support was provided in Wörgl through a civil society organisation.

Graph 4. Sampling by federal province



Source: FIMAS data

These forms of convenience sampling do not meet the strict standards of random or probability sampling, but are often the only alternative, especially when working with hard-to-reach target groups. In order to be able to test possible sample biases that result from this, the two phases of the field work were recorded as a variable. This enabled methodical comparisons of the two sub-samples to be carried out.

It is indeed possible that distortions arose precisely because of these supplementary strategies specific to each federal province, for example, in educational qualifications or labour market status in the sample, such as when the persons being interviewed were currently in education or training, meaning the sub-sample for that federal province contained a particularly high number of persons in education or training. The sampling variable can be used to control this effect.

Table 6. FIMAS net sample

Phase 2: Convenience Sampling		
Non-neutral non-response		
interview abortions	4	
Realised interviews	717	
Total net sample	**1,231**	
Data Cleaning		
communication problems	10	0.8%
missing basic data (non-response)	7	0.6%
outside sampling criteria	16	1.3%
total	33	2.7%
Data set for analysis	**1,198**	

Source: authors' calculations

Overall, the remaining 717 interviews were the result of the second survey phase. Together with the 514 interviews conducted during the first phase, the total net sample comprised 1,231 cases (Table 6).

The average interview duration was 30 minutes. One quarter of the interviews lasted 22 minutes or less (first quartile), one quarter lasted 37 minutes or longer (third quartile), and nine out of ten interviews lasted 47 minutes or less (ninth decile). Discontinued interviews played a minimal role – only five persons ended the interview early.

3.6 Data processing

All paper questionnaires were collected and manually entered into a data file. During this step, extra care was taken to separate the personal contact details also collected (on a supplemental page) from the personal questionnaire after ensuring the possibility of linking them later on for subsequent surveys using a unique key.

During data entry and data cleaning, consistency checks were carried out several times to ensure the completeness and accuracy of the data. Of the total net sample, 33 cases were ultimately excluded, which means the analysis dataset therefore contains 1,198 individuals (Table 6).

Not included in the dataset were 16 interviews because the persons interviewed clearly did not belong to the target group based on their attributes. This was only the case for persons aged 60 or over. A further ten interviews were excluded because there were serious communication problems during the interview, which cast doubt on the quality of the data collected. In addition, seven interviews were not included because they were missing several key data points on socio-demographic attributes such as gender, age, citizenship or residence status.

Some borderline cases were retained in the analysis dataset and excluded only for certain analyses. For example, the dataset includes an interview with a 14-year-old and three interviews with 60-year-olds – through an apprenticeship or prior to retiring, these persons could by all means be available to the labour market, and follow-up interviews in additional surveys are still of relevance here. In addition, the data also included 31 asylum seekers, 13 of whom stated that their asylum procedure had been

completed. In principle, this is possible if the procedure continues to the second instance (the questionnaire did not differentiate in this regard). Broken down by nationality, there were 55 stateless persons, 13 Austrian nationals, four Iranian citizens, four Somali citizens and three other nationalities included in the data beyond the four target groups. This does not necessarily mean that these individuals were not part of the intended group, since citizenship can change or dual citizenship can be attained. This was one of the reasons why a combined variable analytic group was created for most of the analyses in this report, which contains information on citizenship, country of birth, and country of education (see section 4).

Strictly speaking, persons who came to Austria prior to 2006 (primarily Russian citizens) were also outside the definition of the target group. Considerations relating to the analysis of integration processes over time meant that these cases were retained in the dataset despite the original restriction of the sample to those arriving between 2006 and 2017.

4 Analysis

The following section focuses on which resources help refugees in Austria to find a job as quickly as possible. It also looks at the economic sectors in which refugees find work and what future prospects arise for them. The views and hopes of the interviewees themselves are also included.

All analyses below refer to the sample data and, depending on the makeup of this group as a whole or its sub-groups, should be understood as estimates that contain a margin of error.

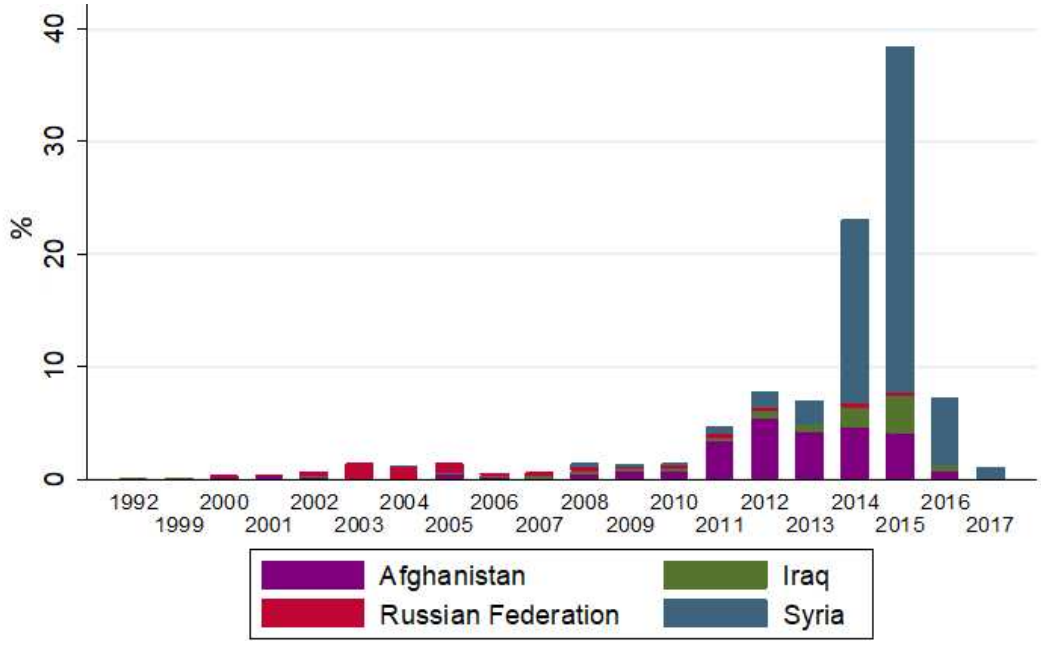
This can be clearly illustrated using the example of employment status. In the overall group of respondents, the proportion of employed persons is 15.3%. The 95% confidence interval (not taking into account possible bias) for this proportion ranges from 13.4% to 17.5%. This is equivalent to a margin of error of +/- 2% for the overall results, and for numbers (shares) in the vicinity of 50% even up to +/- 3%. Analyses specific to the federal provinces of Tyrol, Salzburg, Upper Austria and Styria include a margin of error of up to +/- 7%, whilst analyses for Vienna include a margin of error of up to +/- 5%.

4.1 Socio-demographics

According to the target group definition, the survey interviewed respondents who had come to Austria in the past several years. The arrival years 2014 (23% of all cases) and 2015 (38%) provided the most cases. The cohorts of 2011 to 2013, as well as 2016, each account for less than 8% of cases; and the cohorts for the years prior to 2011 account for less than 2% each. This distribution reflects the overall asylum trends in Austria for the same period. Annual asylum applications had reached their peak level in 2015, with the number of applications declining sharply by 2016 (see Graph 71 in the Appendix). The same is true of positive decisions to grant asylum. Here, too, peaks were reached in 2014 (8,734), 2015 (14,413) and 2016 (22,307) (Statistics Austria, 2017c). The distribution within the groups surveyed in terms of **arrival cohorts** is highly heterogeneous, since the date of arrival varies due to war and conflict situations in the various countries of origin. Many interviewees were persons who had left Syria and arrived in Austria since 2014 and after; persons from Afghanistan were much more broadly dispersed in terms of their date of arrival (Graph 5Table 5). The sample primarily includes persons who have entered Austria since 2011 – this, too, corresponds to the trend in positive asylum decisions, which rose by 41% in 2011 compared to the previous year (Statistics Austria, 2017c).

Significantly fewer cases were included in the sub-samples of individuals from Iraq and the Russian Federation. Nevertheless, there are also specific distributions that roughly correspond to the years with the highest levels of arrivals from these states. Respondents from Iraq primarily entered in 2014 and 2015, and the following years, 2015 and 2016, are also among those with the highest number of positive decisions for this group in the statistics on asylum decisions. The same applies to persons from the Russian Federation, most of whom represented in the sample entered between 2003 and 2005. The asylum decision statistics for 2003 and 2004 also show large leaps in decision figures. The highest number of positive decisions for individuals from the Russian Federation was issued between 2004 and 2007. Since then, those numbers have been declining, as reflected in the sample. In sum, the number of respondents by immigration cohort is similarly distributed as in the statistics on positive asylum decisions, as is to be expected in a random drawing.

Graph 5. Immigration cohorts by analytic group (n=1,160)

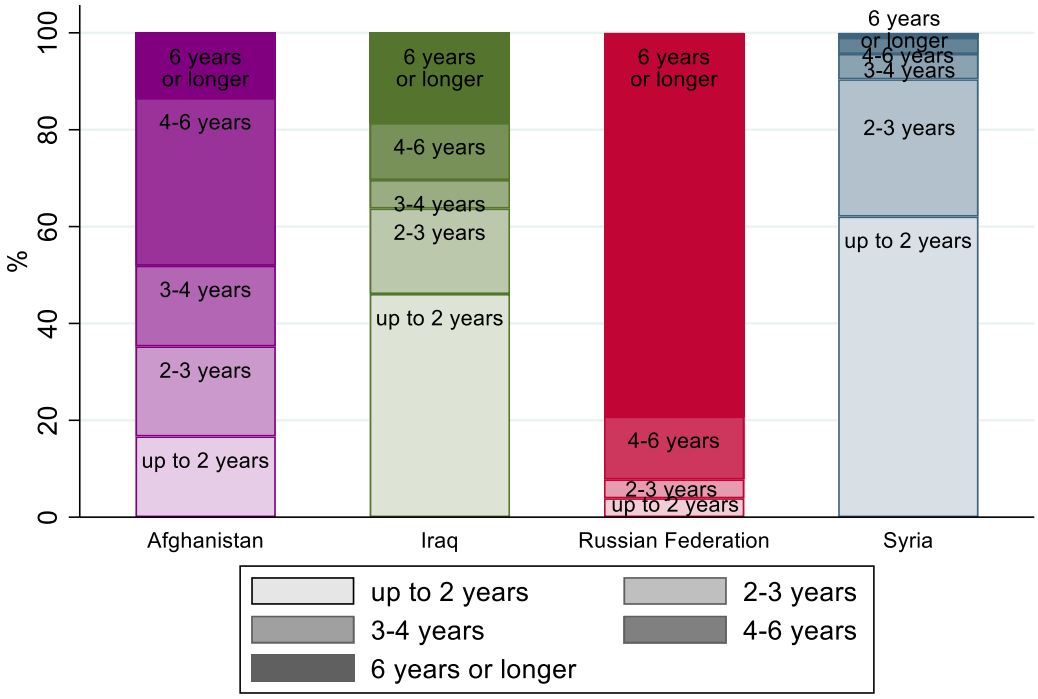


Source: FIMAS data

The result is a highly heterogeneous composition of the groups by **duration of stay**. For the most part, persons from the Russian Federation have been living in Austria for more than six years (79%, Graph 6); considerably longer than the other groups. Of the persons from Afghanistan, 13% had already been living in Austria for six years or longer, 35% had already been living here for four to six years, 17% had stayed in Austria for three to four years, 19% for two to three years and 17% for less than two years.

The duration of stay for persons originating from Iraq was shorter than for the first two groups. The largest share comprised persons who had stayed in Austria for less than two years (46%). The duration of stay was even shorter for persons originating from Syria – nearly all of them (90%) had arrived in Austria in the three years prior to the survey, i.e. between the summer of 2014 and 2017.

Graph 6. Analytic groups by duration of stay in Austria (n=1,179)



Source: FIMAS data

It should be noted that Graphs 5 and 6, as well as the remainder of the report, present **analytic groups** based on the citizenship attribute but supplemented with further information (country of birth and country of education) for cases where values were missing or changes to citizenship occurred.

For example, 55 of the persons interviewed were stateless or indicated that they held "Palestinian" citizenship (Table 7). Many of these persons were born in Syria and were, therefore, classified under the analytic group of refugees from Syria. In total, 49 stateless persons were assigned to one of the four target groups based on their country of birth. Similarly, those who held Austrian citizenship, but were born in one of the four target group countries, were also categorised in these groups. Another striking feature of the cross-classified table for nationality by country of birth is that 20 persons (7%) of Afghan citizenship were born in Iran and eight (3%) were born in Pakistan. A number of Afghans may have moved to Austria directly from Iran.

Table 7. Nationality by respondents' country of birth

Country of birth / Citizenship	Afgha-nistan	Iraq	Russ. Fed.	Syria	Iran	Pakistan	Other/ missing	Total
Afghanistan	274	0	0	2	20	8	1	**305**
Iraq	1	95	1	0	0	0	0	**97**
Russ. Federation	0	0	66	0	0	0	3	**69**
Syria	10	2	0	596	0	0	26	**634**
Stateless/Palestine	3	3	2	41	0	0	6	**55**
Austria	0	5	4	3	0	0	1	**13**
Other/missing	7	0	1	7	4	0	6	**25**
Total	**295**	**105**	**74**	**649**	**24**	**8**	**43**	**1,198**

Source: FIMAS data

The cross-classification of interviewees' country of birth and country of education, i.e. where interviewees were asked which country they learned to read and write in, offered a more differentiated picture, especially for persons born in Afghanistan. Only 52% of those born in Afghanistan (152 persons in total) had achieved literacy in Afghanistan (Table 8). Of this group, 17% learned to read and write in Austria, whilst 11% learned that in Iran. Of respondents born in Afghanistan, 12% stated that they were illiterate.

Table 8. Country of education by respondents' country of birth (n)

Country of birth / Country of education	Afgha-nistan	Iraq	Russ. Fed.	Syria	Iran	Pakistan	Other/ missing	Total
Afghanistan	152	0	0	0	3	3	0	**158**
Iran	33	0	0	0	20	0	0	**53**
Iraq	1	100	1	0	0	0	0	**102**
Russ. Federation	1	0	71	3	0	0	2	**77**
Syria	11	1	0	602	0	0	28	**642**
Austria	50	2	2	1	0	1	0	**56**
Other/missing	13	0	0	5	0	4	8	**30**
Illiterate	34	2	0	38	1	0	5	**80**
Total	**295**	**105**	**74**	**649**	**24**	**8**	**43**	**1,198**

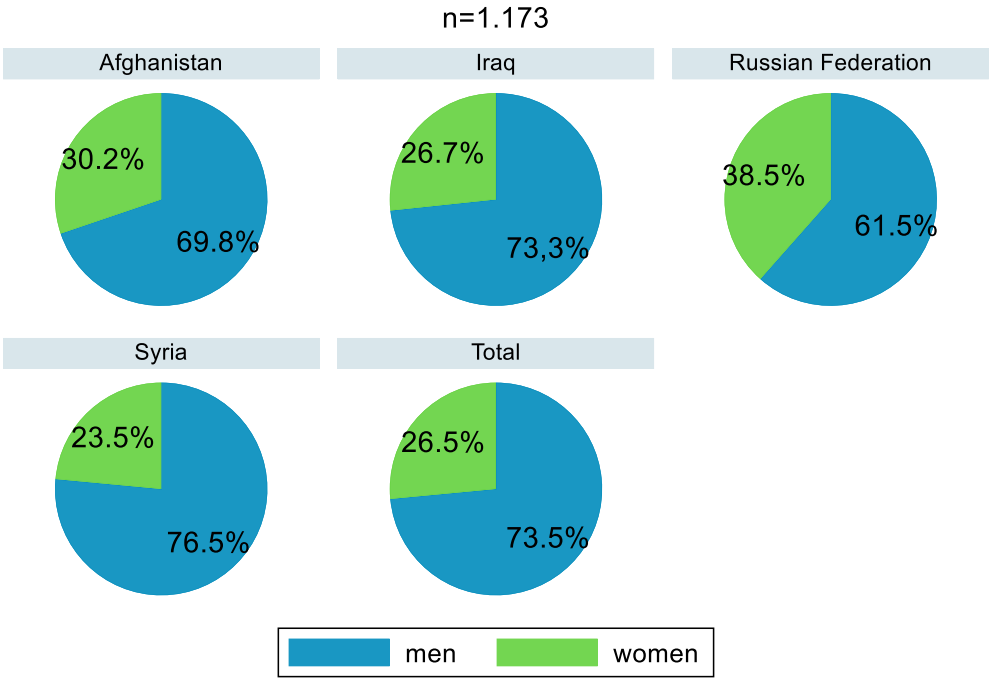
Source: FIMAS data

It should also be noted that the recognised refugees and beneficiaries of subsidiary protection who originated from the Russian Federation and were interviewed in the survey were almost exclusively ethnic Chechens. This became visible in the breakdown of the analytic groups by first language (for an example, see Section 4.3).

Taking the four analytic groups together, 73.5% of the interviews were conducted with men and 26.5% with women (Graph 7). These shares are almost identical to those of the AMS statistics (72.9% men, 27.1% women, see Section 2.1). Broken down into analytic groups, the relatively higher proportion of women among Chechens/Russians stands out (38.5%), as this rate is well above the average for the other groups. This also corresponds to the higher proportion of women among the positive asylum decisions received among this group compared to the other three groups.

The proportion of women among respondents from Afghanistan was 30.2%, 26.7% among those from Iraq and 23.5% among those from Syria. Methodological analyses indicate that women were under-reported in all groups, as the numbers reflected in asylum decision statistics are higher.

Graph 7. Analytic groups by gender

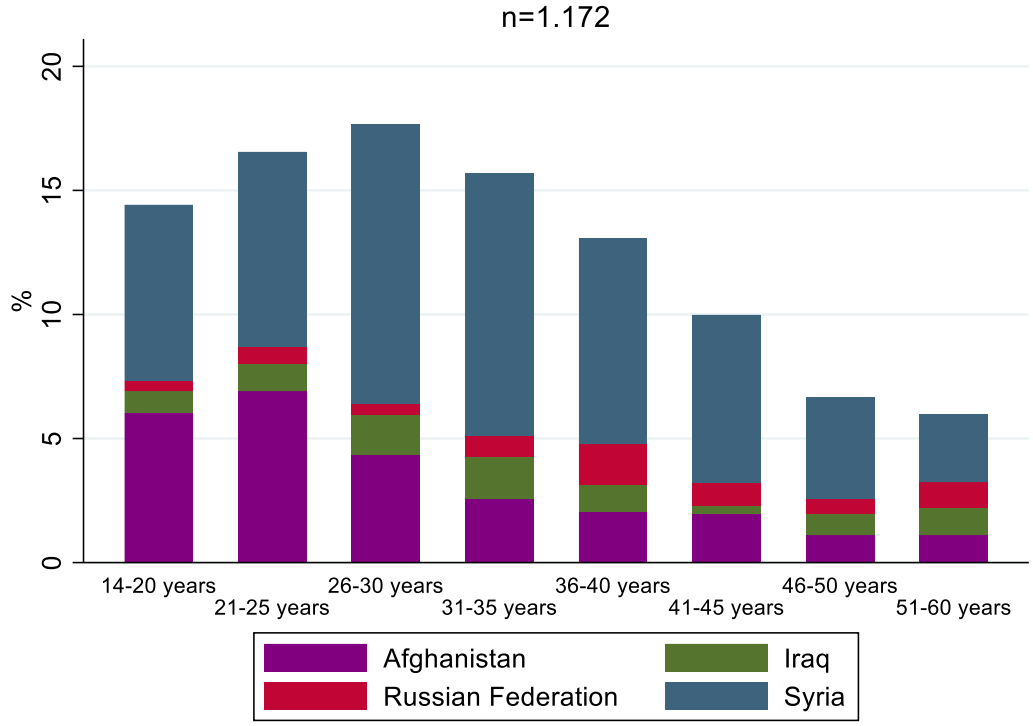


Source: FIMAS data

The **median age** of respondents was 31 years. There were clear differences between the analytic groups. Respondents from the Russian Federation were markedly older (median age of 38.5) and respondents from Afghanistan were younger (median age of 26). Respondents from Syria (median age of 32) and Iraq (median age of 33.5) were only slightly above the overall average.

This also became evident when examining age categories by analytic group (0). Afghans were overrepresented in the younger age groups, whilst Chechens/Russians tended to be overrepresented in the age groups between 36 and 45 and between 50 and 60. Overall, half of all respondents fell in the group between 21 and 35 years of age: 16.6% were between 21 and 25 years old, 17.7% between 26 and 30 years old and 15.7% between 31 and 35 years old. Only 14.4% of respondents fell in the age group of 14 to 20 years, whilst 6.0% fell in the 51 to 60 years age group.

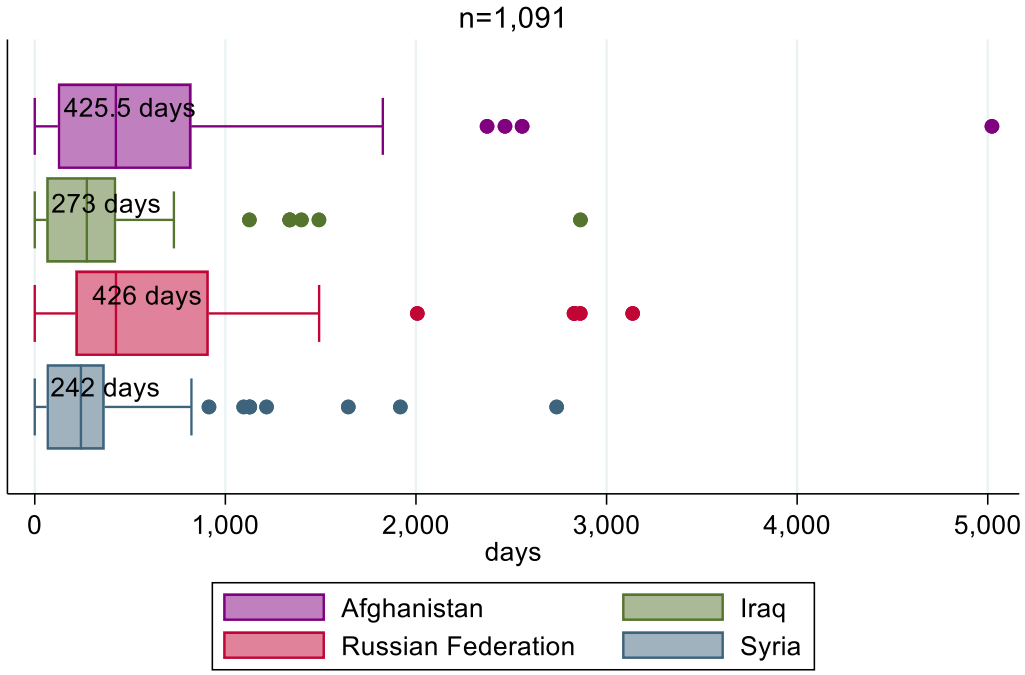
Graph 8. Age categories by analytic group



Source: FIMAS data

An estimated **length of asylum procedures** can be calculated based on the date of arrival in Austria and the date on which the asylum processes were concluded (Graph 9). For persons from Afghanistan and the Russian Federation, median processing times were more than 420 days (around 14 months). This was markedly different than the, on average, much shorter processing times for Iraqis (273 days) and Syrians (242 days). Three quarters of Syrians (78%) and 67% of Iraqis, 45% of Chechens/Russians and 48% of Afghans received a positive decision within the first year of stay in Austria.

Graph 9. Estimated length of asylum procedure by analytic group



Source: FIMAS data

4.2 Qualifications

This section describes which qualifications refugees and beneficiaries of subsidiary protection in the sample had obtained in different countries. The analysis takes into account both the highest level of formal qualifications as well as the employment experience of refugees upon their immigration to Austria.

Because the education and training systems in the different countries of origin vary widely, information on education and training was collected in the FIMAS project using country-specific questionnaires. Specific questionnaires were designed for Afghanistan, Iraq, Iran, Syria and the Russian Federation. Educational qualifications from other countries were recorded in line with the 2011 International Standard Classification of Education (ISCED). Country-specific information was then recoded into categories commensurate to the 2011 ISCED during the course of data preparation. This step was based on the reference tables issued by the UNESCO Institute for Statistics (UIS, 2017). The country-specific overviews on the information portal for foreign vocational education systems compiled on behalf of the German Federal Ministry for Economic Affairs and Energy (Cologne Institute for Economic Research, 2017) were used to classify the qualifications. Using the portal, qualifications obtained abroad were divided up into primary and lower secondary education, upper secondary and post-secondary education, and tertiary qualifications. Vocational education and training was also included in the classification if the duration of professional training indicated by respondents was at least three years.[6] On average, this corresponds to the duration of an apprenticeship in Austria. The duration of professional training was assessed separately for each form of employment carried out abroad. If an interviewee had worked in two different occupations, the occupation with the longer training time was used to calculate the duration of training.

The chosen approach of categorising respondents into three groups based on levels of education largely corresponds to the aggregated presentation according to ISCED categories, as is often used in international surveys. The three categories used were: low level qualifications (ISCED 0-2), intermediate level qualifications (ISCED 3-4) and higher level qualifications (ISCED 5-8) (EUROSTAT, 2017b). ISCED level 3 includes upper secondary education relating to general education or post-secondary vocational education and training. Level 4 includes post-secondary non-tertiary qualifications that, although they had been completed following secondary education, are not yet considered tertiary. The fifth level of ISCED classification includes post-secondary qualifications that impart job-specific knowledge, such as vocational schools from the fourth year onwards, colleges or master schools (OeAD, 2014).

Such aggregated presentation is preferred in international surveys because the ISCED classification does not provide information on the value of individual, country-specific qualifications. Classifications along the ISCED scheme are based on information on admission requirements, minimum age of matriculation, duration of education and the sequence of education, and the type and designation of the educational institution, but not on the comparison of curricula and content (Schneeberger, 2010, p. 11). Furthermore, country-specific educational qualifications do not always lend themselves for clear equivalency to an ISCED qualifications level (UIS, 2017). As such, despite the use of this reference system, one cannot assume that educational qualifications are directly comparable.

For example, the measurability of educational qualifications obtained from Afghan religious schools is debatable. This kind of education takes up to 12 years but focuses primarily on the study of the Qur'an and, later on, Islamic studies (Cologne Institute for Economic Research, 2017). Against this backdrop, an educational qualification acquired from a religious school would be assessed as secondary

[6] Participants were asked how they learned a trade they were engaged in abroad and for how long (blocks I and J of the questionnaire).

education. However, essential content that is required for an upper secondary degree (Matura) in Austria would not have been learned by Afghans who attended religious schools.

Assessing levels of acquired knowledge is also problematic, since learning in other countries is often less formal than it is in Austria, and is not necessarily accompanied by attendance at a vocational school.[7] The sector-specific subsystems of national qualification systems (in vocational schools, colleges, etc.) that are at the core of the Austrian educational system significantly limited the comparability of vocational qualifications. For the reasons stated above, more detailed representations often suggest comparability, which is, in fact, not the case, nor is comparability reflected in the transferability of qualifications and marketability of professional experience in the country of immigration (Biffl et al., 2016, p. 26).

In this report, professional experience is presented on the basis of up to two instances of gainful employment exercised prior to immigrating to Austria. The information on the professional field and the activities carried out as part of respondents' occupations was determined in each case through the use of two open questions, which were then categorised during data processing. The International Standard Classification of Occupations (ISCO 08) was used to classify professional fields. Information on professional fields was included at the minor occupational group level (three-digit code). Assessments were primarily undertaken at the major occupational group level (one-digit code). Persons working in different professional fields were attributed to both professional fields.

Furthermore, information on the scope of employment abroad and the duration of professional training was used to estimate employment experience. The scope was calculated as the sum total of the length of employment stated in both professions worked abroad. If the start dates of the two occupations worked abroad overlapped, the job that lasted longer was used to calculate the scope of employment experience. Duration of professional training was calculated in addition for each occupation.

4.2.1 Qualifications obtained prior to arrival in Austria

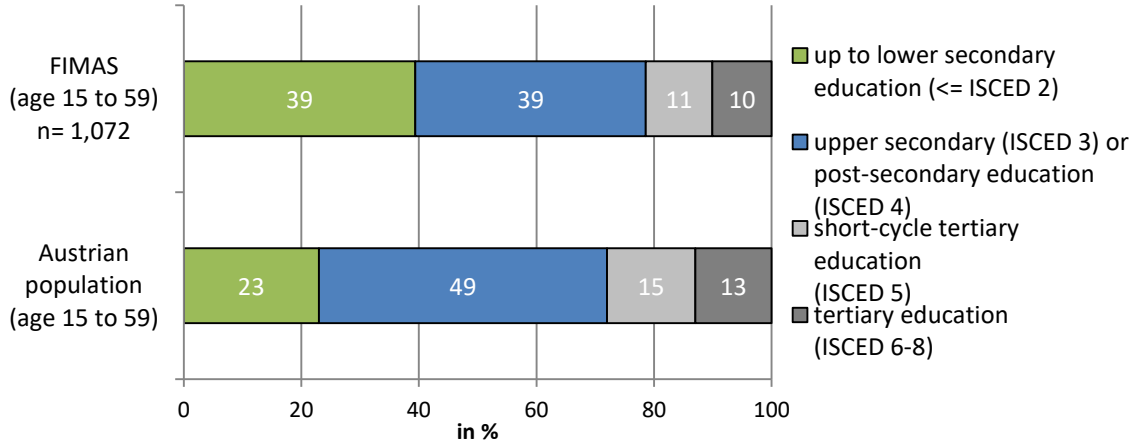
Four out of ten respondents (39%) aged between 15 and 59 had a low level of educational qualifications (up to ISCED 2) at the time they arrived in Austria. 39% had intermediate educational qualification level (ISCED 3-4), and slightly less than 21% had reached a high educational qualification level (ISCED 5-8) (0). This indicates that the refugees and beneficiaries of subsidiary protection included in the sample had a **lower overall level of qualification** when entering Austria compared to the Austrian population. The proportion of persons who had achieved compulsory education as the highest educational level, in particular, was higher among the 15 to 59-year-olds included in the FIMAS sample than in the comparable age group among the total Austrian population in 2015.

The proportion of compulsory school graduates among respondents was above average especially in Salzburg.[8] In Vienna, the high share of university graduates was striking (Graph 11). At 21%, the proportion of university graduates in Vienna was roughly equivalent to that of the city's overall population (Statistics Austria, 2017f). The differences in the educational levels of respondents in different federal states reflected the heterogeneity of the group as a whole.

[7] Although these reasons were taken into account in the ISCED 2011, classification remains difficult.

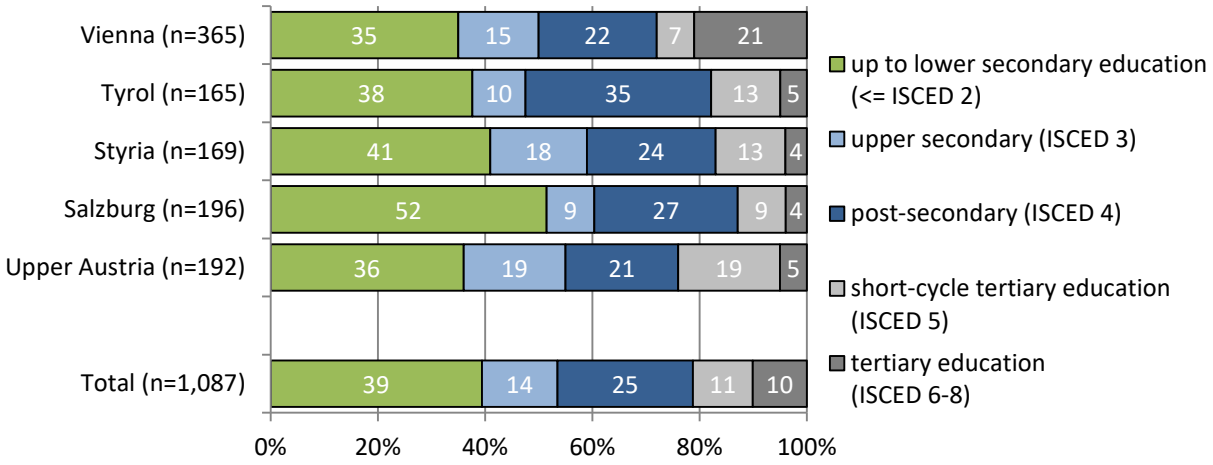
[8] This may be an effect of the sample generation since many respondents could only be reached during the second survey phase.

Graph 10. Educational level (ISCED 2011) of the FIMAS sample and the general population between 15 and 59 years of age in Austria in 2015



Source: FIMAS data; Statistics Austria, 2017f

Graph 11. Educational level by federal province



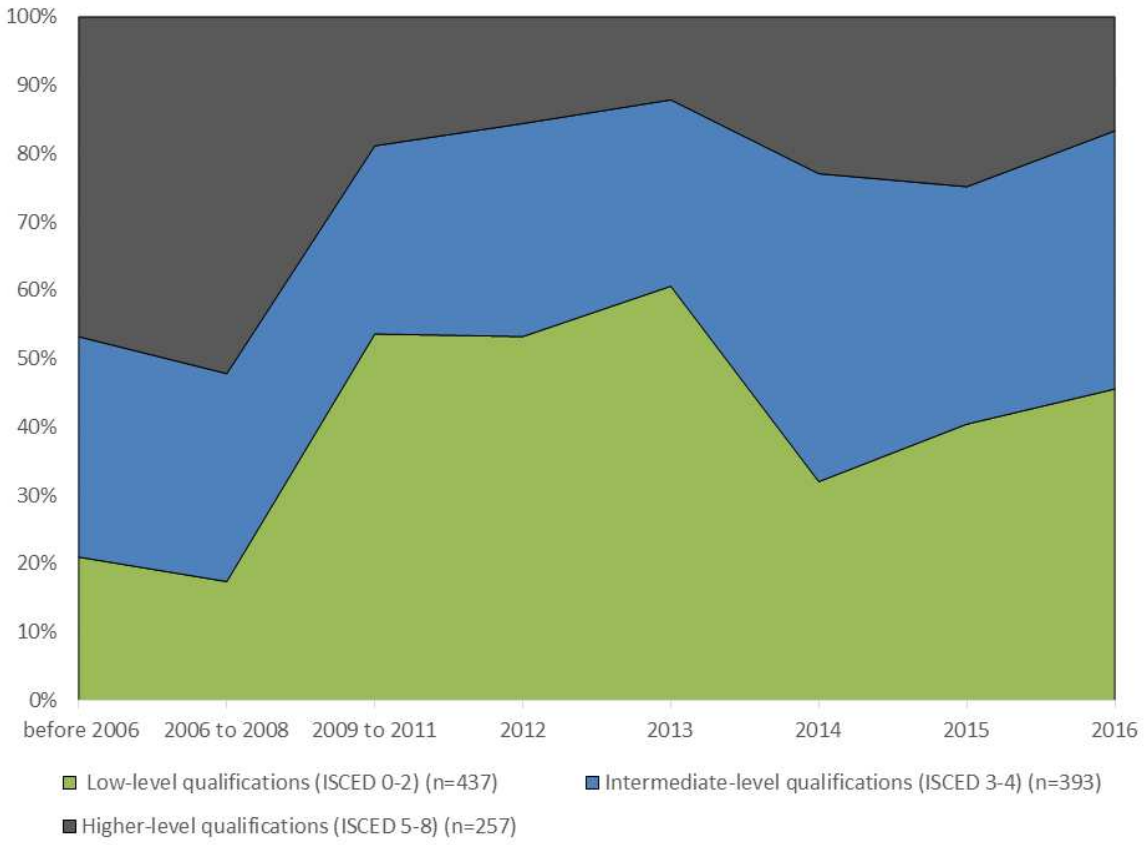
Source: FIMAS data

In order to analyse the differences between qualifications by country of education and country of origin, information on respondents' citizenship and country of education was consolidated. For those with Austrian citizenship, information about their country of education was used for comparison. This applied, in particular, to Russian citizens. The qualifications held by the four largest refugee groups in the sample are described in greater detail: Syrians, Iraqis, Chechens/Russians and Afghans.

Respondents who had arrived in Austria prior to 2008 had the **highest levels of education** (0). Among them, Russians (most of whom were **Chechens**) made up the largest group (see Table 16 in the Appendix). Together with Iraqis, they accounted for the **best-educated** group among the FIMAS sample. About half of the respondents from each of these two countries had already obtained a higher qualification level (ISCED 5-8) upon their arrival in Austria. Detailed analysis shows that persons with academic qualifications are predominant in both groups.[9]

[9] Chechens were difficult to reach for the survey. Cf. the comments on sampling in Section 3. It can be assumed that university graduates are over-represented in this sub-sample.

Graph 12. Educational level by year of arrival in Austria



Source: FIMAS data

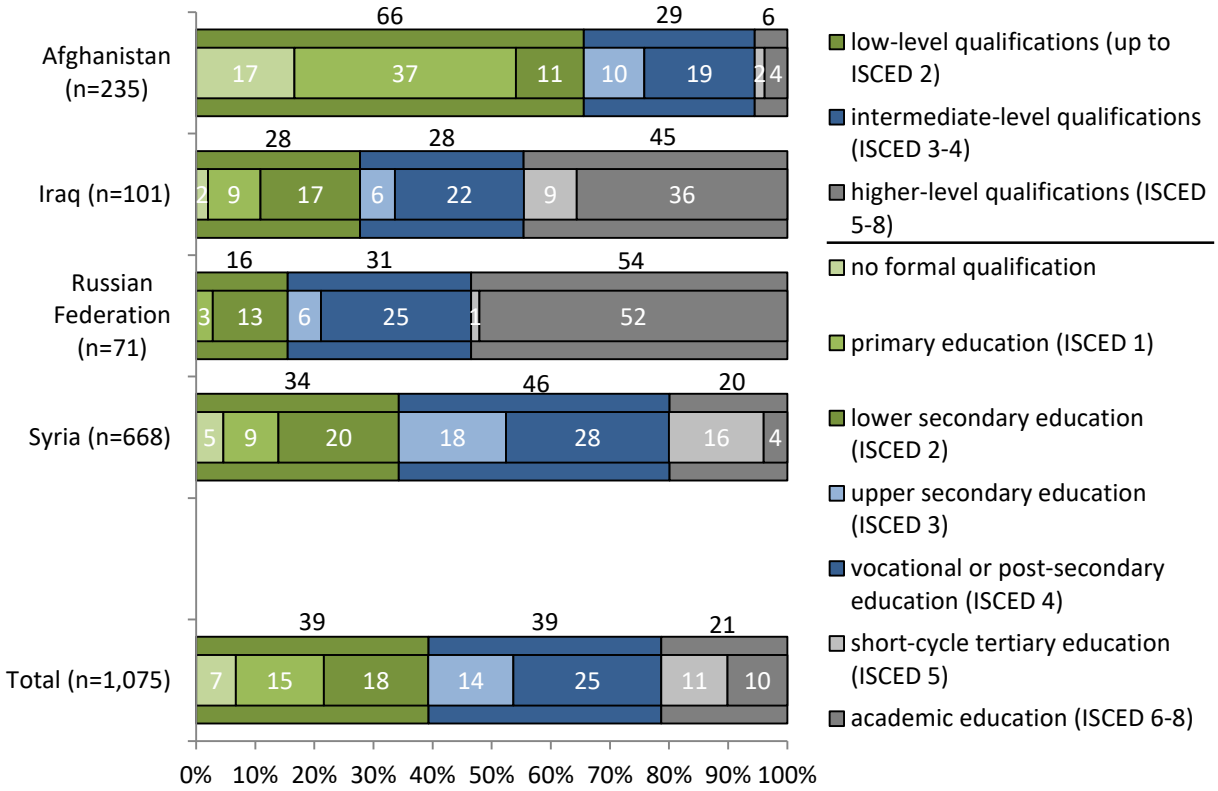
From 2008 onwards, the share of **persons with low qualifications** among the new asylum seekers arriving in Austria started to increase. This effect was mainly caused by the high proportion of Afghans among the cohorts of asylum seekers arriving in Austria during this period, as they had generally obtained the lowest qualifications relative to the other groups of origin in the FIMAS sample.

Of the **Afghans**, 66% had a low qualification level prior to entering Austria, meaning they had earned at most a school leaving certificate for the compulsory level of education (**Fehler! Verweisquelle konnte nicht gefunden werden.**). The proportion of Afghan respondents with formal tertiary or academic qualification was 6%. Afghans educated outside Afghanistan generally had higher educational qualifications. Whilst the majority of respondents educated in Afghanistan completed at most primary education (45%), at least 20% of those who attended school outside of Afghanistan prior to arriving in Austria had completed compulsory education, and 22% had completed upper secondary education. As of 2014, the proportion of **Syrians** among the group of recognised refugees and beneficiaries of subsidiary protection increased, which also raised the proportion of persons with **intermediate qualification levels**.

Information on age at the time of completion of their most recent educational level points to differences between educational systems. For example, 20% of refugees claiming to have post-secondary non-tertiary qualifications were under the age of 19 at the time of completion.[10] It was not possible to differentiate by country due to the small number of cases. However, the trend shows that, especially in the Russian Federation, a comparatively high proportion (33%) of respondents earned post-secondary non-tertiary qualifications between the ages of 16 and 18 years old.

[10] Afghanistan was not taken into account here.

Graph 13. Educational level by analytic groups



Source: FIMAS data

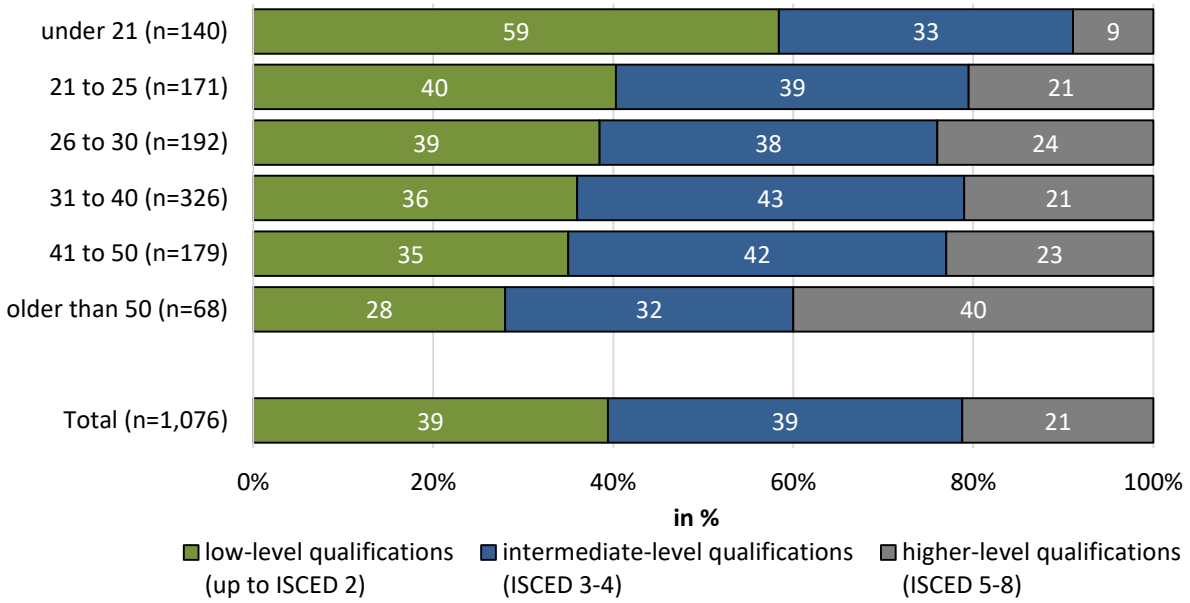
Although the proportion of persons with low levels of qualification is high among recognised refugees and beneficiaries of subsidiary protection, one can assume that those who come to Austria are nevertheless part of a group with comparatively higher educational qualifications in their countries of origin. Also persons from Afghanistan show this **positive selection of more highly qualified persons** (Buber-Ennser et al., 2016). Notwithstanding this, overall qualification levels have decreased over time. Highly qualified persons were most strongly represented among the earlier arrivals during the observation period, regardless of their country of origin. For example, among the Afghans sampled, the proportion of low-skilled workers who had come to Austria before 2009 was 55%, but increased to 71% for those who came to Austria after 2010. The same trend was evident for Syrians. Among them, the proportion of persons with a low level of qualifications rose from 30% to 40% after 2015.

The reason for this positive selection of more highly educated individuals and the initially slow alignment of refugees' educational attainment to the average level in their country of origin (Berger et al., 2016, p. 13) is that wealthier and, therefore, often more educated parts of the population tend to be the first to leave the country. Those who had little or no access to formal education in their country of origin often move to neighbouring countries or other regions within their country of origin because of the costs associated with fleeing to Europe. This trend only changes once an increasing number of refugees start leaving a conflict country or region. What is more, ongoing wars and conflicts in refugees' countries of origin lead to fewer educational opportunities (Al Hessan, 2016, p. 4) and subsequently also to a decline in the qualification levels of those who come to Europe in general, or Austria in particular, at a later point in time.

In addition to the differences in respondents' level of education based on their country of origin, gender and age are also distinctive attributes. Younger refugees, in particular, who were unable to complete their education in their country of origin and who only sometimes managed to continue their initial education after fleeing (Al Hessan, 2016, p. 32), on average, had low levels of qualification upon their arrival in Austria. Afghans are an exception to this. Among them, the proportion of persons with low-level qualifications in the older generation (over 41 years old) is higher than in the younger generations

of refugees, who often had attended school outside of Afghanistan. Compared to the total number of respondents included in the FIMAS survey, respondents older than 50 years old, in particular, tended to have higher qualifications (Graph 14). This group was predominantly made up of Chechens/Russians. The majority of these persons had been living in Austria for a longer time than the other groups in the sample.

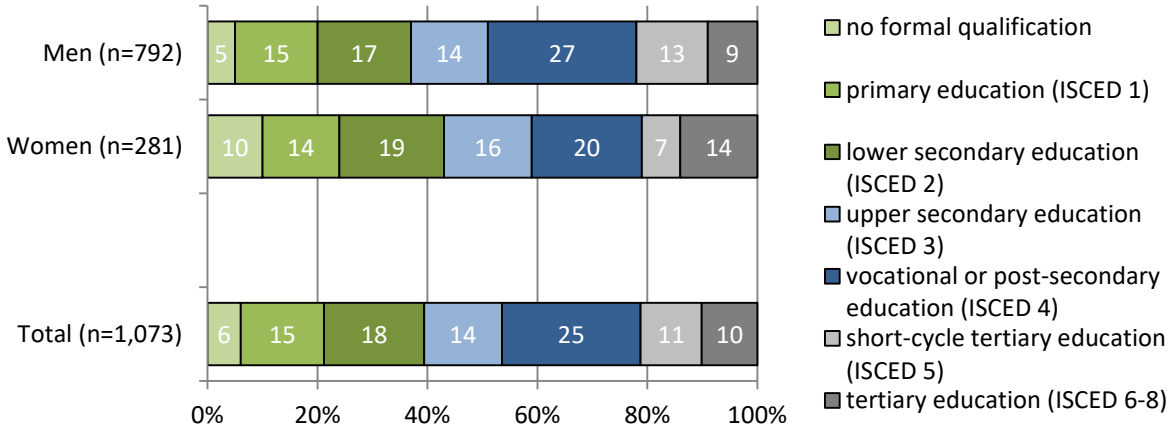
Graph 14. Educational levels by age groups



Source: FIMAS data

There were also differences in educational levels by gender. Women tended to have lower levels of education than men. Every tenth woman surveyed responded that they had not formal qualification, whilst the percentage of men who without any formal qualification was only 5% (Graph 15).

Graph 15. Educational levels by gender



Source: FIMAS data

Overall, the women interviewed were not necessarily less educated than men, but had been educated in different ways. **Women were more likely to have obtained upper secondary school leaving certificates (ISCED 3) or academic degrees (ISCED 6-8)**. By contrast, men tended to be more likely to hold professional qualifications, such as post-secondary (ISCED 4) and short tertiary (ISCED 5) educational qualifications.

Of all the respondents who worked abroad, only about half said that they had been gainfully employed prior to their arrival in Austria. Highly educated persons were able to make use of their qualifications prior to entering Austria, but even among university graduates, only 69% stated that their occupation prior to immigrating to Austria was related to their education.

The **low level of marketability of qualifications in countries hit by crisis** means that respondents' fragile employment histories had already devalued their qualifications prior to seeking asylum. This suggests that most refugees worked in semi-skilled occupations prior to their arrival in Austria.

This was also confirmed by the comparatively short training period stated for these occupations. Across all qualification groups, the average duration of training was 1.4 years in occupations for which information was available (78%). Only around 15% of those who learned a craft or trade had trained in this field for more than three years. In the service sector or in sales, the average training time was even shorter. Of respondents working in these areas, 71% said their job training lasted less than a year. Taking into account their level of education and training time, one can assume that a good part of stated occupations would be considered semi-skilled in Austria.

4.2.2 Recognition and evaluation of qualifications obtained abroad

Only a **small proportion (14%) of refugees** who completed a qualification abroad applied for recognition or evaluation of their foreign qualifications in Austria. As Kirilova (2016, p. 17) points out, this is often explained by the hurdles or ambiguities associated with the recognition of qualifications. Missing documents, the duration of recognition procedures and the refugees' wariness of these procedures are further barriers (Biffl et al., 2016, p. 43).

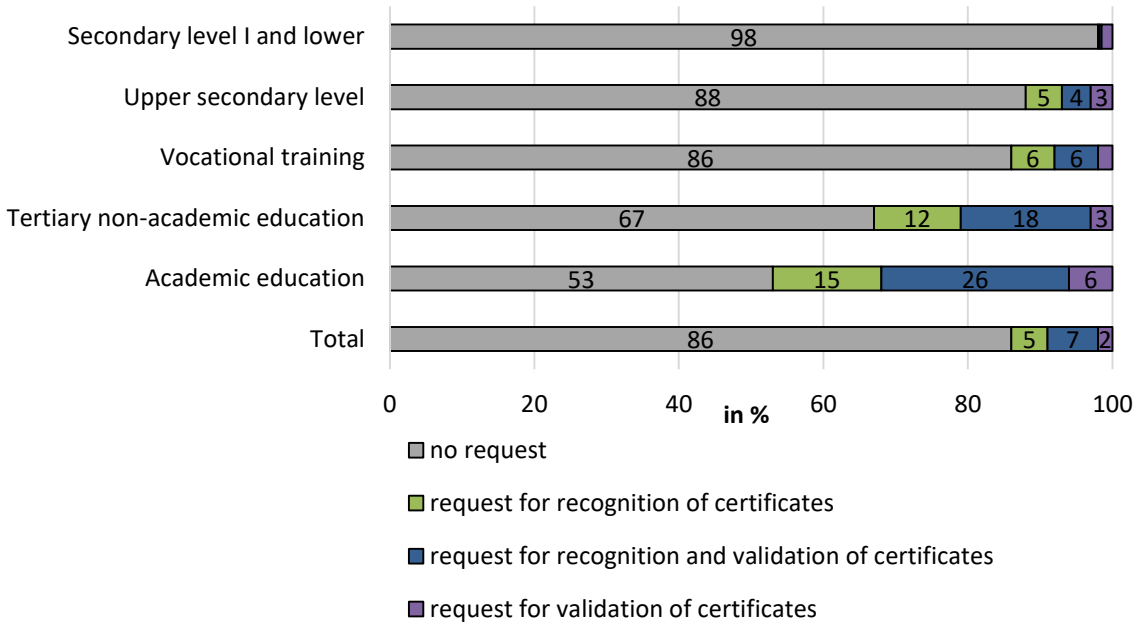
The Austrian Recognition and Evaluation Act (*Anerkennungs- und Bewertungsgesetz* [AuBG]), which entered into force in July 2016, has made it much easier for refugees and beneficiaries of subsidiary protection to have their foreign qualifications recognised. For example, if their documents are not available, supplementary procedures are in place to determine qualifications. An online portal and advice centres set up for every federal province provide relevant information. This could simplify the recognition of foreign qualifications in the future.

Looking at the survey data, **university graduates** were more likely to use the opportunity to have their qualifications recognised or evaluated. Of these, 47% submitted a request for their qualifications to be recognised or evaluated (0). The congruence between educational background and country of origin means that the proportion of Iraqis and Chechens/Russians, among whom university graduates were strongly represented, was also particularly high among those who sought recognition or evaluation of their qualifications.

According to respondents, 60% of applications for recognition of educational qualifications received a positive outcome. Only a small number of respondents (8%) reported that their request for recognition was rejected. The remaining recognition procedures were still ongoing at the time of the FIMAS study. Those whose education had been recognised had been living in Austria for nearly four years.[11] The expenses involved in applying for a recognition procedure were identified as one of the biggest hurdles. However, once an application was submitted, the chances for recognition were comparatively high.

[11] Due to the high margin of error with respect to the average duration of stay and the overall low number of persons seeking recognition of their qualifications, this value is only of limited validity, which is why differences between the groups were not considered here in greater detail.

Graph 16. Application for recognition/evaluation by highest qualification earned abroad (n=1,015)



Source: FIMAS data

4.2.3 Qualifications obtained in Austria

The educational participation of refugees of different origins and with different educational backgrounds is presented below, including information on the highest overall qualifications completed.

A total of **6% of respondents said they were unable to read and write** at the time of the survey. At the time of the survey, 79% had **one or more foreign qualifications**, of which 5% were in education or training in Austria at the time.

10% said they had earned a qualification in Austria. The majority of respondents had already obtained degrees before entering Austria. 4% of respondents listed Austria as their primary country of education and 17% of respondents had started or successfully completed education or training in Austria (those who listed Austria as their primary country of education were not taken into account here). Admission requirements and language skills (Koppenberg, 2016, p 41), as well as the short duration of stay initially may function as barriers to entering education in Austria. Another main reason for low participation in education might be the desire to work in order to earn an independent income, which precludes (further) education or training.

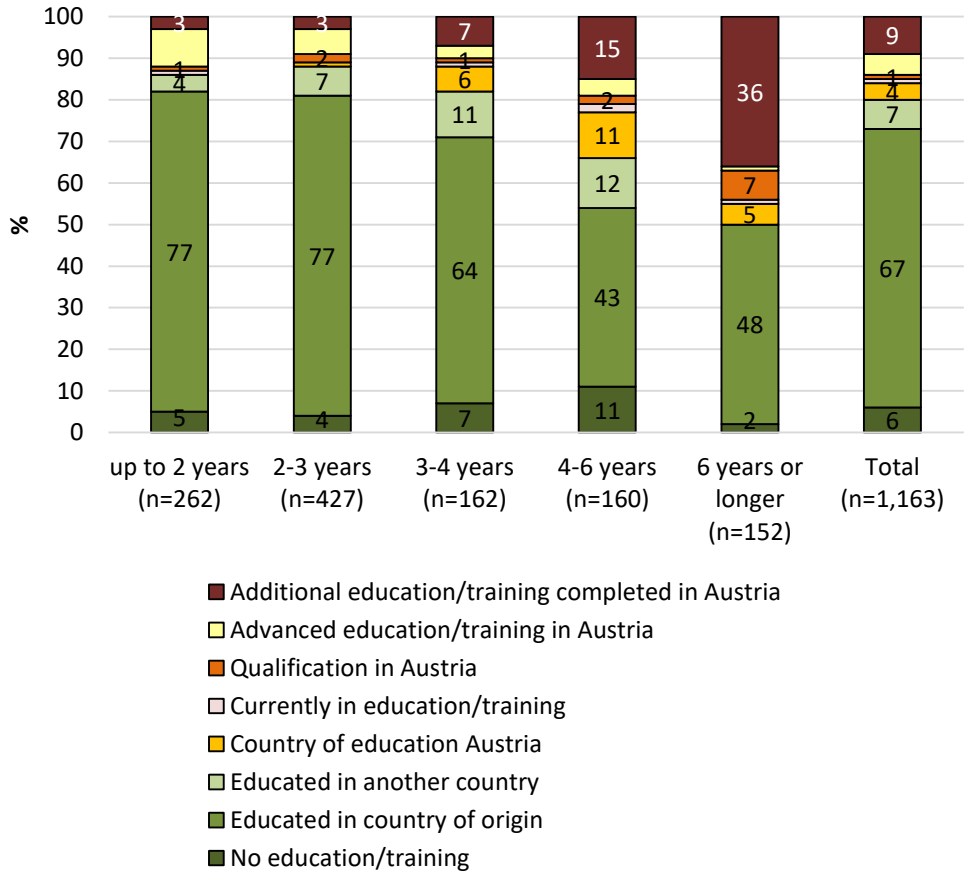
The longer respondents had lived in Austria, the more they had found the opportunity to continue their education. As such, around 43% of respondents who had already been living in Austria for six years or more, completed an Austrian educational qualification. Of those who had only been living in Austria for two years or less, around three per-cent had completed training or education in Austria (Graph 17).

The **formal qualification most often stated** among refugees and beneficiaries of subsidiary protection was a **certificate of secondary education**. A total 4% of respondents over the age of 18 had completed secondary school in Austria. A further 4% had completed an apprenticeship, vocational college certificate or general qualification for university entrance (*Matura*) in Austria (education corresponding to ISCED category 3). University degrees were comparatively rare (1%).

The above implies that recognised refugees and beneficiaries of subsidiary protection in Austria primarily acquire basic qualifications and attend the training programmes that are increasingly sponsored by public institutions in Austria, such as the AMS. Despite increased funding for basic

qualifications, university graduates were more likely to pursue or complete education or training in Austria than those who had acquired low educational qualifications in their country of origin. Of the group of respondents who already held a university degree at the time of the survey, every third person had attended or completed education or training in Austria. By contrast, only 15% of those who had only completed compulsory schooling were in education or training or had already obtained a qualification at the time of the survey. Only participation by persons with upper secondary education was comparatively higher. These respondents, who often had been unable to complete their education in their country of origin, seemed to have tried to find a way to continue their education in Austria. At the time of the survey, 23% of respondents had begun or had already completed a course in Austria.

Graph 17. Educational history by duration of stay



Source: FIMAS data

In addition to the duration of stay and income situation, the age of respondents also had a significant influence on their educational participation in Austria. **Younger persons,** in particular, attempted to expand upon their education or lay the foundation for further qualifications through basic training. Of respondents under 21 years of age, 19% pursued education or training, and 18% had already completed education or training in Austria. Most of the under-21s had obtained a secondary school qualification or wanted to complete a secondary school qualification or apprenticeship.

Of the 21- to 25-year-olds, 17% pursued education or training in Austria and 19% had already completed a qualification. There were more persons in this age group with an ISCED category 3 degree – university entrance qualification, apprenticeship or vocational college – than there were among the younger age group.

Table 9. Qualifications started (attending) and highest qualification earned (completed) in Austria by age (row percentage)

Age		Secondary school	Apprentice-ship	Vocational or technical school	University entrance qualification	Academic	Training	Percentage overall	Total
		%						%	(n)
under 21	Attending	5	6	5	3	2		19	151
	Completed	14	1		1		3	18	145
21-25	Attending	7	4	1	4	2		17	177
	Completed	9	2	2	5	1	1	19	168
26-30	Attending	2	2	1	1	2		8	184
	Completed	4	1	1	2	1	2	10	184
31-40	Attending	1		1		2		4	299
	Completed	1		2	1	1	2	8	305
41-50	Attending				1	1		2	160
	Completed			1	2	1	2	6	156
over 50	Attending					3		3	59
	Completed	2			3	2	10	17	60
All	Attending	3	2	1	1	2		9	1,030
	Completed	5	1	1	2	1	3	10	1,018

Multiple answers were possible for education/training started; total row percentages may differ due to rounding.
Source: FIMAS data

By comparison, older respondents focused more on academic degrees, with some attempting to expand upon their foreign educational qualifications. Qualifications and education begun by those over 50, in particular, tended to be additional qualifications. Their comparatively high educational participation – and the high number of qualifications already attained – was primarily related to the duration of their stay in Austria. The majority of over-50-year-old persons who had completed an education or training (70%) had lived in Austria for more than ten years.

4.3 Language skills and German courses

Very few respondents were able to speak German when they entered Austria. A total of 57% were unable to communicate in German or English. 8% had at least basic German skills and 37% had at least basic English skills. For around 32%, English was the language of communication upon their arrival because they did not have any German language skills. At the time the survey was conducted, around one third of respondents had only basic or no German or English skills.

According to the sample selection, most persons were Arabic-speaking and spoke other regional languages, such as Kurdish or Dari. Among Syrians, 30% spoke only their first language when they arrived in Austria. Among Iraqis, this was around one quarter. Among the Afghans, 80% spoke more than one language, but very few spoke German or English. Their language skills resulted from the variety of languages spoken in Afghanistan. Dari was the first language for approximately 74% of Afghans included in the sample (Table 10), 11% spoke Farsi as their first language, and 8% spoke Pashto. Russians were also overwhelmingly bilingual (77%), although nearly all of them stated their first language was Chechen (92%). It should be noted that, among respondents from Iraq and Syria, 12% and 29%, respectively, stated that their first language was Kurdish. Thus, additional analyses on possible features of Kurds in terms of labour market integration are recommended for further study.

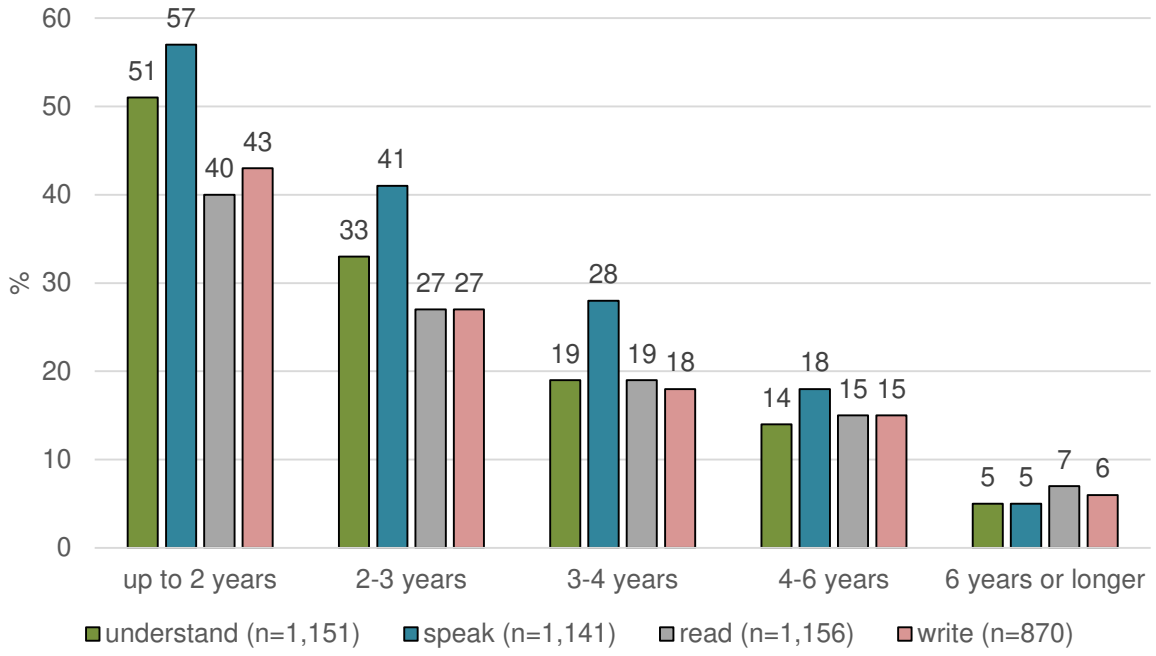
Table 10. First language by analytic group

Afghanistan (n=301)	%	Iraq (n=103)	%	Russian Federation (n=78)	%	Syria (n=678)	%
Dari	74.1%	Arabic	85.4%	Chechen	92.3%	Arabic	70.2%
Farsi	11.3%	Kurdish	11.7%	Arabic	3.9%	Kurdish	28.9%
Pashto	8.3%	Armenian	1.0%	Russian	2.6%	Armenian	0.6%
Punjabi	2.7%	Turkish	1.0%	Lak	1.3%	Turkish	0.2%
Arabic	1.0%	Dutch	1.0%			Greek	0.2%

Source: FIMAS data

According to respondents' own statements, their **German skills** rapidly improved once they had arrived in Austria. Among those who had been living in Austria for up to two years, only around half stated they had a limited understanding of German. Among those who had been living in Austria for between two and three years, one third said they understood German only to a limited extent (Graph 18). Respondents rated their ability to speak German as rather poor initially. Among those who had already been living in Austria for between two and three years, 41% stated that they only had limited skills in speaking German.

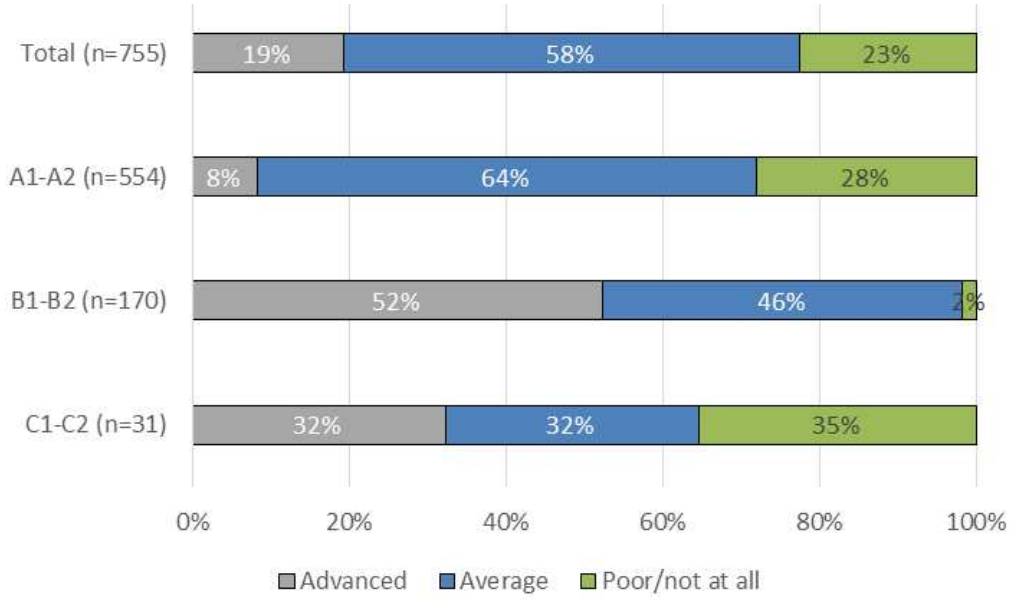
Graph 18. Share of respondents with limited or no German language skills by duration of stay



Source: FIMAS data

However, whether they rated their language skills positively or negatively also depended on the skills they had already developed, as shown by the information provided on the level of German language courses attended. Respondents' own assessments of their language skills may have been distorted. Information on the level of the most recently attended German course was available for around 64% of respondents. Of respondents that had recently taken a C1 or C2-level course (a total of 31 persons), only around one third believed they had reached an advanced level (0). Of those that had taken a B1 or B2-level course, however, a little over half claimed to understand German at an advanced level. Self-assessment by respondents, therefore, provides only a rough guide to the assessment of the actual skills they possess, as the information regarding the level of the courses taken shows.

Graph 19. Self-assessment of language proficiency by recently attended course level



Source: FIMAS data

The following three categories can be distinguished to assess language proficiency:[12] basic or no language proficiency (corresponds to A), independent user (corresponds to B) and proficient user (corresponds to C). If information on the level of the most recent language course taken by respondents was available, this information was used to assess language proficiency. If this information was not available, the information provided by means of self-assessment was used. Taking into account the information on the level of the most recent German course respondents completed, the level of communication (not taking into account German reading and writing skills) at the time of the interview was that of an **independent user (B)** for just over **half of respondents**.

The availability of language courses is crucial for language acquisition. As such, the majority of respondents had already completed at least one **German course** or attended a German course at the time of the interview (77%). Those who had not attended any German courses often believed that they did not have any German language skills whatsoever (39%). However, one quarter also stated that consuming media or learning through everyday life were opportunities for initial language acquisition. The respondents' age and the level of education/training refugees had prior to their arrival in Austria was key to acquiring language skills, together with attending German language courses and the duration of respondents' stay in Austria. Younger persons, in particular, tended to rate their language skills comparatively better. With respect to German proficiency, no gender-specific differences were identified.

4.4 Participation in values courses and competence checks

Austria has developed a number of integration courses and programmes with a special focus on recognised refugees. Among others, they include orientation and values courses, competence checks and so-called Youth Colleges and Women's Colleges.

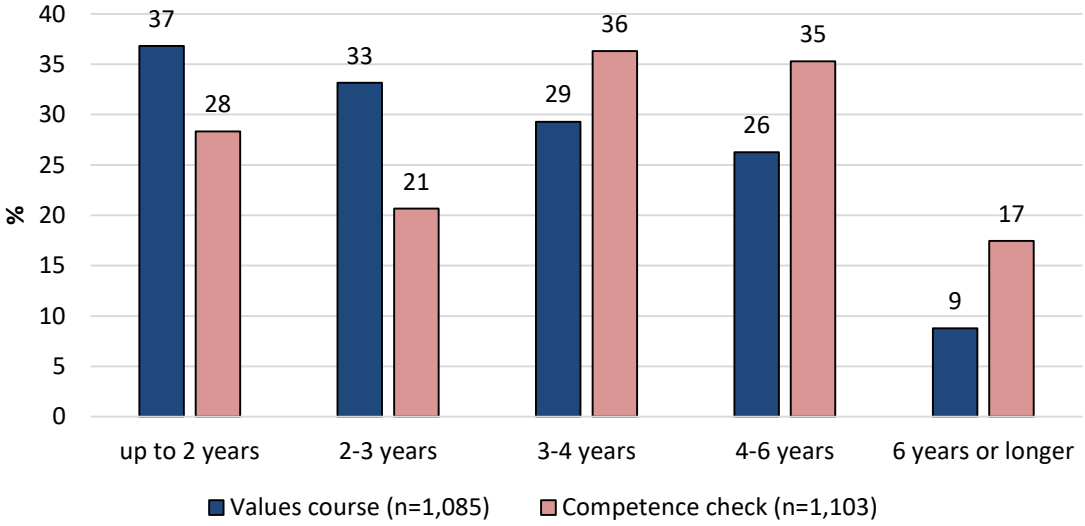
So-called orientation and values courses were first offered in 2015 and included in the AMS portfolio as of 2016. These courses are led by trainers from the Austrian Integration Fund. When the Austrian Integration Act came into force in 2017, completion of these courses, designed as integration measures, became compulsory. According to the 2017 Integration Report, some 30,000 refugees have participated in a values and orientation course since the measure was started in 2017 (ÖIF, 2017).

[12] According to the Common European Framework of Reference for Languages (CEFRL).

The competence checks offered by the AMS serve to ascertain refugees' qualifications, which oftentimes have not been formally documented, and to compare the skills involved in foreign and Austrian job profiles. Competence checks were offered as part of a pilot project in Vienna in 2015 (AMS, 2016c), and were introduced nationwide in 2016. According to the 2017 Integration Report, that year, 7,144 persons took part in these competence checks (Expert Council for Integration, 2017, p. 47).

Among those included in the sample, around **30% had attended a values course**, where the participation rate for **men was higher at 34% than it was for women (18%).**

Graph 20. Rate of participation in values courses and AMS competence checks by duration of stay



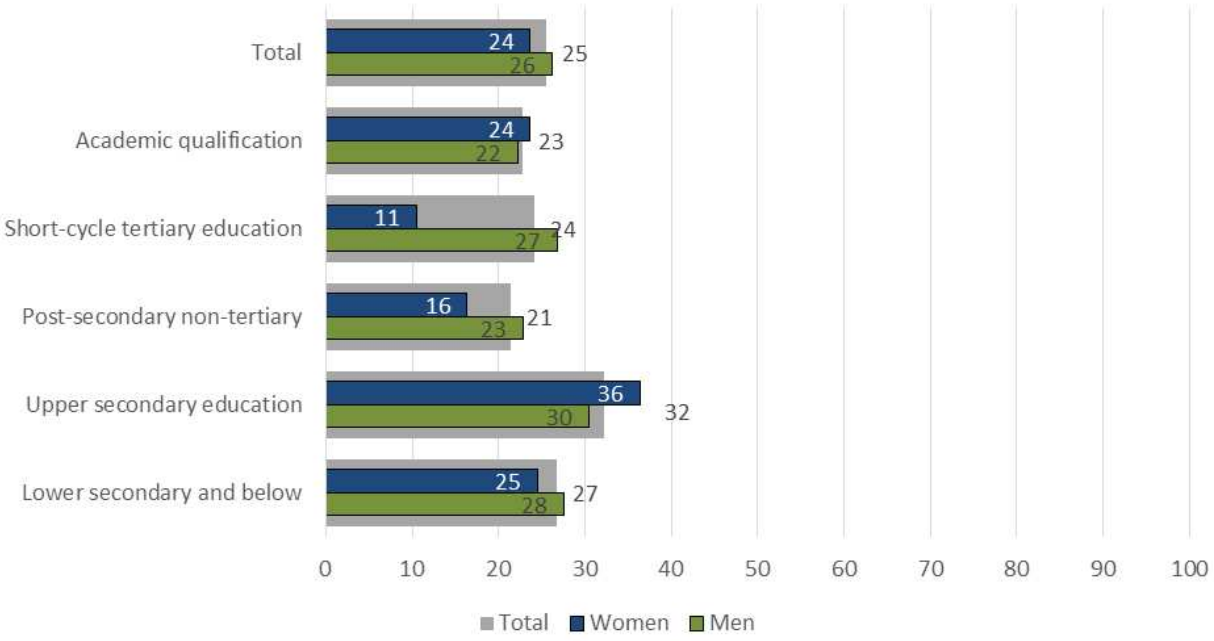
Source: FIMAS data

What is striking is that the duration of residence in Austria only had a limited influence on participation in a competence check. These were even taken by persons who had already been living in Austria for three to six years – primarily by Afghans – more frequently than those who had only been living in Austria for two to three years. Among respondents who had already been in Austria for six years or longer, however, only 17% had participated in a competence check.

The lowest rate of participation was found for Chechens/Russians, who were most likely to prove formal qualifications, most of whom had been living in Austria for longer time periods and, to a large extent, held qualifications earned in Austria. There were no specific differences in terms of participation in competence checks between the other groups. Out of the **Afghan, Syrian and Iraqi respondents in the sample, 31-32% of each group had completed a competence check**.

Among all respondents, **25% had completed a competence check** (0). Respondents with a foreign upper secondary qualification more frequently (32%) took part in competence checks (0). The participation rate for persons with post-secondary non-tertiary qualifications was the lowest. This included qualifications that, although they are classed as secondary education, are not yet considered tertiary, such as nursing schools.

Graph 21. Rate of participation in competence checks by education status, in % (n=1,008)



Source: FIMAS data

4.5 Professional history abroad

4.5.1 Professional experience

A total of **87% of respondents who provided information on their employment were gainfully employed prior to immigrating to Austria.** Around 16% of them stated that they had previously been gainfully self-employed at least once prior to arriving in Austria.

There were major differences in terms of employment experience by gender. Only **around 59% of women reported to have been gainfully employed abroad** (compared to 95% of men). Slightly less than one third of women did not provide any information about occupations abroad, or stated that they had been unemployed, 25% of whom listed "homemaker" as their profession. At 45%, the number of homemakers was particularly high among women with only compulsory level of education. Among this group, only 39% had already gained employment experience prior to moving to Austria. By contrast, the majority of women holding an academic degree had been gainfully employed prior to their arrival in Austria (85%). Notably, such impact of education on labour force participation prior to immigrating to Austria could not be determined among the men.

Men and women with an upper secondary qualification also had comparatively little experience of being gainfully employed. Of these men and women, many had been pursuing their education or had been in training when they fled their countries. Furthermore, among those who had completed upper secondary education, the number of homemakers (women) was above average at 29%.

Buber-Ennser et al. (2016), who in their study draw a very similar conclusion, attribute the reasons for women's lower labour market participation to the **precarious labour market situation in their countries of origin** on the one hand, whilst also pointing out **cultural reservations** about women's participation in the labour market on the other. One key factor influencing the degree of labour market participation by women prior to arriving in Austria is, therefore, linked to their country of origin: whilst only around half of women from both Afghanistan and Syria had been gainfully employed prior to their arrival in Austria, labour market participation among Iraqi women was 65%, and 86% among Chechen/Russian women. Including the women who stated that they were unemployed, every Chechen/Russian woman included in the sample demonstrated a labour market orientation.

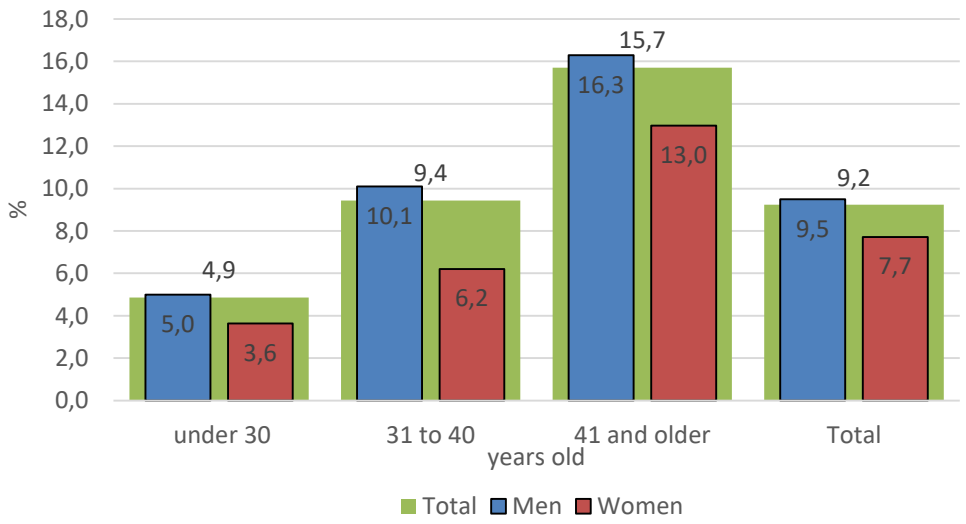
Table 11. Employment experience by qualification prior to arriving in Austria, in %

		Employment experience	In training/ education	Home-maker	Un-employed	Total
Men	at most compulsory education	98	2		0	100
	upper secondary education	84	15		1	100
	post-secondary non-tertiary	99	1		0	100
	short-cycle tertiary education	88	10		2	100
	academic qualification	92	8		0	100
Men	%	95	5		0	100
	N	691	37		3	731
Women	at most compulsory education	39	9	45	7	100
	upper secondary education	46	25	29	0	100
	post-secondary non-tertiary	80	2	12	6	100
	short-cycle tertiary education	68	21	5	5	100
	academic qualification	85	9	3	3	100
Women	%	59	10	25	5	100
	N	126	22	54	11	213
Total	at most compulsory education	83	4	11	2	100
	upper secondary education	76	17	6	1	100
	post-secondary non-tertiary	95	2	2	1	100
	short-cycle tertiary education	85	12	1	3	100
	academic qualification	90	8	1	1	100
Total	%	87	6	6	1	100
	N	817	59	54	14	944

Source: FIMAS data

Even women who had been gainfully employed prior to immigrating to Austria tended, on average, to be have less labour market experience than men. The average length of employment for women prior to immigrating was 7.7 years (Graph 22), whilst for men it was 9.5 years. This difference remains unchanged, even among those in the same age cohort, and increases among the over-30-year-olds. This could be attributed to the impact of care responsibilities (cf. Liebau, Salikutluk, 2016, p. 735), which leads to a further differentiation of labour force participation between men and women.

Graph 22. Average length of employment abroad in years by gender and age (n=777)

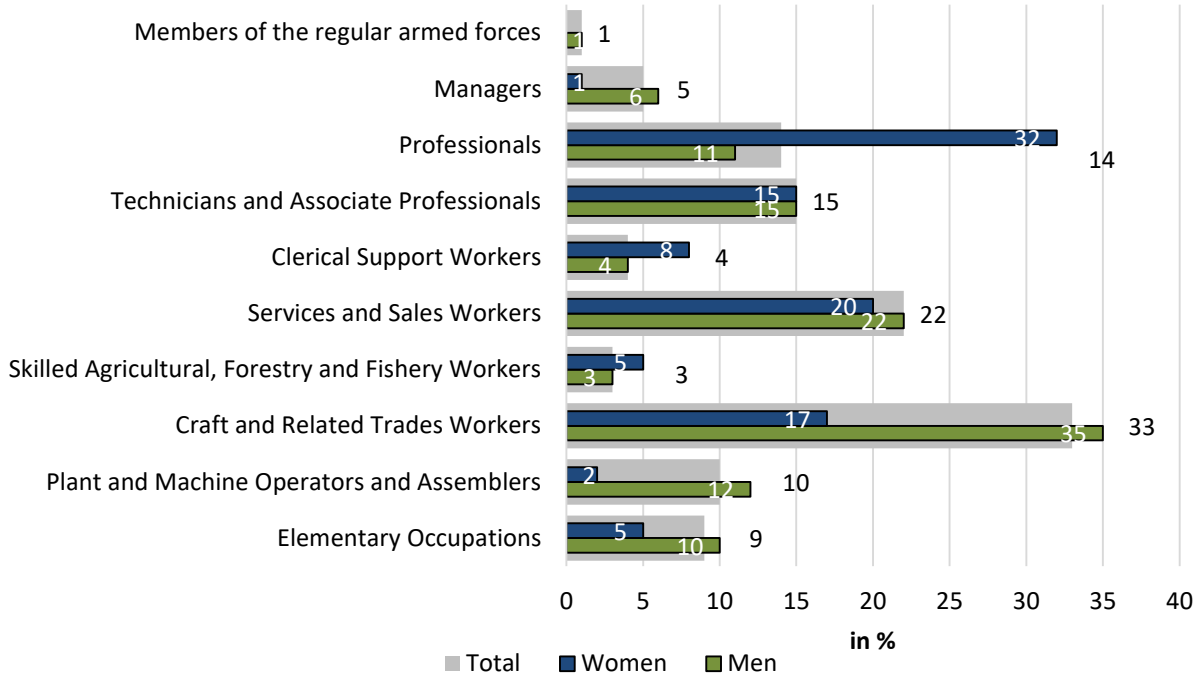


Source: FIMAS data

4.5.2 Occupational groups

Taking into account all professional fields respondents were engaged in abroad, the most important fields of employment for refugees and beneficiaries of subsidiary protection prior to their arrival in Austria were **crafts and trades** (33%), as well as **service sector and sales-related occupations** (22%, Graph 23). Women often worked in academic professions abroad, primarily as **teachers** (cf. also the findings by Buber-Ennser et al., 2016).

Graph 23. Fields of employment abroad (multiple answers), major occupational groups, Ö-ISCO 2008 (n=1,031)



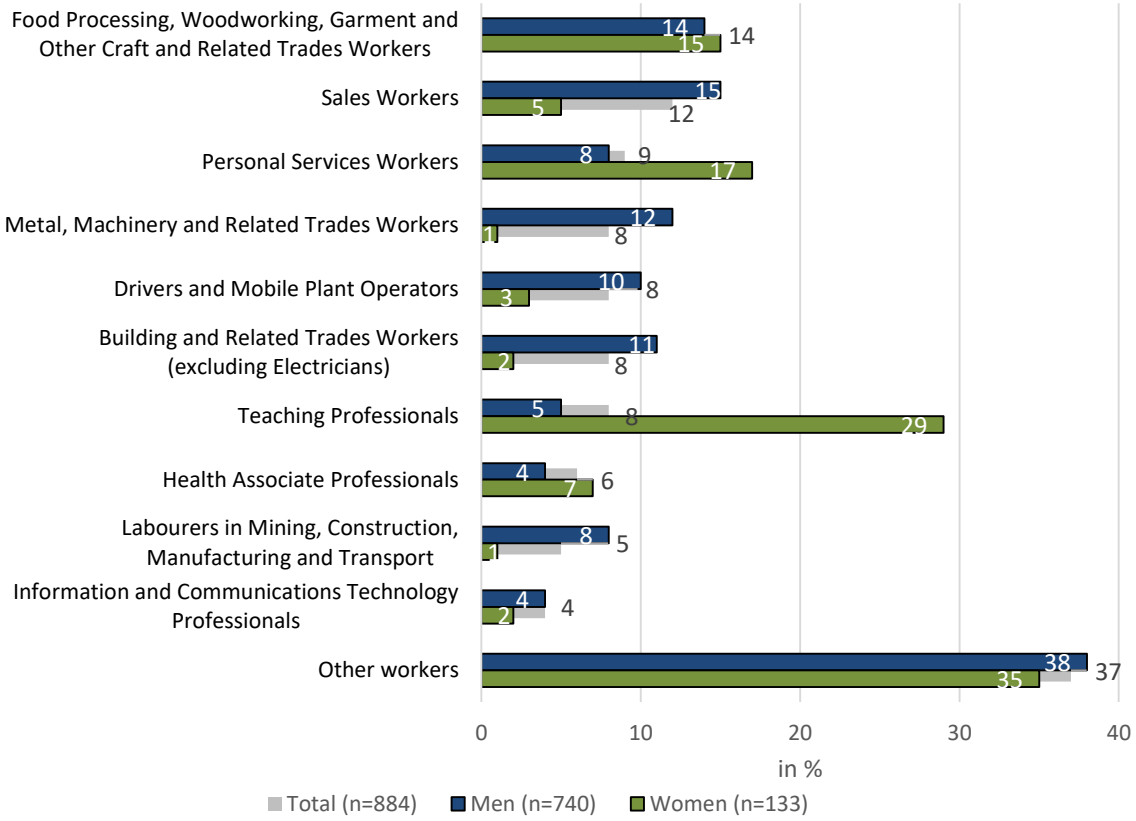
Source: FIMAS data

Men's fields of activity in the crafts and trades sector were broadly diversified. Of these men, around 34% worked in the construction industry as builders, electricians, painters and other structural design-related professions. Around one quarter of men worked in the metal industry or as locksmiths or auto mechanics. Only 18% of men had experience in garment manufacturing, while women who worked in manual occupations primarily concentrated on garment manufacturing (70%).

Experience in the field of service and sales professions was also differentiated by gender. Women were particularly experienced in personal care services in this sector (74%). Men, however, were often employed in retail (60%) or the restaurant industry (20%).

The fact that men's professional fields prior to coming to Austria were more highly differentiated than those of women was also supported by the detailed analysis by occupational group (0). The largest professional group overall included jobs in food processing, woodwork and clothing manufacture (14%), followed by sales (12%) and personal services (9%). Teaching (8%) was especially prevalent among women (29%, compared to 5% among men).

Graph 24. Fields of employment abroad (multiple answers), ten largest occupational groups, Ö-ISCO 2008, in %, multiple answers



Source: FIMAS data

Nearly one quarter of respondents had worked in multiple different fields of employment before coming to Austria. Elementary workers, in particular, often mentioned more than one field of employment. The total percentage of elementary workers abroad was 9%. Among those who specified more than one profession they worked in, the share of those who had taken up elementary occupations at least once was nearly 17%. Elementary workers were also more likely to state that they had held at least one job abroad that did not correspond to their qualifications.

4.6 Marketability of knowledge in Austria

This section delves into the question of whether, and if yes, to which extent, recognised refugees and beneficiaries of subsidiary protection were able to utilise skills acquired abroad (through work and formal qualifications), and what effect their qualifications had with regard to labour market participation.

Education is one of the important resources for gaining a foothold in the labour market. This becomes evident when differentiating unemployment rates by education level. To provide just one example: the unemployment rate among persons with at most compulsory schooling was 26% across Austria in 2016, and only 3.6% among university graduates (AMS, 2016, p. 2). Pursuant to this, the literature argues that opportunities for refugees to participate in the Austrian labour market also depend on their educational background and the professional skills they bring with them (Buber-Ennser et al., 2016).

However, it is also a fact that professional experience and formal educational qualifications acquired are devalued when a person flees their country. Normally, labour markets in destination countries offer only limited opportunities to work in areas where refugees had acquired their skills. Refugees are often confronted with their skills not matching those required by the labour market in their country of immigration, face discrimination issues (Bonin/Rinne, 2017, p. 20), and encounter difficulties getting their qualifications formally recognised.

Breaks in employment history, even before entering Austria, as well as long waiting periods during asylum procedures and unemployment in Austria also contribute to loss of expertise and devalued qualifications (Johansson et al., 2015, p. 21). Against this backdrop, the finding that education and work experience have a positive impact on labour market participation in Austria must be put into perspective. Indeed, it appears that other factors have a stronger influence on labour market integration in Austria; such as qualifications acquired in Austria (Huber et al., 2017) as well as the duration of stay in Austria, gender, care responsibilities, housing situation and possible limitations on employability due to physical and mental disabilities and illness (Johansson et al., 2015, p. 22), which only partially correlate to a person's educational background.

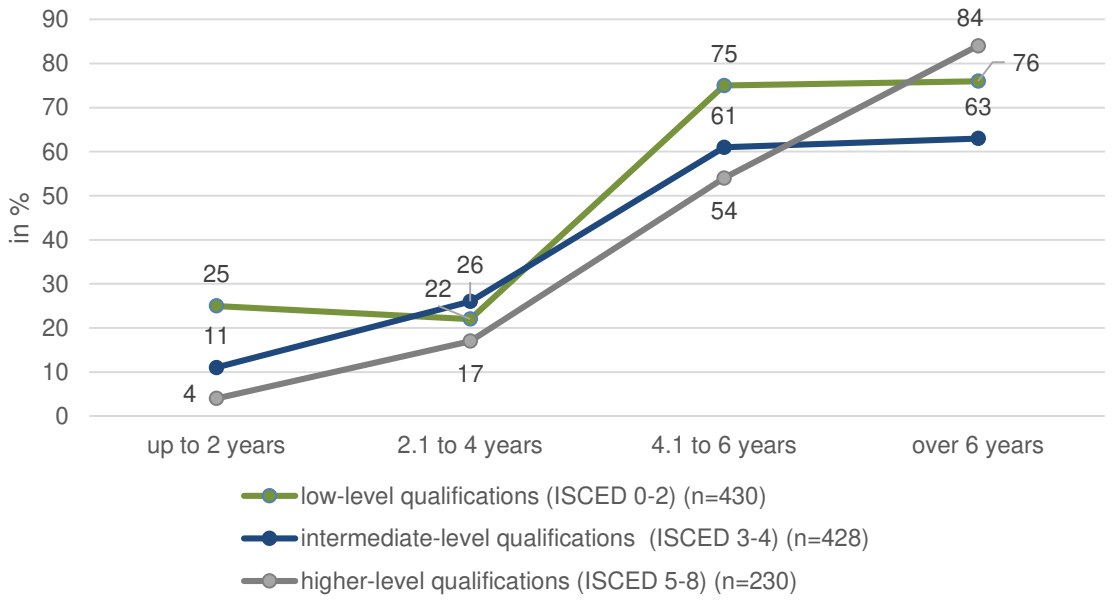
In addition, language acquisition is particularly important for participation in the Austrian labour market and represents a significant hurdle for migrants (Wössmann, 2016, p. 22). As Koppenberg (2016, p. 22) argues, the minimum German skills required of least B1 level are absolutely crucial for participating in the Austrian labour market, especially in the highly skilled sector. The over-qualification rate among migrants (Smoliner, 2011, p. 96) in Austria is correspondingly high once they manage to find gainful employment.

Indications that **certain educational qualifications are not marketable in Austria** include the higher rate of persons who have never been employed in Austria among respondents with higher educational qualifications (ISCED 5 to 8), relative to those with lower qualification levels (ISCED 3 to 4). Corresponding differences are evident in almost all immigration cohorts, making them largely independent of duration of stay in Austria (0).

One quarter of those with low qualifications who had been living in Austria for two years or less had already worked in Austria at least once. This group was the fastest to enter the Austrian labour market.

Among those with higher qualifications, the proportion of those who had already worked in Austria at least once was consistently lower than it was among those with lower or intermediate qualifications. Only after residing in Austria for a longer period of time (six years and more), did the large majority (84%) of persons with higher education backgrounds state that they had already worked in Austria at least once. For recognised refugees and beneficiaries of subsidiary protection with **higher levels of qualification**, **entry into the labour market seems to be particularly difficult when they seek jobs corresponding to their educational level**.

Graph 25. Share of respondents who had already worked in Austria (at least once) by duration of stay and level of education



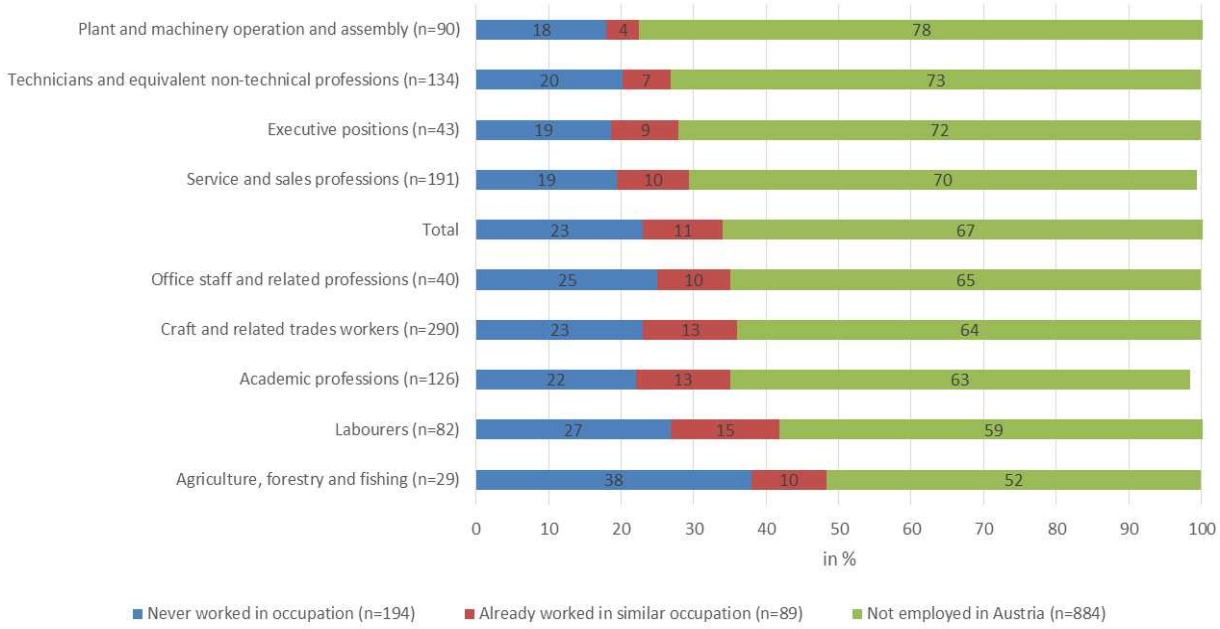
Source: FIMAS data

Furthermore, the **lack of continuity in fields of employment relevant to respondents' qualifications** points to the devaluation of experiential knowledge as a result of seeking asylum. Of those refugees who came to Austria with employment experience, only around 30% had previously been employed (unpaid work not included). Only one in ten persons who were gainfully employed prior to arriving in Austria stated they had already held a position in Austria corresponding to held positions before coming to Austria. However, around one in four stated that the job they held in Austria was the first time they had worked in that profession.

Of those who had worked as **labourers** or in **agriculture** prior to their arrival in Austria, slightly less than half were already employed in Austria when the survey was conducted (0). Recognised refugees who had worked in **assembly jobs** and as **technicians** before fleeing their home countries were especially unlikely to be gainfully employed in Austria. Of persons working in assembly jobs, 78%, and 72% of technicians had yet to find a job in Austria. Only 4% to 7% of these groups were able to work in a professional environment they were familiar with. Partly, the low levels of labour market integration of these skilled workers at the time of the survey can be attributed to the short amount of time they have been in Austria, a maximum of three years. Predominantly, they were specialists from Syria who, compared to other groups, had been living in Austria only for a short time. The average duration of stay in Austria among former skilled agricultural workers was around five years. Those who had worked in academic professions prior to immigrating to Austria tended to have been in Austria for an average of 4.5 years, and former office workers for 4.3 years.

The fact that technicians and executives in specialised occupations had only been able to gain limited employment experience in Austria is an indication for how difficult it is for refugees to market the experience they gained abroad in Austria. Nevertheless, the effects of employment experience on labour market integration cannot be accurately measured without information on duration of stay and other influencing factors. Personal living conditions in Austria and the specifics of the Austrian labour market in different regions must be taken into account as well when assessing refugees' chances of finding gainful employment. Multivariate analysis is particularly suitable for this, as discussed in chapter 5.

Graph 26. Marketability of professional experience by job prior to arrival in Austria in % (multiple answers possible), major occupational groups, Ö-ISCO 2008



Source: FIMAS data

4.7 Labour market integration in Austria

4.7.1 Current employment status

The information provided on current primary occupation, labour market availability, courses currently being taken and educational recognition was used to illustrate the current employment status of respondents in Austria. Based on the ILO's Labour Force Concept (1982), which is used for international labour market statistics, a distinction is made here between persons who are **employed, unemployed** and **outside the labour force**.

Persons who are **employed** were those who pursued paid work at the time of the interview or were temporarily not working (due to holiday leave, sick leave, parental leave, etc.), but were effectively employed.

Unemployed persons were those who were not effectively employed, were available on the labour market (meaning they were able to start working within two weeks) and were actively looking for work. Included in this group were persons with job offers and those applying to have their qualifications recognised.

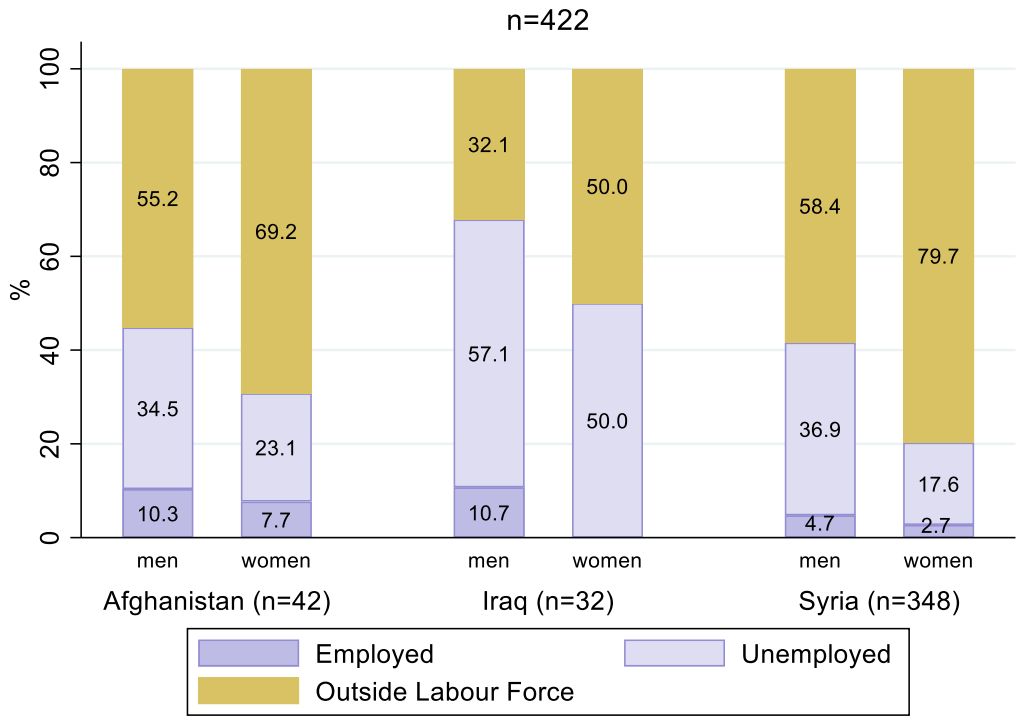
Persons outside the labour force were defined as those respondents who were neither employed nor unemployed. This group included persons who worked exclusively on a voluntary basis, those in education or training, those who were unable to work due to illness or disability, those who were responsible for childcare, homemakers and any other persons of an employable age who were not employed and not seeking work at the time of the study.

The applied definitions result in distinct differences between self-classification as unemployed and categorisation under the Labour Force Concept. Although 623 respondents indicated that they were unemployed, according to the Labour Force Concept, this was true for only 364 of them, whilst 255 were classed as outside the labour force. Subjective assessments of unemployment sometimes included underemployment or educational or training periods.

There is little sense in drawing conclusions from the overall distribution of employment status among respondents, since duration of stay, training and education, as well as other structural attributes vary widely between them. In order to account for the impact of factors like duration of stay, the language skills or the availability of more established social networks, it was deemed useful to analyse the situation by cohorts of arrival. For the 2015 immigration cohort, which was the largest in the sample, this approach made obvious clear differences for the three analytic groups of persons from Afghanistan, Iraq and Syria, also by gender (0).

In all groups analysed, men were more likely to be employed than women and were less likely not to be active in the labour market. This difference is striking, especially among respondents from Iraq, although so few women were interviewed that the gender difference cannot be generalised. The same was true for persons from Afghanistan. There was only a weaker effect (V=0.181) for the group from Syria: 80% of women but only 58% who arrived in 2015 were not active on the labour market. Chechens/Russians were barely represented among this cohort, which is why they are not included.

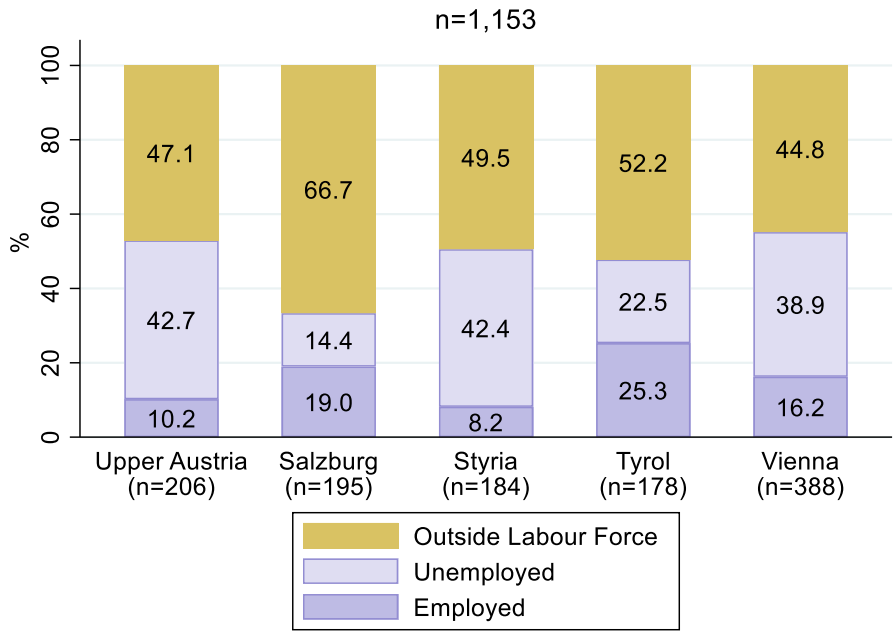
Graph 27. Employment status by analytic group and gender, 2015 immigration cohort, Labour Force Concept



Source: FIMAS data, excluding asylum seekers

When looking at employment status by federal province, clear differences became evident, especially in Salzburg. Whilst the proportion of non-active persons did not fall above 52% in any other federal province, this figure was 67% in Salzburg (Graph 28). However, this higher rate can be explained by the form in which sampling was carried out. In the non-randomised part of the sample in Salzburg, this proportion was around 70%, with Diakonie refugee service clients being interviewed. It may therefore be the case that there is sampling bias towards persons outside the labour force. In the other four federal provinces, the percentage of non-active persons does not differ distinctively.

Graph 28. Current employment status by federal province, Labour Force Concept



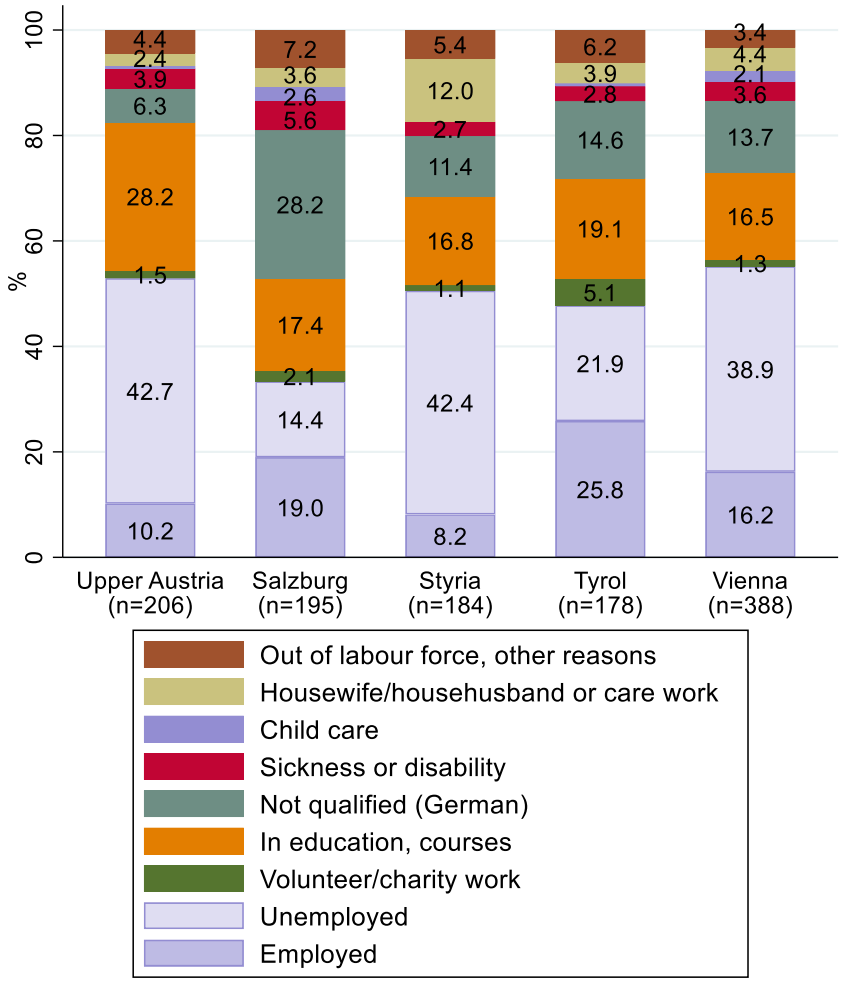
Source: FIMAS data, excluding asylum seekers

The proportion of unemployed persons was lowest in Salzburg (14%) and Tyrol (23%) – much lower than in Upper Austria, Styria and Vienna. Associated with this is the fact that the chance of gainful employment was highest relative to the working population in Salzburg and Tyrol. These strikingly high chances of employment are due to the particularly heavy tourism and service sectors in the western provinces (see the following section, Graph 40).

An even more detailed breakdown of reasons for non-employment shows that the two main factors – education and lack of knowledge of German, according to self-assessment – are unevenly distributed across the federal provinces (Graph 29). At 28%, the Salzburg sample had the highest share of respondents who identified their lack of language skills as a reason for not being able to work. This share was the lowest in Upper Austria (6%).

In Upper Austria a high proportion of respondents (28%) stated that they were currently in education or training and therefore not available for the labour market. Other reasons, including illness, disability and unpaid work such as childcare, housework or volunteer work, were only minor barriers labour market participation, but were stated most often in Styria.

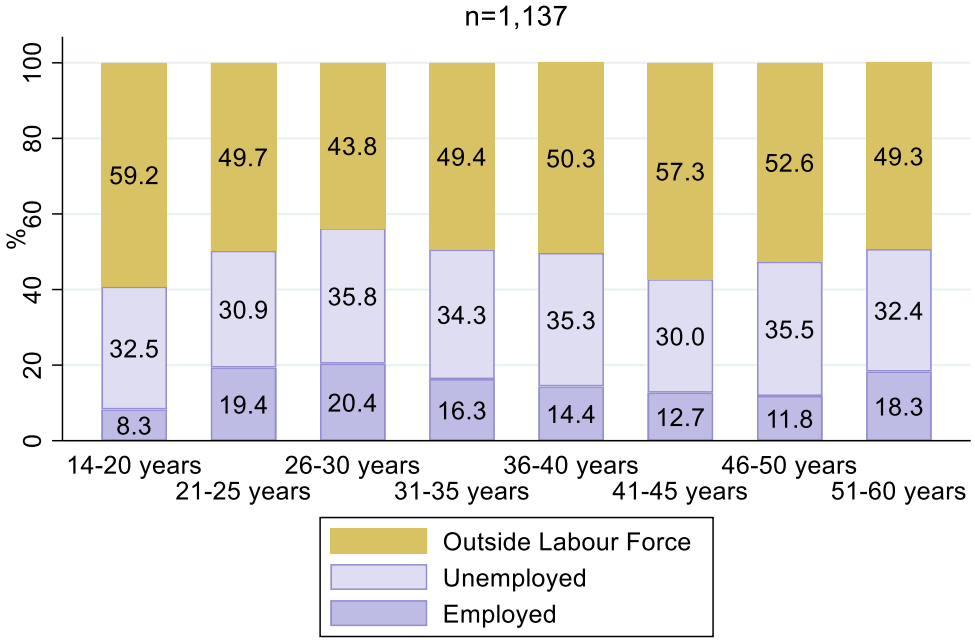
Graph 29. Current employment status (detailed) by federal province



Source: FIMAS data, excluding asylum seekers

When analysed by age group, only minor differences in employment status became evident. As expected, the labour force participation rate was lower for the age group 14 to 20, who were mostly in education or training. In this group, 8% of respondents were in employment, 33% were unemployed and 59% were not active in the labour market (Graph 30). Thus the proportion of persons not active in the labour market in this age group was higher than that in all the others. In the 21 to 25 age group, the non-active rate was 50%, and 44% for those aged between 26 and 30. Similarly, the proportion of employed persons in the 21 to 25 age group exceeded 19%, and over 20% of respondents aged 26 to 30 were in employment. In the older age groups, however, the share of persons who were employed and unemployed did not increase. It seems that age alone cannot account for differences in employment status. What is also striking in the distribution is that labour force participation among 50 to 60-year-olds was comparatively high – 18% were employed and 32% were unemployed. However, the numbers did not fundamentally differ from those of the other age groups.

Graph 30. Current employment status by age group, Labour Force Concept



Source: FIMAS data, excluding asylum seekers

A more detailed breakdown of those not active in the labour market by age group confirms the interpretation that participation in education and training are the primary reason for lower labour market participation among young refugees. Thus, they are not available for the labour market during this period (0).

In the youngest age group (14 to 20 years old), the proportion of persons in education was higher than in all other groups (45%) and almost twice as high as in the 21 to 26 age group (24%). This logic persisted throughout the whole sample; in older age groups, the number of persons in education gradually decreased. At the same time, the proportion of those who reported illness or disability as a reason for not working increased with increasing age. In the highest age group, this was true of 16% of respondents. Insufficient knowledge of German played a role in all age groups; the proportions ranged from 9% for 14 to 20-year-olds to 19% for 26 to 30-year-olds. Childcare, other care and household responsibilities were only relevant categories for over-30s.

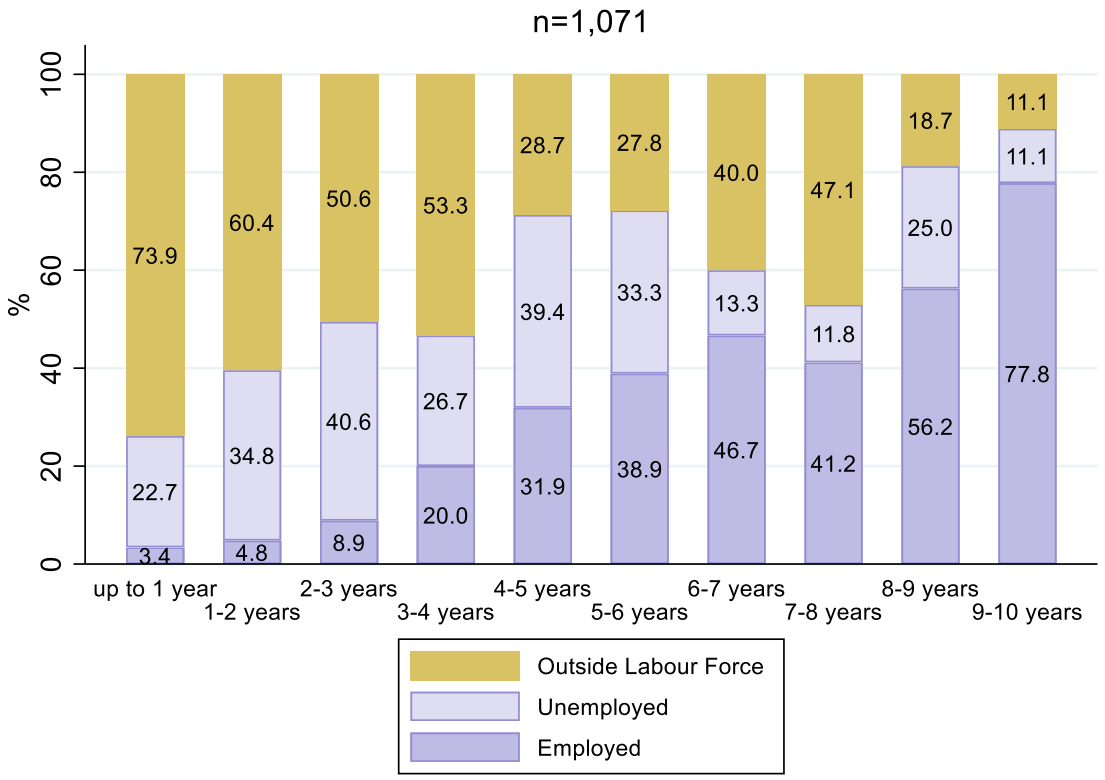
Graph 31. Current employment/non-employment status (detailed) by age group

Source: FIMAS data, excluding asylum seekers

The most informative illustration is the breakdown of employment status by duration of stay in Austria (0). Language acquisition, recognition of foreign qualifications, educational qualifications in Austria, the generation of social networks and several other factors that can facilitate labour market integration consolidate over time. This notion is reflected in the figures as well. The labour force participation rate and the employment rate of recognised refugees and beneficiaries of subsidiary protection increased almost continuously the longer they lived in Austria.

Among those who had been in Austria for less than one year, only 3% were in paid work at the time of the survey. Around 23% were unemployed and 74% were not active in the labour market. Among those who had been in Austria for one to two years, the proportion of employed persons was 5% and unemployed persons accounted for 35%. After two to three years of residence, the proportion of employed persons had reached 9%, and the share of unemployed persons was higher (41%) than in the later cohorts. After three to four years of residence, 20% were employed and the share of unemployed had reduced to 27%.

Graph 32. Current employment status by duration of stay in Austria, Labour Force Concept



Source: FIMAS data, excluding asylum seekers

With each additional year of residence – or, to put it more precisely, with each previous cohort – the number of persons in employment continued to increase. Of those who had arrived six to seven years ago, 47% said they were engaged in paid work. However, there was a remarkable break in this trend among those who had arrived seven to eight years ago. In this group, only 41% were employed, considerably fewer than in the more recent cohort of those who had arrived six to seven years ago.

There is also a break concerning those who reported themselves as either unemployed or outside the labour force. Their respective shares decreased progressively in the cohorts who had stayed in Austria between one and six years, but increased again for the cohort that had stayed for seven to eight years, and dropped for those cohorts who had stayed for eight to ten years in the country.

The most obvious explanation is the financial and economic crisis in 2008, which also left its mark on the Austrian labour market. Economic growth in Austria collapsed in 2009 (-3.8%), the number of job vacancies fell by 28%, the number of employees fell by 1.5%, and the unemployment rate (EU concept) rose from 4.1 to 5.3% (Statistics Austria, 2017e; ÖNB, 2017). Those who had arrived in Austria seven to eight years ago had to cope with particularly unfavourable labour market conditions. Although the economic indicators listed above indicate an upward trend one year later, the tense situation of the labour market remained. It is rather likely that the labour market conditions upon arrival in Austria impact labour market participation of the respective cohorts later on, for time periods of up to ten years (Aslund/Rooth, 2007).

The proportion of those in employment was highest (56% and 78%, respectively), and the share of those not active in the labour market lowest, among the two cohorts who had been living in Austria for eight to ten years. Of these cohorts, 80% to 90% were active in the labour market, a higher rate than the overall rate in Austria.[13]

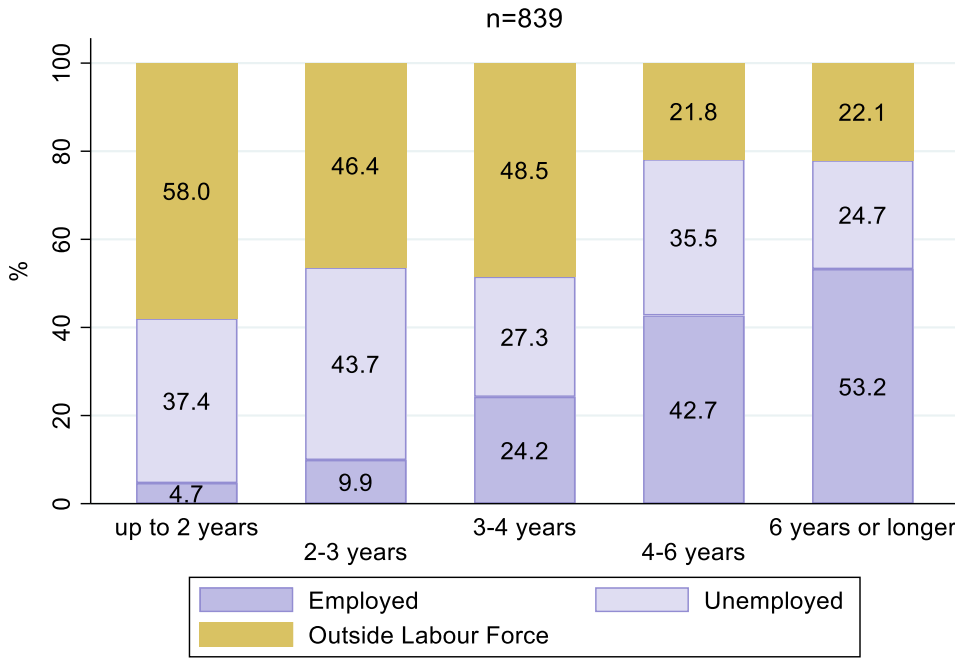
[13] However, the sub-samples of these cohorts were relatively small, which translates into large sampling errors.

When extrapolating these outcomes into the future, i.e. assuming progressive increase of labour market participation of the recent refugee cohorts, it might take a minimum of between **four to eight years** before the **overall level of refugee participation in the labour market reaches the average rate of the Austrian labour market.** Moreover, **six to nine years of residence will be needed until labour force participation by refugees reaches the Austrian average**[14] (more precise information cannot be provided due to the margin of error). Labour market participation is a **gradual process**, also because language acquisition and education and training often are the priority during the first years of stay. However, this does not mean that labour market integration takes this long on an individual basis (further analysis below and in section 4.8 follows). Certain groups of recognised refugees and beneficiaries of subsidiary protection will enter the labour market much earlier.

This may also explain, in part, the relatively low number of recognised refugees and beneficiaries of subsidiary protection currently registered with the AMS (28,552 in August 2017), compared to the high asylum application figures for 2015 and 2016 and the positive asylum decisions granted during these two years. Of the 14,413 persons whose asylum was granted in 2015, 10,070 were aged between 14 and 64; and of the 22,307 persons granted such a decision in 2016, 19,485 were aged between 14 and 64 (Statistics Austria, 2017c; EUROSTAT, 2017a). Between January and July 2017, a further 17,981 positive asylum decisions were issued (Austrian Federal Ministry of the Interior, 2017), of which most were likely for persons of working age. Not all of them seem to be registered with the AMS, especially since the AMS statistics also include many persons from previous cohorts.

The gendered analysis indicates that **men integrated into the labour market more quickly and more broadly than women.** Whilst 5% of men among the population of recognised refugees and beneficiaries of subsidiary protection who had stayed in Austria for up to one year, and 10% of men who arrived one to two years ago were employed at the time of the survey, these figures were only 2-3% % for women (Graph 33 and Graph 34).

Graph 33. Employment status of men by duration of stay, Labour Force Concept



Source: FIMAS data, excluding asylum seekers

[14] The labour force participation rate in Austria in 2016 was 76.2%, and the employment rate was 71.5% (Statistics Austria, 2017d).

After six or more years of residence, the proportion of employed men was 53%, whilst that of women was much lower at 33%. The same was true for labour market participation (the sum of shares for employed and unemployed), which, at 42%, was much higher for men in the first two years than for women at 22%. Even among those who had been living in Austria for six years or more, the participation rate of men was much higher (78%) than it was for women (52%). However, it must be taken into account that **women's participation in the labour market was lower also before immigrating to Austria**.

It is also striking that, among those who were active in the labour market, the relative risk of being unemployed after two years was lower for men than for women. For example, female respondents who had been living in Austria for six years or more had a 20% higher risk of being unemployed than men in the same group.

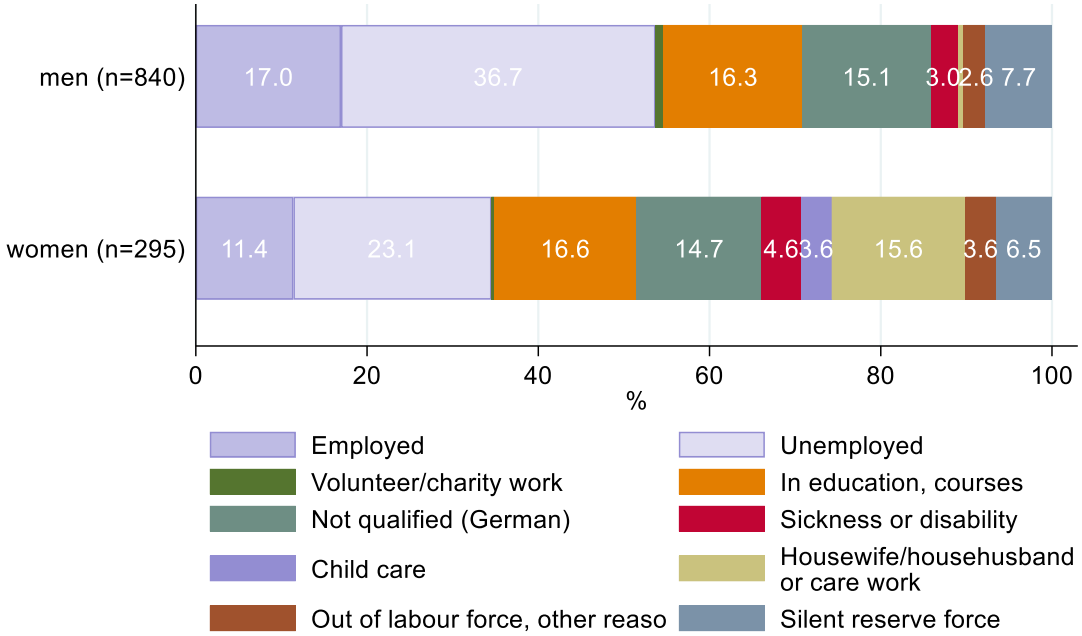
Graph 34. Employment status of women by duration of stay, Labour Force Concept

Source: FIMAS data, excluding asylum seekers

Another way to analyse employment status is to look at the sub-group of persons who are not active in the labour market but could potentially participate, who have expressed a desire to work but have not taken active steps, or are not registered as unemployed (Institute for Employment Research, 2010). This "silent reserve" is particularly interesting from a labour market policy standpoint, in terms of mobilising them through targeted measures. A total of 7% of respondents fell into this category, with the rate slightly higher for men (7.7%) than for women (6.5%).

0 clearly illustrates the roots of gender-specific labour market participation: whilst showed a higher labour force participation rate (men 54%, women 35%), women (19% with childcare and household responsibilities) were much more likely to do unpaid work in domestic settings. Furthermore, women were more likely to state that they were not working or seeking work due to illness or disability (5% compared to 3% among men). In all other categories – education, volunteer work, silent reserves and other reasons – men and women differed only slightly. This suggests that reproductive work among respondents was divided between the genders; whilst men were more likely to financially support the household, women were more likely to look after the household and children. However, the data collected did not provide information about possible further causes, such as gender roles or discrimination.

Graph 35. Labour market status and silent reserve by gender

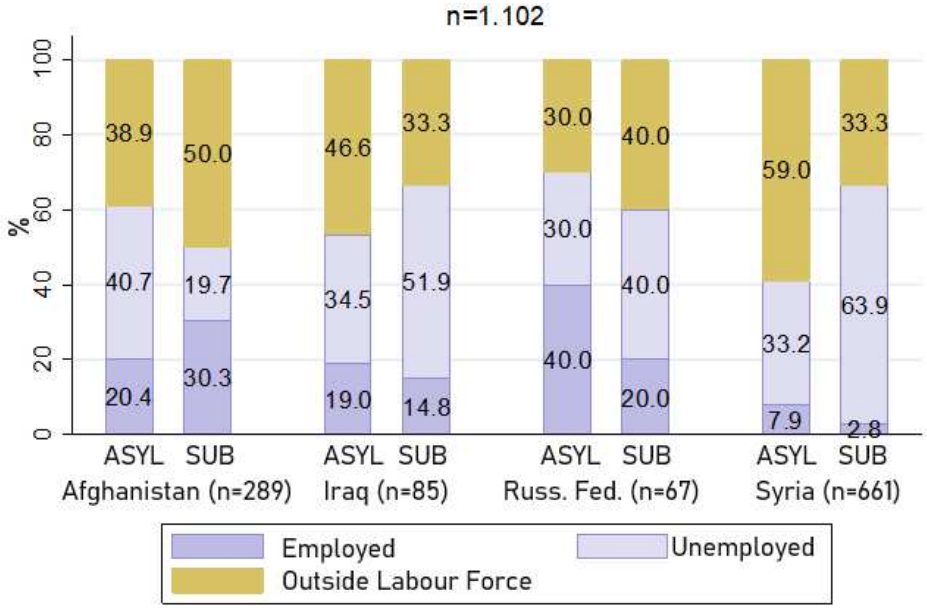


Source: FIMAS data, excluding asylum seekers

When analysing labour market participation according to legal status – **recognised refugees** and **beneficiaries of subsidiary protection** – the results vary as well (Graph 36). Among Afghans, beneficiaries of subsidiary protection were more likely (30%) to be in employment than refugees (20%), but were also less active in the labour market (50%) than refugees (61%). The opposite seems to be the case among Syrians. Beneficiaries of subsidiary protection were less likely to be in employment (3%) than refugees (8%), but were more often active in the labour market (67%) than refugees (41%). The two sub-samples of Iraqis and Chechens/Russians were too small to allow for generalisations.

Overall, beneficiaries of subsidiary protection were more often in employment (23%) than refugees (13%), and less often outside the labour force (45%) than refugees (53%). However, this effect can also be explained by the duration of stay. On average, beneficiaries of subsidiary protection had been living in Austria for longer (mean 3.9 years) than those granted asylum, i.e. refugees, (mean 3.0 years), which at least partly determines integration in the labour market.

Graph 36. Employment status by analytic group and residence status, Labour Force Concept



Source: FIMAS data, excluding asylum seekers

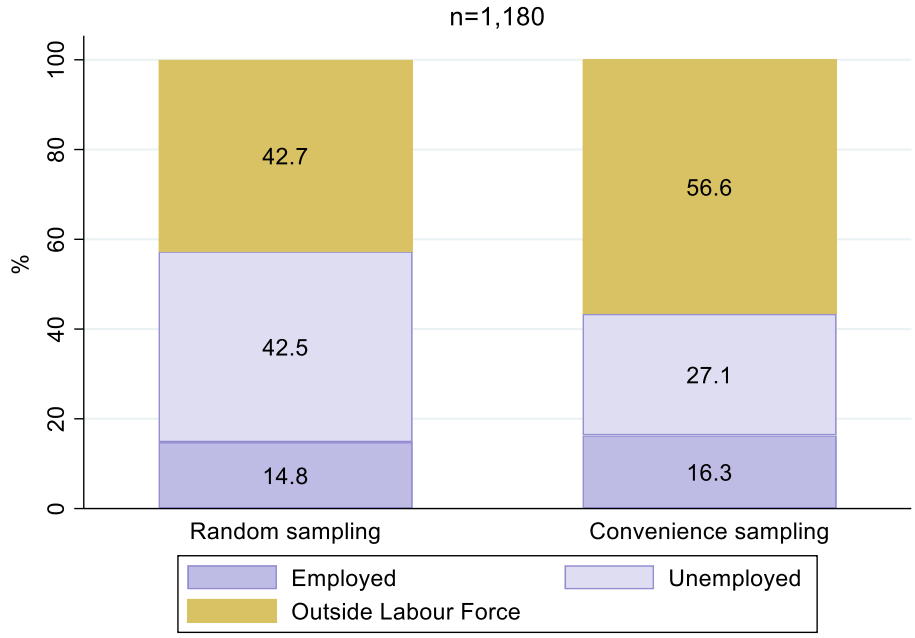
As stated above, the survey was split in two phases, a random sampling phase followed by a non-random sampling phase.

For the random sampling approach, persons registered in the AMS data were drawn. These comprised persons registered as unemployed at the time of sampling as well as persons formerly registered with the AMS. For this latter group, there are two pathways out of unemployment. Either they enter into employment or they become inactive (such as is the case for pensioners or persons receiving maternity benefits). This also explains why interviews with persons who were unemployed (43%) and persons who were inactive (43%) were equally frequent during the first survey phase (Graph 37).

During the second survey phase, which used convenience sampling, a much higher proportion of persons who were inactive (57%) was recorded, and a lower proportion of persons who were registered as unemployed (27%). This suggests that the sample from the AMS data is less likely to include persons not available for the labour market. This under-coverage of inactive persons during the first survey phase was probably offset during the second survey phase. As such, one can assume that the second survey phase is less likely to distort the structure of the population, but rather to serve as a useful complement to the sample.

Nonetheless, it seems that **employed persons were under-reported** throughout the entire sample, resulting from the interview process (see section 3 as well as Graph 69 and Graph 70 in the Appendix). It proved difficult to reach these individuals for a personal interview on a weekday during working hours. Compared with analyses of the labour market database (Prettenthaler et al., 2017), the FIMAS data show higher rates of unemployment and lower rates of employment (-14.8% on average).

Graph 37. Employment status by sampling method, Labour Force Concept



Source: FIMAS data, not including asylum seekers

4.7.2 Current professions among those who are employed

One third of those in employment (33%) worked in **elementary occupations**, for example as food preparation assistants (14% of all employed persons), cleaning staff (6%), in manufacturing (4%), in transport and storage (3%) and other elementary jobs (4%) (Graph 38 and 0).

Just under a quarter of respondents (24%) worked in a **service and sales professions**. The most frequent responses within this major group were occupations in retail sales (5%), waitressing and as bartenders (4%), chefs or cooks (4%) or security staff (3%).

About 16% worked in **craft and related trades**, mainly as mechanics (4%) and welders (3%).

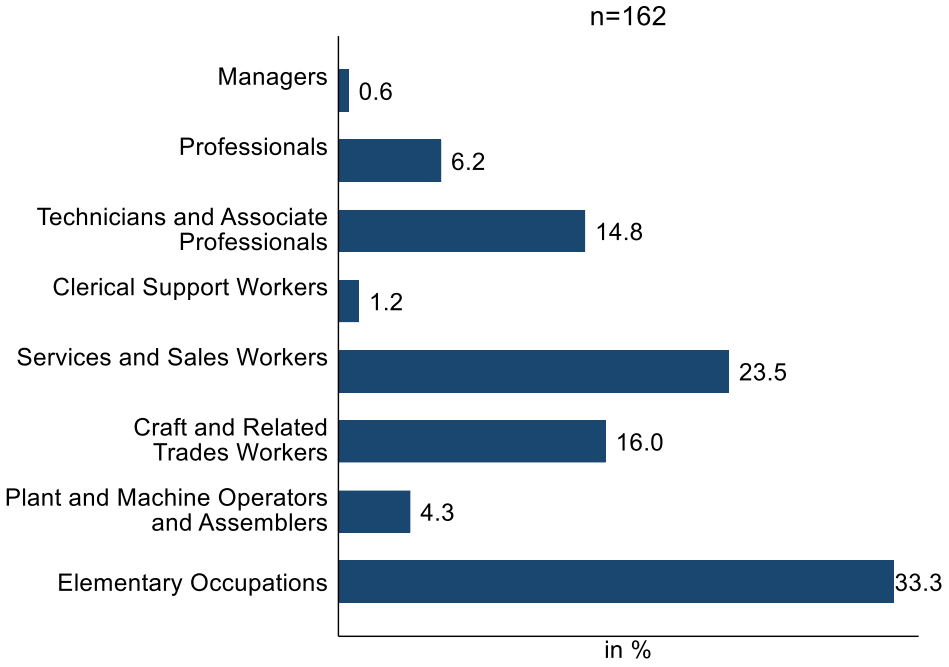
Technical professions (15%) were listed almost as often, but referred to quite heterogeneous occupations, including non-academic legal, social, cultural and related professions. **Social workers,** therefore, account for the largest group within technical occupations (9%), with the second-largest group being materials and engineering professions (2%).

Professionals were rare among those currently employed (6%), which included mainly interpreters/translators (3%).

The major groups of **assemblers, machine operators and drivers** (4%), **office clerks** (1%) and **management positions** (1%) were those mentioned the least. Not a single person surveyed worked in agriculture or forestry.

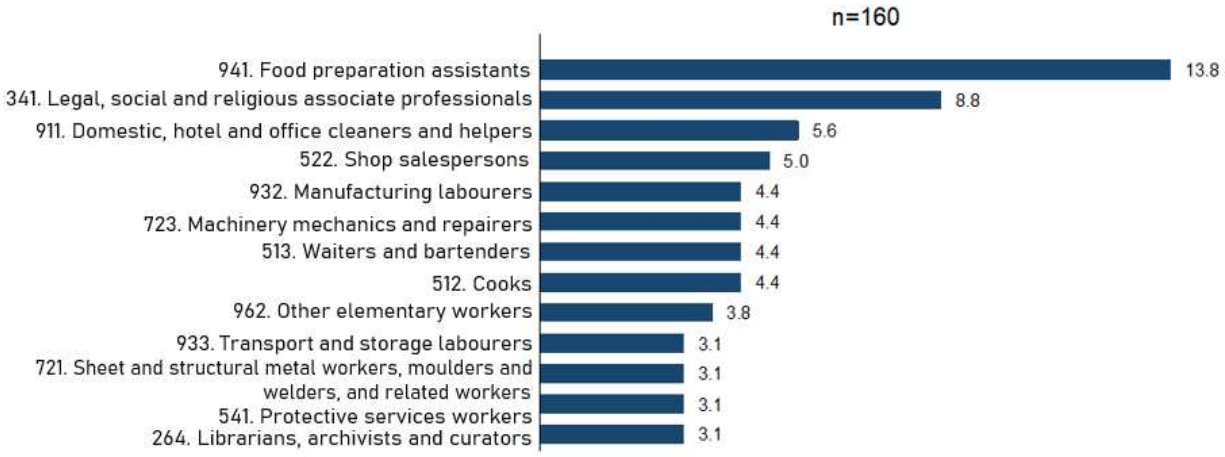
Broken down by gender, it was striking that nearly four out of ten women (39%) worked as **social workers**, with elementary workers making up the second largest group (26%). Among men, the distribution was almost the same as in the overall sample covering both sexes, with elementary jobs (35%) and service and sales professions (26%) being the most common. However, the sub-group of women employed in Austria was too small to allow for generalisations.

Graph 38. Current professions in Austria, major occupational group, Ö-ISCO 2008



Source: FIMAS data, excluding asylum seekers

Graph 39. Current professions in Austria, most frequent minor occupational groups (as % of employed), Ö-ISCO 2008

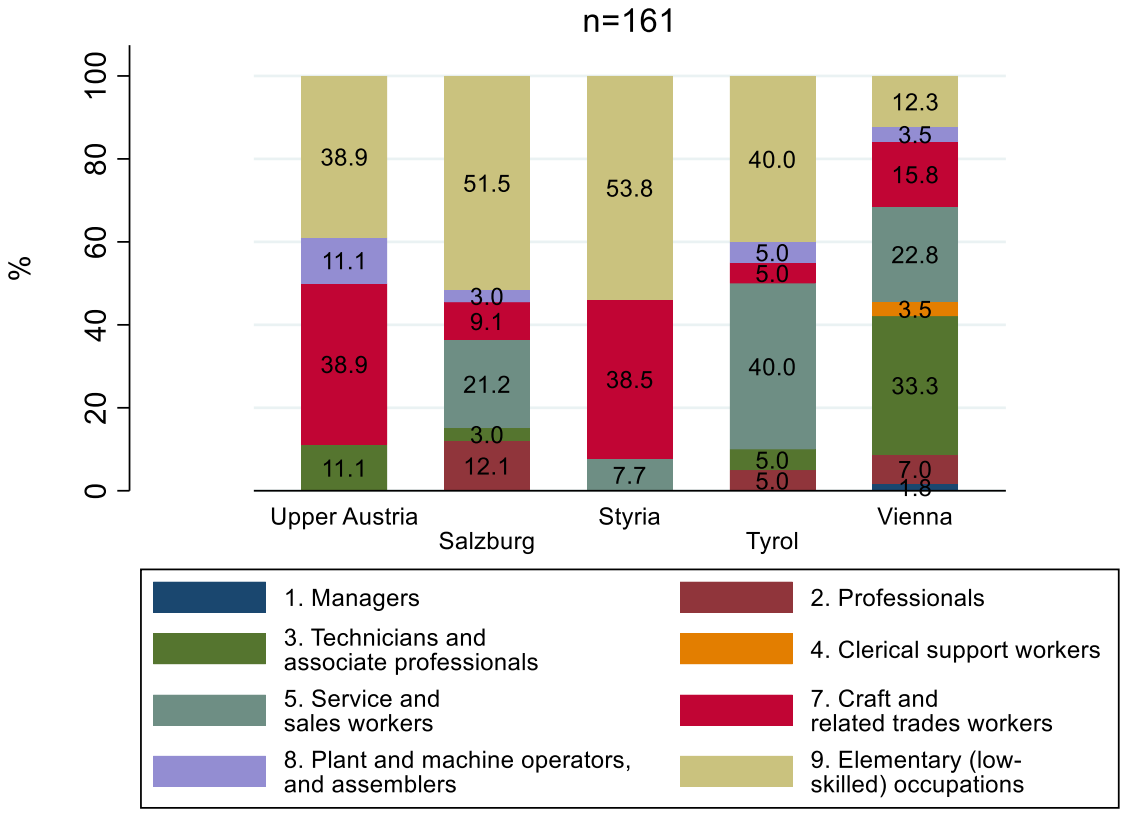


Source: FIMAS data, excluding asylum seekers

The occupational structure in the federal provinces was highly heterogeneous. Whilst nearly all major occupational groups were represented among those who were employed in Vienna, the number of low-skilled workers was considerably higher in the other four federal provinces, from 39% in Upper Austria to 54% in Styria (Graph 40). In Vienna, only 12% of recognised refugees and beneficiaries of subsidiary protection worked in elementary occupations.

Crafts and trades were most frequently listed among respondents in Upper Austria (39%) and in Styria (39%). Service and sales professions were most frequently listed in Tyrol (40%). They were also above average in Salzburg (21%) and Vienna (23%). Social workers and technical professions, in turn, were mainly found in the capital Vienna (33%). Also here, generalisations are not possible due to the small sub-sample sizes by federal province.

Graph 40. Current professions by federal province, major occupational groups, Ö-ISCO 2008

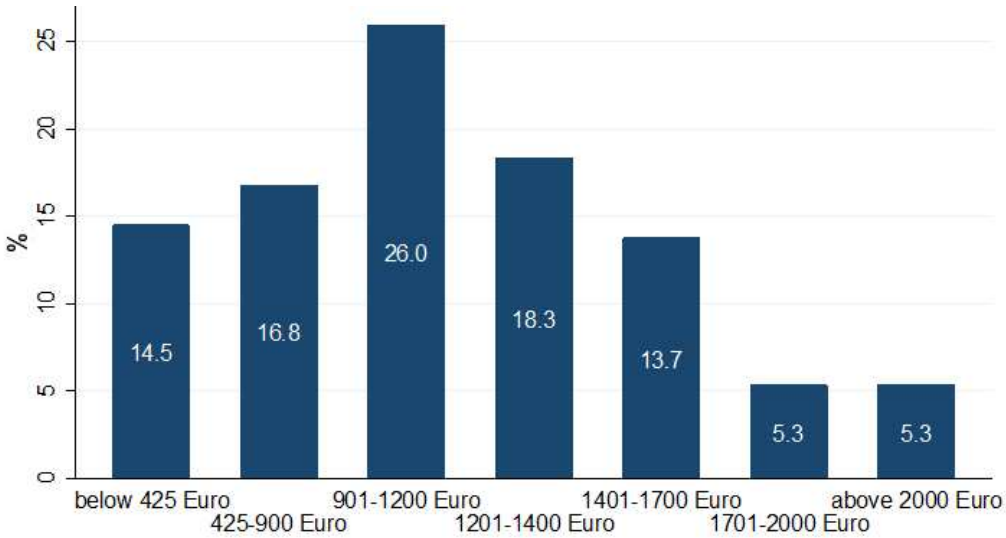


Source: FIMAS data, excluding asylum seekers

The **average duration of employment in Austria was 14 months** (one quarter of respondents worked for less than 5.3 months, one quarter for more than 2.5 years) with a clearly left-leaning distribution; the arithmetic mean was thus higher at 2.0 years. The average duration of employment was **slightly higher among women (1.5 years) than among men (1.1 years)**. As expected, the duration of employment correlated with both the duration of stay and age of respondents.

The average **monthly net income of those who were employed was 1,200 EUR for full-time and part-time workers combined; 1,300 EUR for full-time workers and 640 EUR for part-time workers**. Of the respondents, 15% earned less than 425 EUR a month and 17% earned between 425 and 900 EUR a month (Graph 41). Slightly more than one quarter of respondents (26%) reported a monthly net income of between 900 and 1,200 EUR. A further 18% of the respondents fell in the category of between 1,201 and 1,400 EUR per month, and 14% earned between 1,401 and 1,700 EUR per month. There were hardly any respondents who earned over 1,700 and up to 2,000 EUR a month net (5%) or even over 2,000 EUR a month (5%).

Graph 41. Distribution of monthly net income among the employed (n=131)



Source: FIMAS data, excluding asylum seekers

Here, the highly similar distribution of income among male and female employed persons was striking (Graph 42). The average monthly net income for men was 1,200 EUR, whilst that of women was only slightly lower at 1,150 EUR. The quartiles for distribution of income by gender were also close, with women's incomes even spreading slightly higher; possibly an effect of industry-specific earnings differentials. Women were more often working in academic jobs, less frequently as elementary workers.

Graph 42. Distribution of monthly net income of employed persons by gender (n=130)

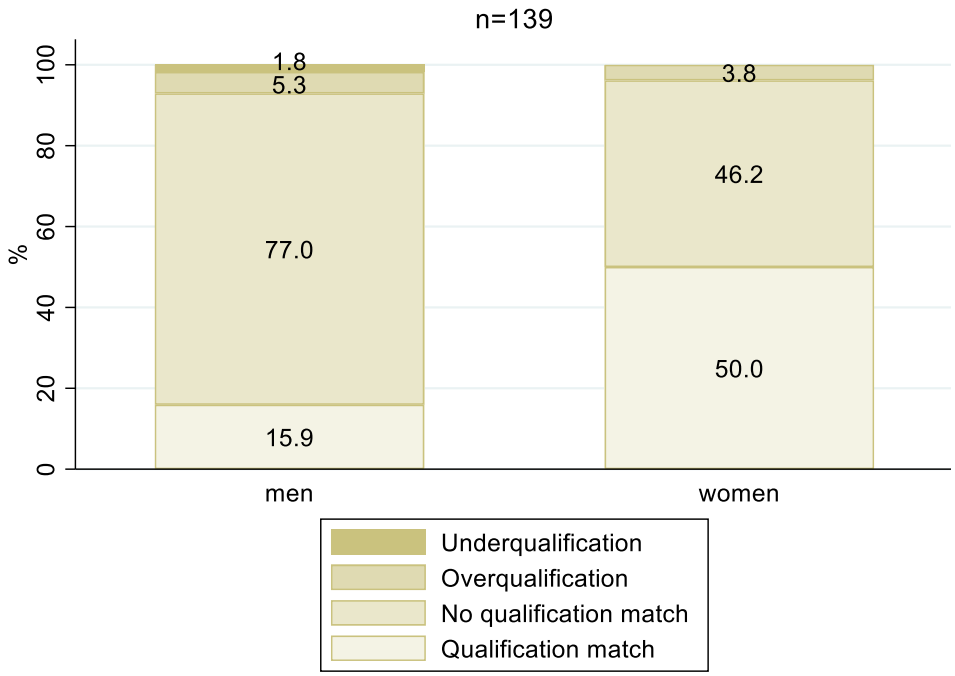
Source: FIMAS data, excluding asylum seekers

4.7.3 Mismatch and over-qualification

Those respondents who stated that they were currently employed were asked whether their current job was related to their education and, if so, whether they perceived themselves underqualified, overqualified or suitably qualified for their job. Of the respondents, 78% stated that, in their own view, qualification mismatch, or an insufficient match between education and occupation, affected their labour market situation. This means that, in their own view, less than one quarter of respondents held an occupation in Austria that corresponded to their qualifications. Overall, over- and under-qualification did not play much of a role: only 5% of the employed stated that they were overqualified and 1% stated that they were underqualified. Of all the employed respondents, 71% stated that their occupations did not relate at all to their educational or training background, whilst only 22% indicated that their job was suitably related to their background qualifications.

Once again, there were clear differences in terms of gender (Graph 43), in this case in favour of women: half of women who were employed (50%) worked in a job related to their education at the time of the survey. This was only the case for 16% of men.

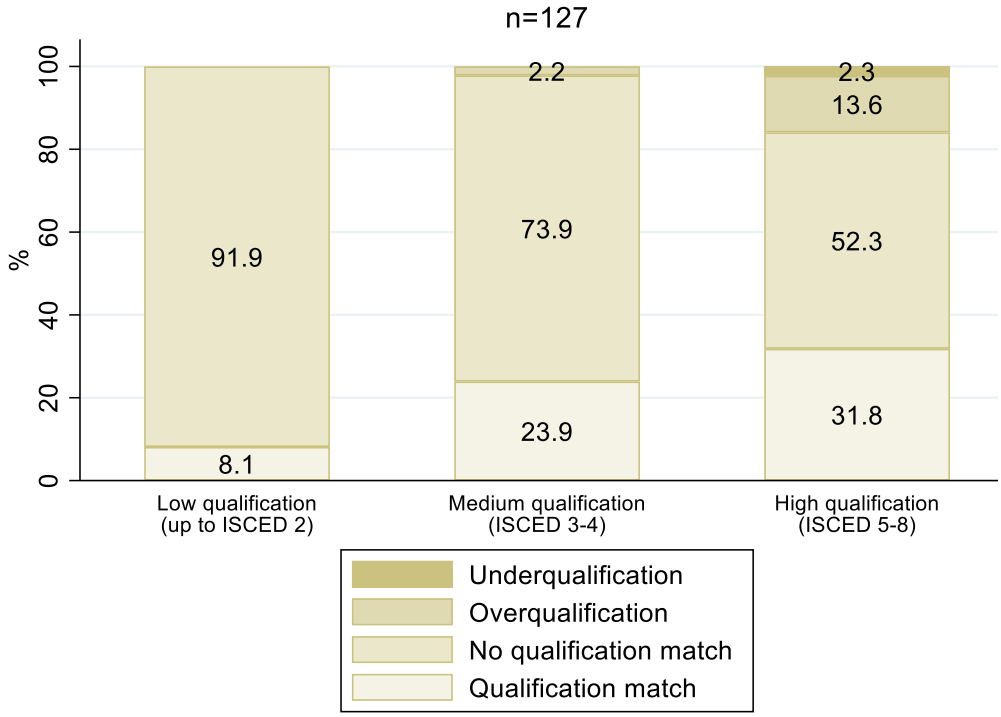
Graph 43. Qualification-related mismatch in current profession by gender



Source: FIMAS data, excluding asylum seekers

To a certain degree, the observed qualification mismatch can be explained by the low qualification levels that almost 40% of recognised refugees and beneficiaries of subsidiary protection brought with them. The mismatch was particularly high among those with low qualification levels, affecting more than nine out of ten respondents (0). Among those with intermediate qualifications, the proportion of mismatch was distinctively lower (74%), and the proportion of persons working jobs that corresponded to their educational background was much higher (24%). Among those with higher qualifications, complete mismatch only affected about half (52%), and almost one third of persons (32%) worked in a job that matched their educational background. It was only in this group that over-qualification came into play; with 14% claiming to have qualifications above what their current job required.

Graph 44. Qualification-related mismatch in current profession by highest educational qualification



Source: FIMAS data, excluding asylum seekers

As the duration of stay increases, there is a clear trend towards a reduction in mismatch and an increase in appropriate employment. There are also a differences according to analytic group as well, with Chechens/Russians more often (48%) reporting that they work in appropriate occupations – an effect that is likely a result of their longer average duration of stay and correspondingly increased likelihood of having obtained qualifications and soft skills that allow for taking up more adequate jobs.

4.7.4 Atypical employment

Atypical employment, especially in the form of part-time work, was quite common among respondents. Of the respondents, 39% worked part time, i.e. less than 36 hours a week, 25% of respondents worked 20 hours or less, and 15% worked 10 hours a week or less. 6% worked 45 to 52 hours per week. In most cases, respondents **wanted a full-time job**, with 62% of part-time workers claiming they wanted to work more than 36 hours a week.

The part-time rate was much higher among women than among men: 70% worked less than 36 hours a week, 33% worked 20 hours or less and 13% worked 10 hours a week or less. Part-time work was less common among men, of who 32% spent less than 36 hours a week in paid work. Of the employed male respondents, 24% worked less than 20 hours a week, and 15% worked less than 10 hours a week. The share of **part-time work decreases with duration of stay**: with respect to the first employment contract respondents held in Austria, 50% said that they had worked less than 36 hours a week, and for the job they held prior to the current job, this figure was 49%.

Temporary employment accounted for 27% of respondents who were employed, for men (29%) more frequently than for women (19%). Elementary workers (44%), academic professionals (16%) and those in service professions (16%), in particular, often had temporary contracts.

The survey also asked if the employment relationship was that of regular employment, was unregistered, was as assisting with the family business or took the form of an internship, or whether the person was self-employed. Of those in employment, 85% said they were in **regular employment**, 5% were **self-employed** or working on a freelance basis, 5% were **interns** and 2% **helped out with a family**

business. Only 3% stated that they were engaged in informal work (social desirability or concerns related to their status may also account for this low rate).

Only 3% of those employed had one or more **side jobs**. **Sub-contracted labour** was also a very rare phenomenon – only 1% of respondents worked through a sub-contracting company.

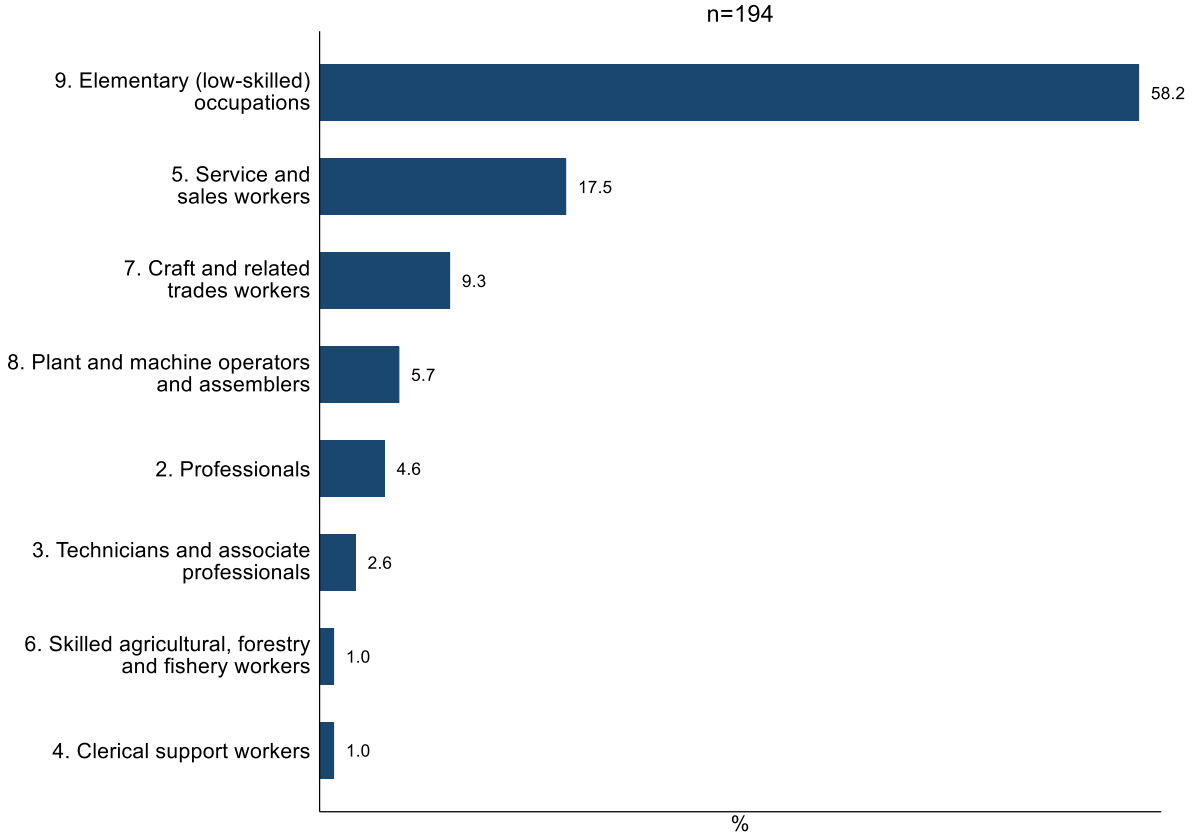
4.8 Labour market integration processes

The FIMAS survey covered up to three employment episodes in Austria as part of the interview. In addition to their current job, respondents were asked about other previous occupations and the first job they held in Austria. Of the respondents, 78% said they had never previously been employed in Austria.

Details regarding **previously held occupations** were, therefore, only available for the sub-group who provided that information. More than half, **nearly six out of ten respondents** (58%) who were previously employed in Austria worked in **elementary occupations**: for example, in food production, transport and storage, or as cleaning staff. Nearly 18% worked in service professions, 9% in craft trades and 6% as machinery operators or assemblers (Graph 45).

Among men, the proportion of persons working in low-skilled jobs was significantly higher (62%), followed by service and sales professions (15%). Among women, only 42% worked in elementary occupations and 29% worked in service and sales.

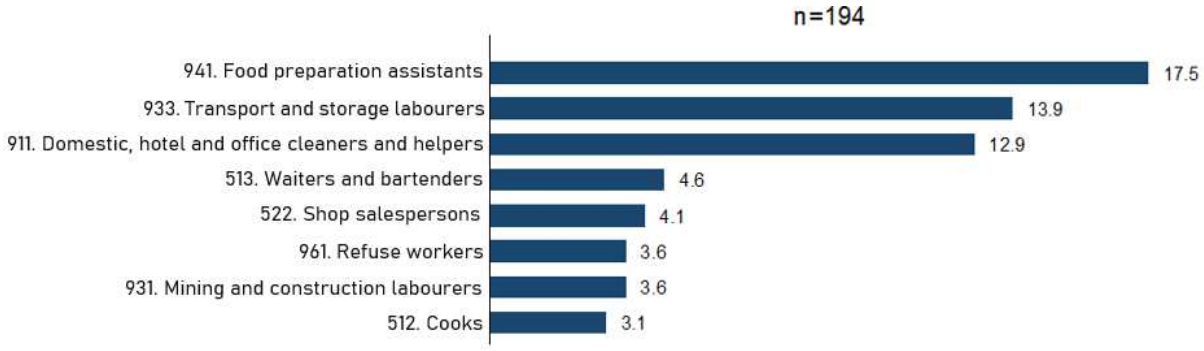
Graph 45. Previous profession in Austria, major occupational groups, Ö-ISCO 2008



Source: FIMAS data

A more detailed analysis by minor occupational group (Graph 46) shows the main types of low-skilled work and service professions. 18% were employed as elementary workers in food production (often as food preparation assistants), 14% as transport labourers, 13% as cleaners, 5% as waiters, waitresses and bartenders, 4% as shop salespersons, 4% in waste disposal, 4% as construction labourers and 3% as chefs or cooks. Many of these occupations were low paid and required neither high levels of qualification nor advanced German language skills.

Graph 46. Previous profession in Austria, most frequent minor occupational groups (in %), Ö-ISCO 08



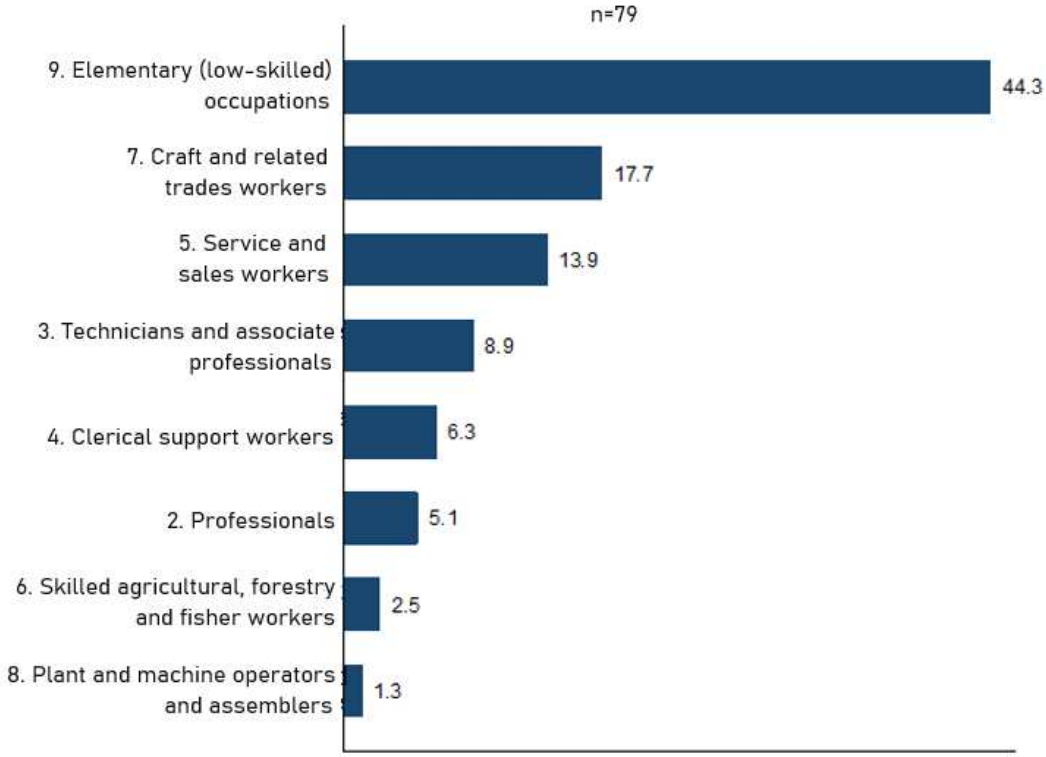
Source: FIMAS data

Another sub-sample contained information on the **first job held by respondents in Austria**. This included both persons who were in employment at the time of the survey (62%), i.e. those who had already switched jobs at least once, as well as persons who were unemployed or inactive persons but who had worked previously (38%).

A breakdown of the major occupational groups of these first employment episodes (Graph 47) provides a similar picture to that of previous employment. **Almost half (44%)** of all first jobs were in **elementary occupations**. Crafts and trades (18%) and service professions (14%) were much rarer, as were technical professions (9%) and office jobs (6%). Women were more frequently employed as technicians and associate professionals (here: social workers) and in the service sector (25%), whilst men more frequently worked in elementary jobs (51%) and in the crafts sector (21%).

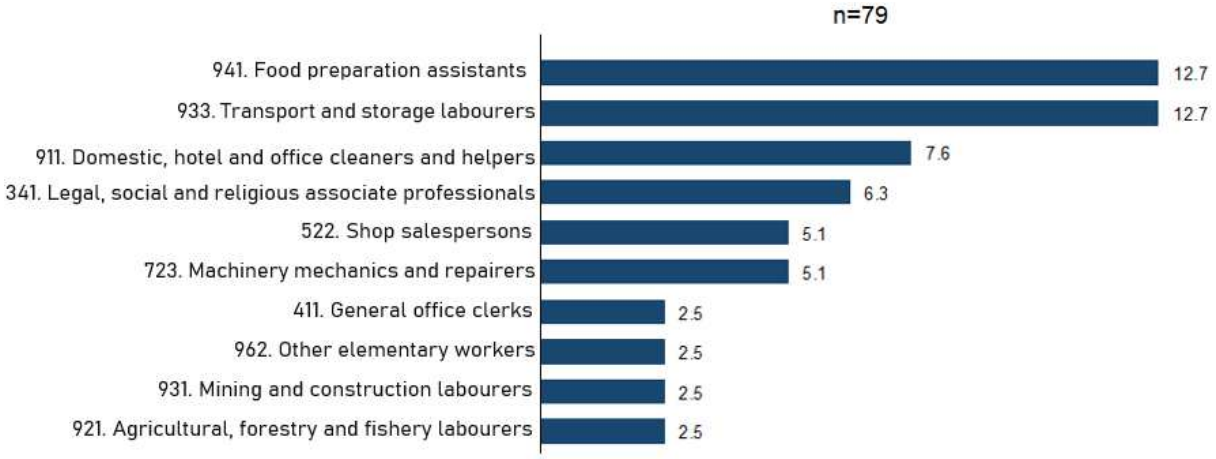
The detailed analysis by minor occupational groups (Graph 48) provides concrete examples. For example, about 13% of those who provided information on their first occupation worked as transport labourers (e.g. packaging), whilst 13% worked as food preparation assistants or dishwashers, 8% as cleaners and 6% as refugee consultants. Office clerks and mechanics accounted for 5% respectively.

Graph 47. First job held in Austria (in %), major occupational groups, Ö-ISCO 2008



Source: FIMAS data

Graph 48. First job held in Austria (in %), most frequent minor occupational groups, Ö-ISCO 2008



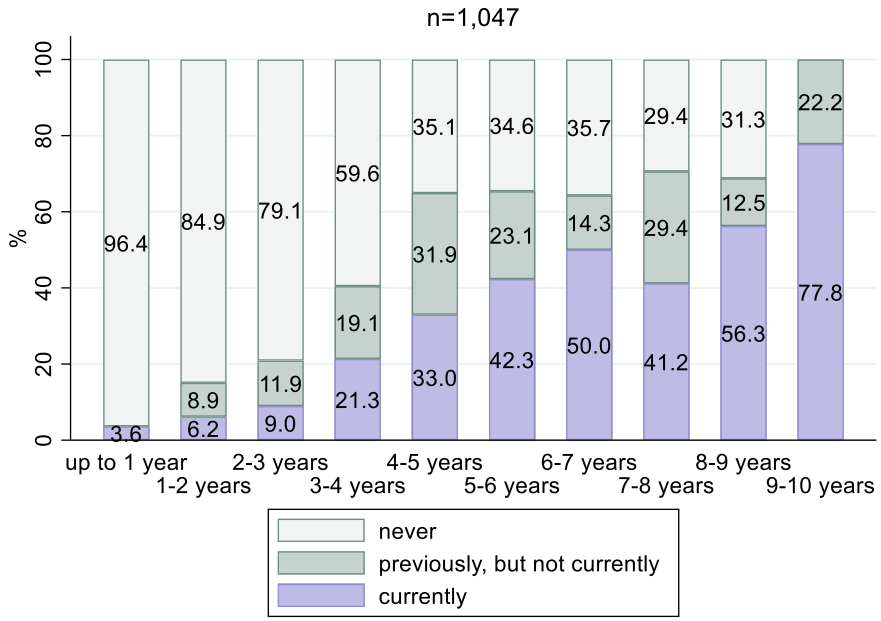
Source: FIMAS data

Comparing these data with the ones on occupations worked abroad (by major occupational group), some discrepancies become apparent. Of those respondents who worked in academic professions in their home country or abroad, only 25% worked in the same occupational group in Austria. Similarly, 25% of those employed abroad in the service industry worked in a service profession in Austria. However, 37% of those involved in crafts and trades were also practising this profession or a similar one in Austria. By contrast, only 20% of technicians worked in the same major occupational group.

Equally revealing is the combination of information from all three questions on current, previously held and first employment in Austria. From the combination of answers, it was able to deduce whether persons were in employment at the time of the interview, whether they had already been employed since arriving in Austria, but were unable to find paid work at the time of the survey, or whether they had not yet been able to find employment (Graph 49).

The trajectory by duration of stay was similar to that of the current employment status (deviations in the proportions for those currently employed were the result of the slightly higher number of missing values for this analysis). Among those who had been in Austria for up to three years, the proportion of those in employment was 4% to 9%, but was about twice as high when taking into account those who had previously been employed at least once. Among those who had been living in Austria for two to three years, the combined rate was 21% (9% currently employed, 12% previously employed).

Graph 49. Previous and current employment (including job offers) in Austria by duration of stay



Source: FIMAS data, excluding asylum seekers

Among respondents who had been in Austria for four to five years, both groups – those currently employed and those previously employed (who were not working for pay at the time of the survey) – were almost equivalent in size: about 33% were in employment and 32% had been employed previously.

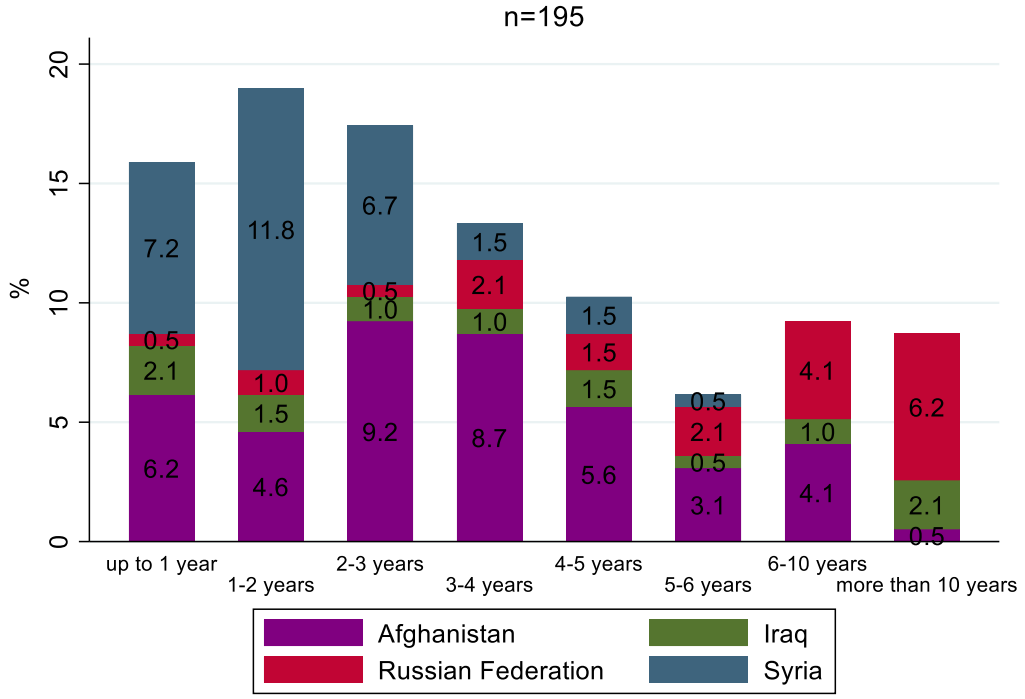
In earlier cohorts, the share of those previously employed decreased compared to those currently employed. The effect of the years of the global economic crisis can also be seen here, although it is not as pronounced as when looking at the current employment status. Only the group of refugees that arrived in Austria in 2009 (and thus represent the group with 7-8 years of stay) breaks the otherwise almost linear relationship between duration of stay and employment status. This also confirms previous conclusions. Active participation in the labour market develops gradually; among those who had spent up to four years in Austria, only up to 40% had gained their first employment experience. For those who had lived in Austria for five years and more, this proportion increased to between 65% and 100%.

Based on the information provided about respondents' date of arrival in Austria, first employment, current employment and other previous employment events, the length of time until refugees were first employed can be calculated for the sub-group that provided information on the start of these employment events. The average duration was 2 years and 11 months; a quarter of those surveyed found employment after 1 year and 5 months at most, a quarter took longer than 4 years and 11 months. For men, the mean amount of time was two years and eight months, whilst for women the mean amount was four years and three months. Thus, **for female refugees, it took on average one-and-a-half years longer to find first employment.**

The results also differed between the analytic groups (Graph 50). **Syrians** had the shortest mean amount of time before finding **first employment (one year and eight months)**, followed by **Afghans (three years)**, **Iraqis (three years and six months)** and **Chechens/Russians (eight years)**. Yet, the validity of these results is limited by the fact that only half of all those in employment provided information on when they began employment. For all for those who were not in employment at the time of the study, it could not be established at all.

The findings as to the mean amount of time it took persons to initially find employment should be interpreted in comparison to the findings described above on current employment status, and current or previous employment, as an indicator of individual lead times for those currently in employment. By definition, no information can be given regarding those who had not yet found employment.

Graph 50. Amount of time until first employment by analytic group



Source: FIMAS data, excluding stateless persons

4.9 Social networks

Participants were also asked about the relevance *of* and their involvement *with* social networks. In social science research, the term "social network" refers to a network of social contacts between persons, which offers their members comparative advantages in different areas. Networks consist of individuals, groups or organisations that are tied together by a variety of links, such as family, friendship or interest. In migration research, the significance of such networks for integrating migrants is well-researched and well-established.

However, especially in studies on migrants' labour market integration, there is no consensus on whether social networks support labour market integration and in which way or if they even inhibit it in the long term. On the one hand, migrants who have already immigrated from the same country of origin can help newly arrived co-nationals find work through their experiences and contacts. On the other hand, however, empirical studies also find social networks to have a negative impact on labour market integration since they can solidify existing qualification mismatches and, therefore, hamper the appropriate placement of migrants on the labour market in the long run.

In order to be able to analyse these possible scenarios for the Austrian context, the FIMAS survey included questions on respondents' experience with social networks in Austria. At the time of writing, the FIMAS dataset was the first of its kind to allow for the analysis of these issues for Austria. The questionnaire included questions about new contacts made since arriving in Austria, how respondents got in contact with them, and about finding employment through this new social network.

Initial descriptive evaluations pointed to gender-specific differences: men were more likely to have made new contacts (91%; 84% of women), and in most cases met these new contacts through courses, training or education (28%; 24% for women). Among women, newly-established contacts were more common with persons from the same country of origin (28%; 25% for men). These findings result, among others, from the different degrees of labour market activity undertaken between the genders: whilst men were more likely to pursue or seek education or employment, women were more likely to take on care and household responsibilities.

Furthermore, the analysis along countries of origin showed that Syrians (85%) and Iraqis (88%) had less frequently made new contacts in Austria. This can be explained by the fact that the other two groups had a longer duration of stay Austria. Compared to Chechens/Russians (99% had made new contacts) and Afghans (97%), respondents from Syria and Iraq were, on average, more likely to have lived in Austria for a shorter period of time (see Graph 6).

4.10 Health and limitations of activity

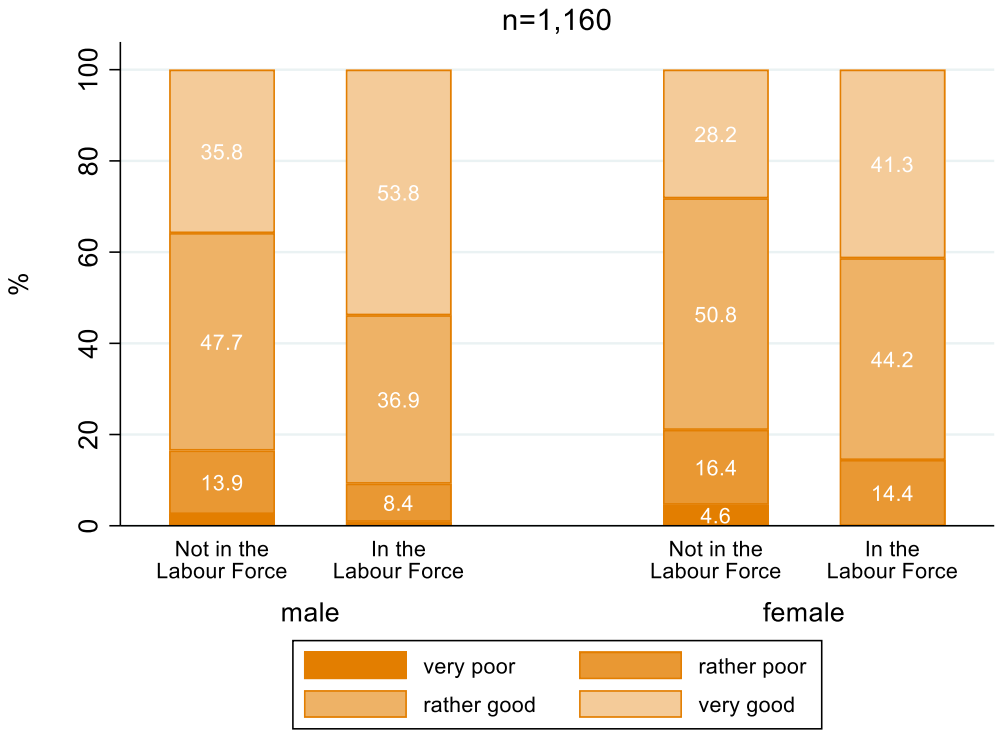
Respondents were also asked about their health situation. The questions related to general health status, as well as specific limitations of activity due to any physical or psychological impairment respondents may be suffering from. Both questions allowed respondents to assess their situation at their own discretion.

Men rated their overall health status higher than women did (87% of men responded very good or somewhat good, whilst this figure was 82% for women). When differentiating between those in employment and those inactive, the same differentiation was evident within gender groups (Graph 51). Of the male respondents who were part of the labour force, 91% reported to be in good or very good health, as did 86% of female respondents in the labour force. However, only 84% of men not active in the labour market reported to be in good or very good health, with 79% of women not active in the labour market reporting the same.

There was also a correlation between labour market participation and health: both male and female respondents in employment considered their health status better than men and women who were not

active in the labour market. Consequently, the proportion of those who assessed their health to be somewhat or very poor was highest among those not active in the labour market, and especially among women. In this group, female refugees who were not active on the labour market, 16% described their general health as rather poor, and 5% as very poor.

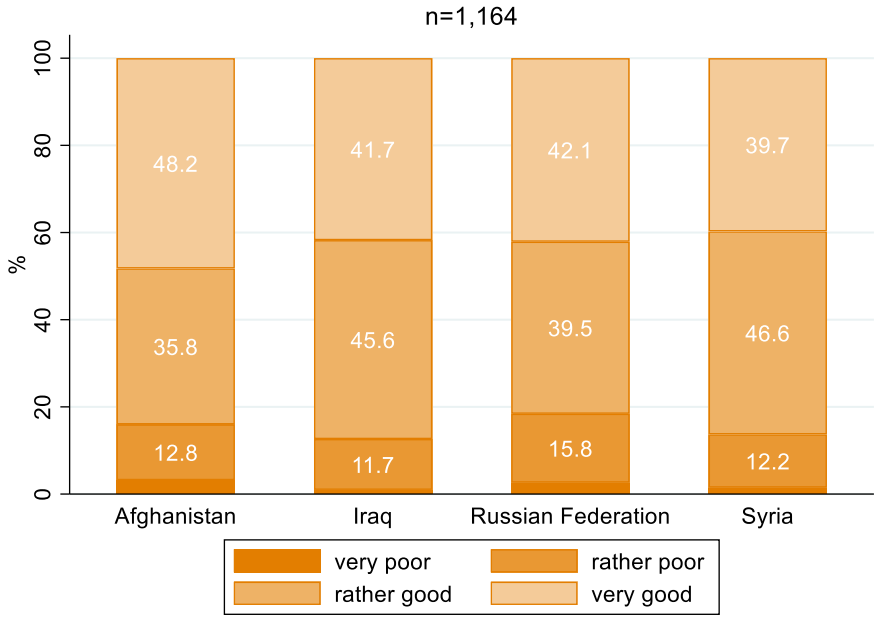
Graph 51. General health status by employment status and gender



Source: FIMAS data

Differentiating by country of origin, Iraqis reported the best health status (88% reported very good or somewhat good health), followed by Syrians (84%) and Afghans (84%). Despite their on average longer duration of stay, Chechens/Russians were at the lower end of the scale in terms of their assessments of their general health at 82% (Graph 52).

Graph 52. General health status by analytic group

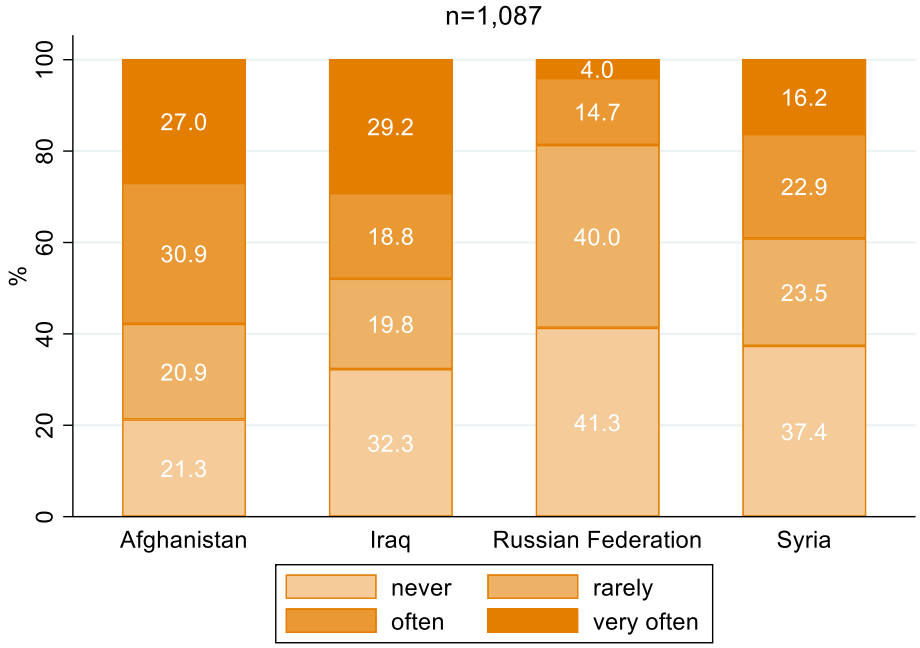


Source: FIMAS data

This result was confirmed by the analysis by duration of stay, which showed no linear relationship between duration of stay and the general health of respondents. However, as age increases, positive self-assessment of the state of respondents' health decreases. Refugees from the Russian Federation were, on average, older than other groups, which may explain the group-specific difference.

Interestingly, gender differences in activity limitation were lower than for general health status. Although 46% of women reported greater limitations of their activity, 43% of men also said they were affected by this. When looking at the situation by analytic group (Graph 53), it becomes clear that **Afghans (58%), Iraqis (48%) and Syrians (39%) were all affected by considerable limitations** (often or very often). By contrast, only 19% of Chechens/Russians stated that they were often or very often restricted by physical or emotional impediments during daily activities.

Graph 53. Health-related activity limitation by analytic group



Source: FIMAS data

Significant differences in activity limitations can be seen between those not active in the labour market and those who are employed, as well as between unemployed and employed persons (0). These differences persist even when excluding those who were not active in the labour market due to illness or disability. This is most evident in those who reported to have never been affected by these kinds of restrictions within the previous four weeks. Whilst 27% of those not active in the labour market reported no restrictions whatsoever, 37% of those who were unemployed and 44% of those who were employed stated they were never affected by these. Of those not active in the labour market, 29% reported being affected by restrictions frequently, while this figure was 21% among those who were unemployed and 14% among those who were employed. This means that **there is a correlation between unemployment and health-related limitations.** Cross-sectional data is not sufficient to clarify with certainty whether the state of refugees' health has a causal effect on their labour market participation or whether work or labour market activity have positive effects on health (a reciprocal effect could also be a possibility).

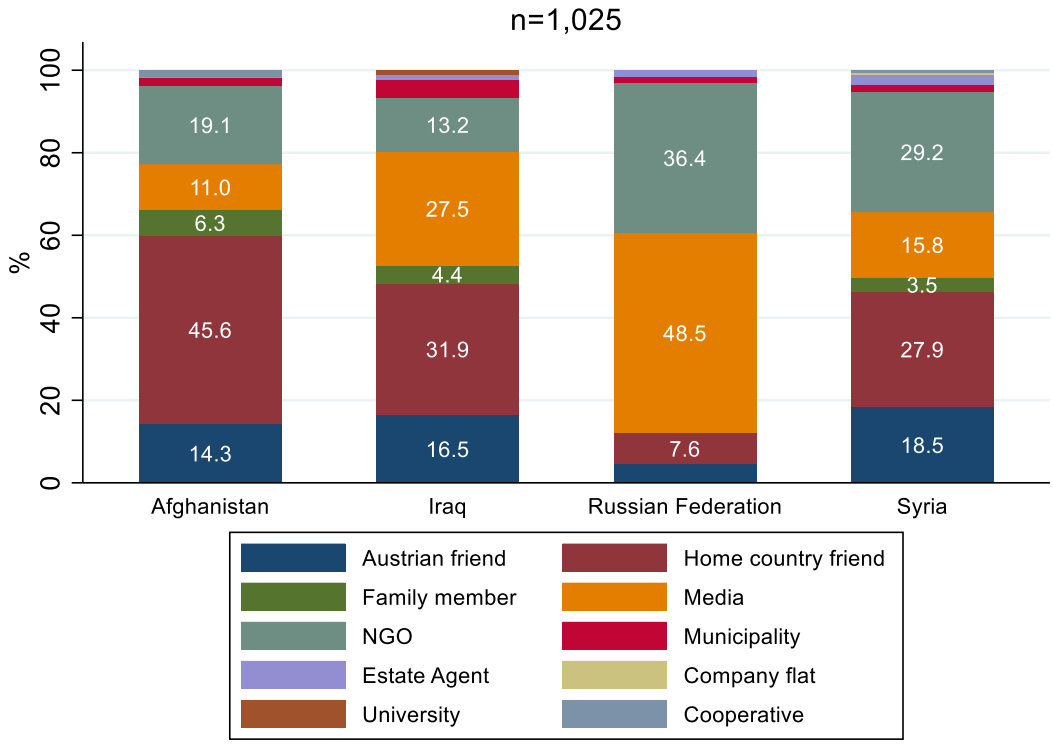
Graph 54. Health-related activity limitation by employment status, Labour Force Concept

Source: FIMAS data

4.11 Accommodation

Understanding how respondents **found their accommodation** and which **type of accommodation** they had managed to find was of particular interest for the FIMAS study. Graph 55 shows the different **types of housing searches by analytic group**.

Graph 55. Housing search by analytic group

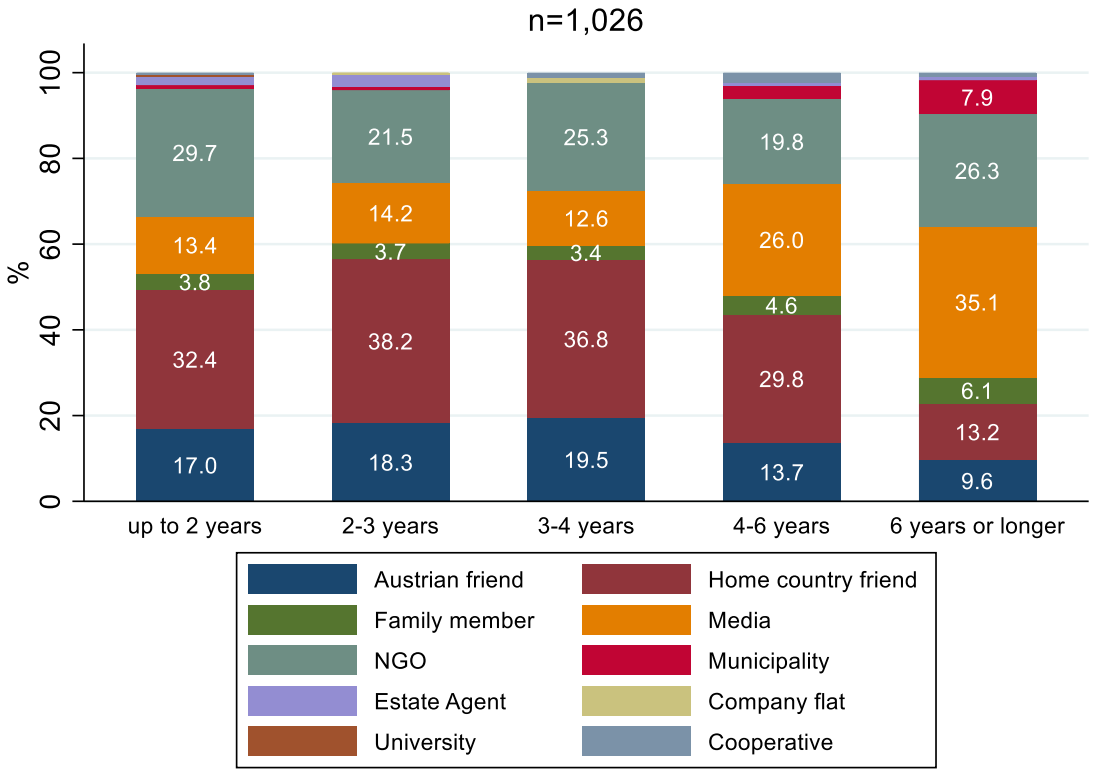


Source: FIMAS data

Whilst Afghans and Iraqis claimed to have found accommodation most often through friends from their country of origin (46% of Afghans, 32% of Iraqis), most Chechens/Russians reported using media such as the newspaper, internet and Facebook (49%) to find suitable housing. Assistance in finding housing by NGOs (29%), as well as friends from their country of origin (28%) played an important role for Syrians. Their relatively short duration of stay may have translated into a widespread use of family and other personal contacts among this group. This support through personal contacts seemed to be especially relevant for Afghans, of who almost 50% were under 24 years old.

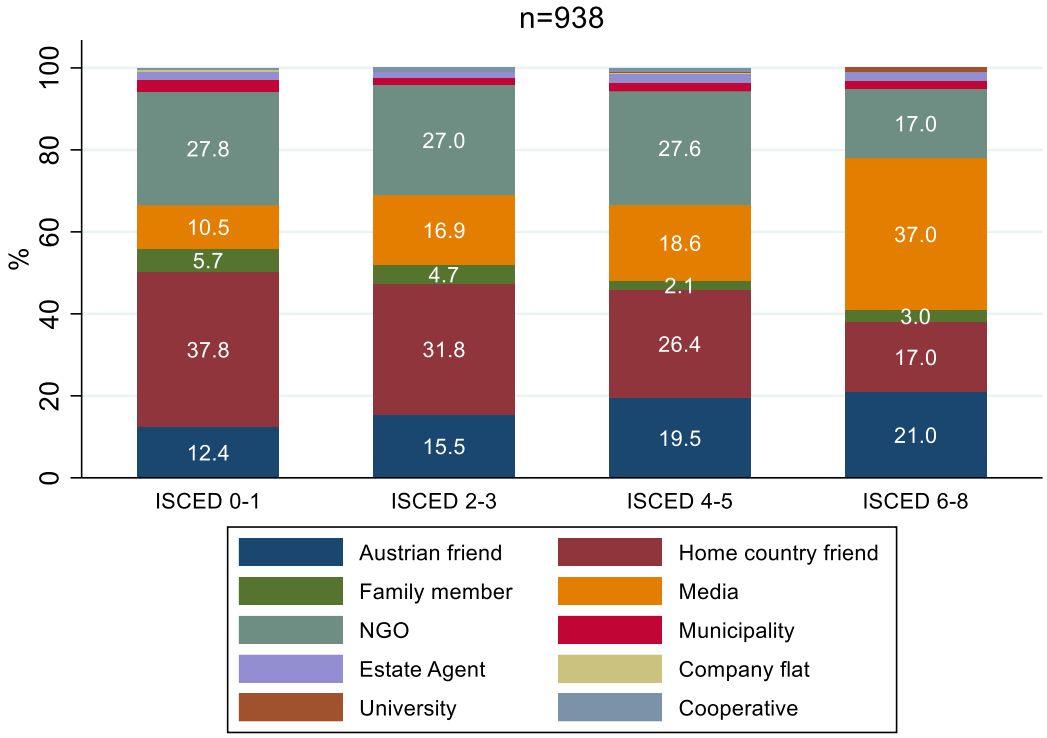
In order to understand the reasons behind these differences, it was necessary to analyse the various housing search options for additional attributes. Those with longer periods of residence (Graph 56) and higher levels of education (0) made greater use of **media** in searching for suitable accommodation. This may explain the findings for Chechens/Russians, in particular, but also for Iraqis. In addition, both male (Graph 58) and younger respondents often found their accommodation through Austrian friends and friends from their respective countries of origin.

Graph 56. Housing search by duration of stay



Source: FIMAS data

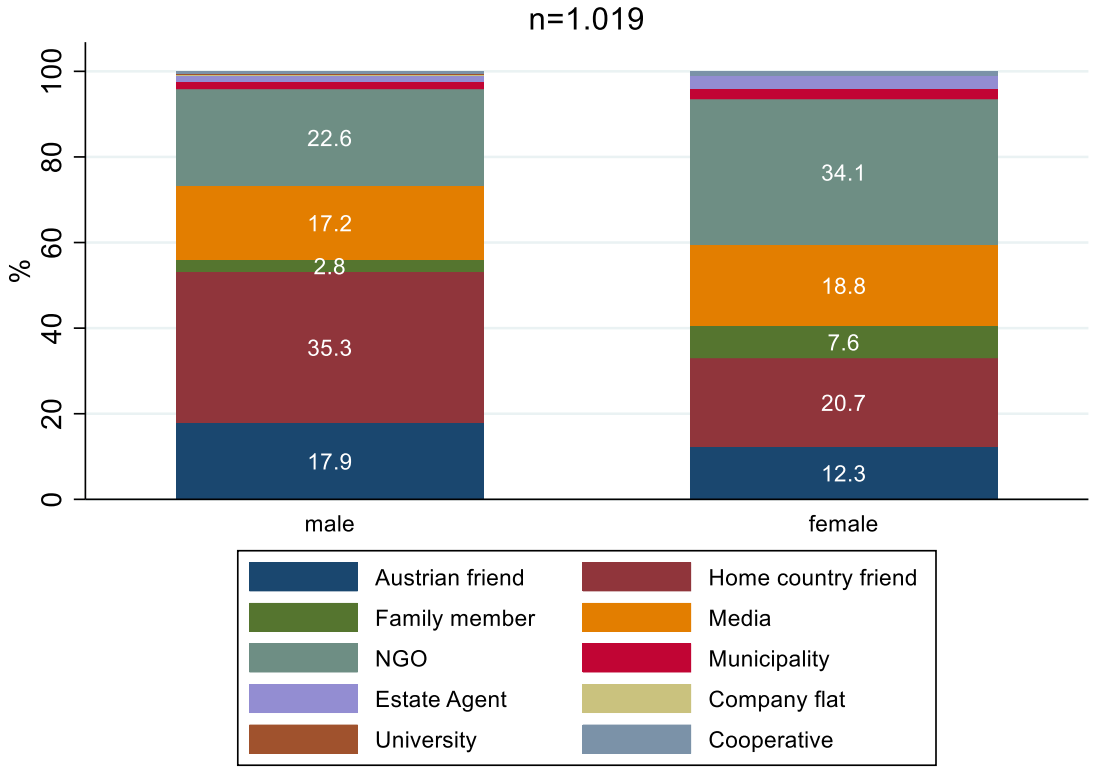
Graph 57. Housing search by education level prior to arriving in Austria



Source: FIMAS data

NGOs have a significant role when it comes to housing. Female respondents used services offered by NGOs more often (34%), whereas men tended more towards relying on contacts with Austrians and persons from their country of origin to find housing (53%), as shown in Graph 58.

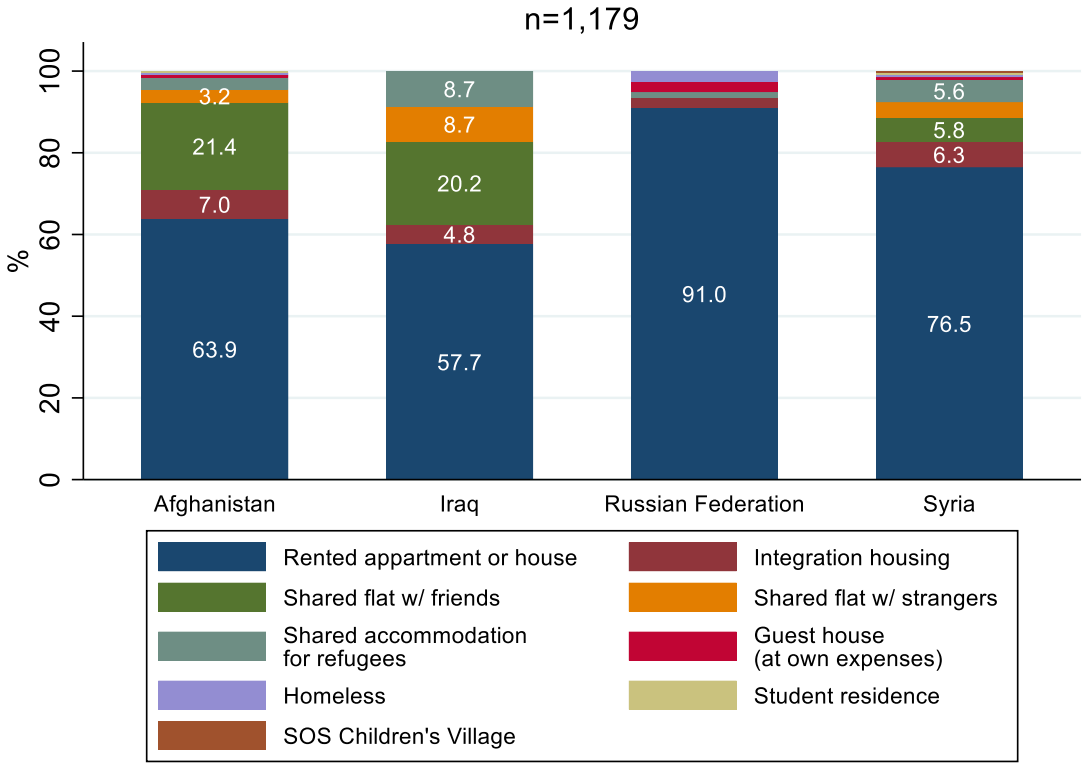
Graph 58. Housing search by gender



Source: FIMAS data

In addition to asking about how respondents found their accommodation, they were also asked about the kind of accommodation they were currently living in. The findings by analytic group are shown in 0. In contrast to the housing search, the differences by analytic group were more limited for this question. In all four country categories, most respondents lived alone or together with their families in a rented apartment or a rented house. This finding was most typical for Chechens/Russians, likely due to their longer durations of stay. Shared accommodation with friends came in as the second most frequent option, especially among Afghans (21%) and Iraqis (20%).

Graph 59. Housing situation by analytic group



Source: FIMAS data

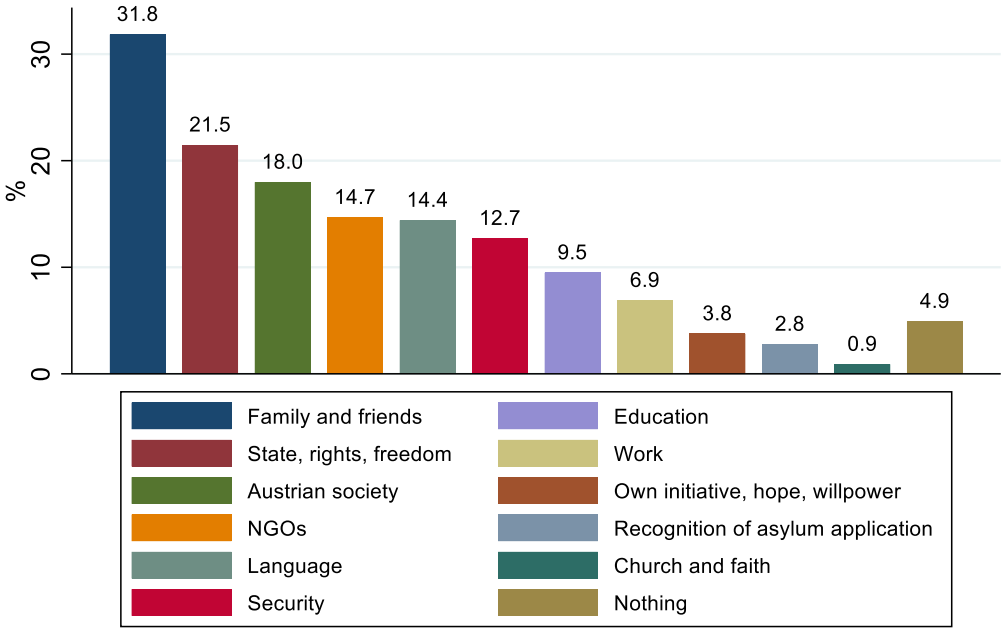
4.12 Experiences and aspirations

At the end of the interviews, respondents were asked open-ended questions about their experiences and aspirations regarding their new lives in Austria. They were asked about the issues that: 1) had helped in building their life in Austria; 2) had made it especially difficult to do so; and 3) what their most urgent wish was for the coming 12 months. Many of these aspects could then be quantified. The answers provided were often quite similar, despite the questions using an open-ended interviewing technique. Obviously there was high congruence between the needs, experiences and aspirations among the group of recognised refugees and beneficiaries of subsidiary protection. This also implies that targeted integration measures addressing these needs, experiences and aspirations could reach out successfully to a large number of persons and are promising in terms of a high rate of return on investments.

The first question asked respondents to state what **helped them most** in **building their lives in Austria**. The two most frequently cited aspects were **family and friends (32%)** and what can be summarised as **the Austrian State (21%)**, which included **rights and freedoms** (0). The third most important aspect mentioned was the welcoming culture and friendliness of the **Austrian society (18%)** – a phenomenon that can be explained, in particular, by the engagement of civil society in 2015 – followed by the support of NGOs (15%) and language acquisition (14%). Although language also represents a barrier, language acquisition was nevertheless perceived as a way to participate in life in Austria.

In addition, safety or security (13%), education/training (10%), work (7%) and intrinsic factors, such as initiative or hope (4%), were identified as particularly helpful aspects for building a life in Austria. **Being granted asylum as such was cited as helpful by only 3% of respondents**. Only a small proportion (5%) of respondents found nothing useful in establishing a life in Austria.

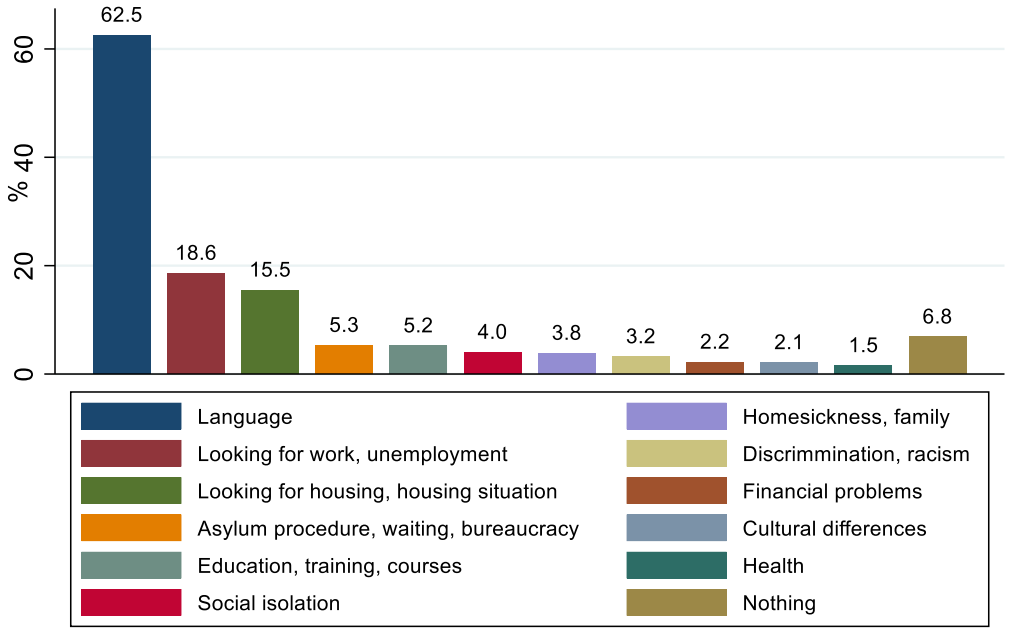
Graph 60. What has helped you most in building your life in Austria? (open question, multiple answers possible, n=1,120)



Source: FIMAS data

The second question referred to specific difficulties respondents had faced in building their lives in Austria (Graph 61). Here, **the most-cited obstacles were the lack of language skills (63%), followed by finding (suitable) work (19%) and searching for accommodation (16%)**. Together with difficulties in education and training (5%), these perceptions confirmed the importance of language and language acquisition for successful integration processes.

Graph 61. What things have made it especially difficult for you to build your life in Austria? (open question, multiple answers possible, n=1,097)



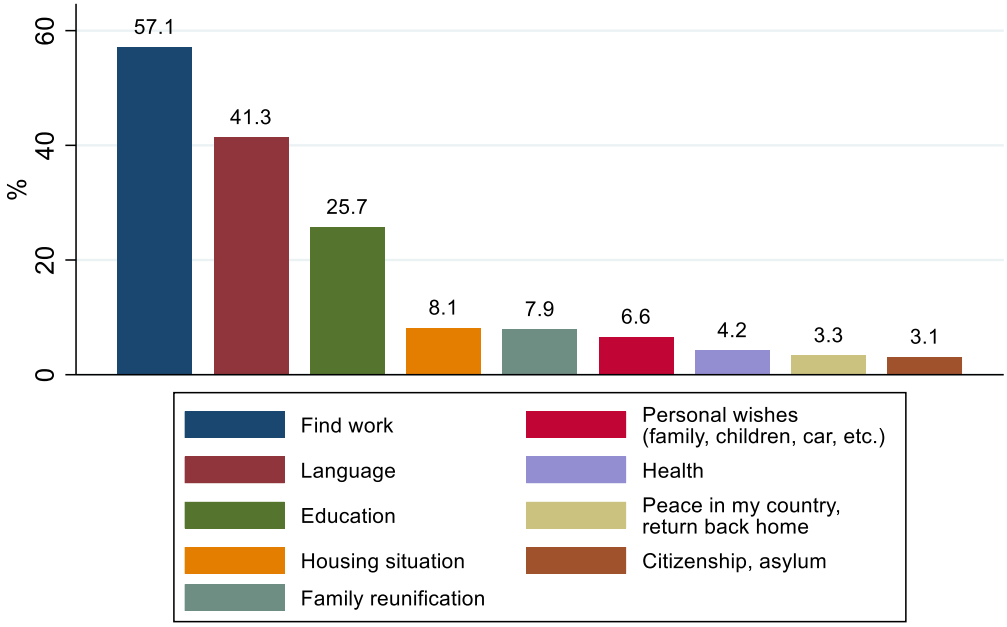
Source: FIMAS data

Other difficulties were stated as well, albeit rarely: the asylum process itself, long waiting times for decisions or bureaucratic hurdles were cited by 5% of respondents, and other responses included social isolation (4%), health problems (2%), and homesickness or missing family (4%). Discrimination and racism (3%) were mentioned even less often. Nevertheless, one should not conclude that discrimination is not relevant. A 2013 study conducted by the Institute for Advanced Studies (Hofer et al., 2013) demonstrated that applicants with non-Austrian profiles were discriminated against in job application processes (using fictitious identities of persons from Serbia, Turkey, China and Nigeria).

The last question concerning respondents' **wishes for the coming 12 months** revealed a picture that corresponded to the difficulties they were facing at the time of the interviews: **finding work (57%), learning the language (41%) and starting or completing training or education (26%)** were the topics most often quoted (Graph 62). These statements also reflected the high degree of motivation of respondents to build a life in their new environment. The wish to find work was particularly prevalent among those who were unemployed (74%), education and training especially prevalent among those under the age of 25 (52%), and language acquisition especially among non-working persons (50%).

Among other wishes mentioned, were those relating to improving their housing situation (8%), family reunification (8%), health (4%) and other personal wishes (7%). The wish for naturalisation (or being granted asylum for the small group of asylum seekers in the sample) was rarely mentioned (3%), as was the case regarding intentions to return home (3%).

Graph 62. What is your most urgent wish for the next 12 months? (open question, multiple answers possible, n=1,144)

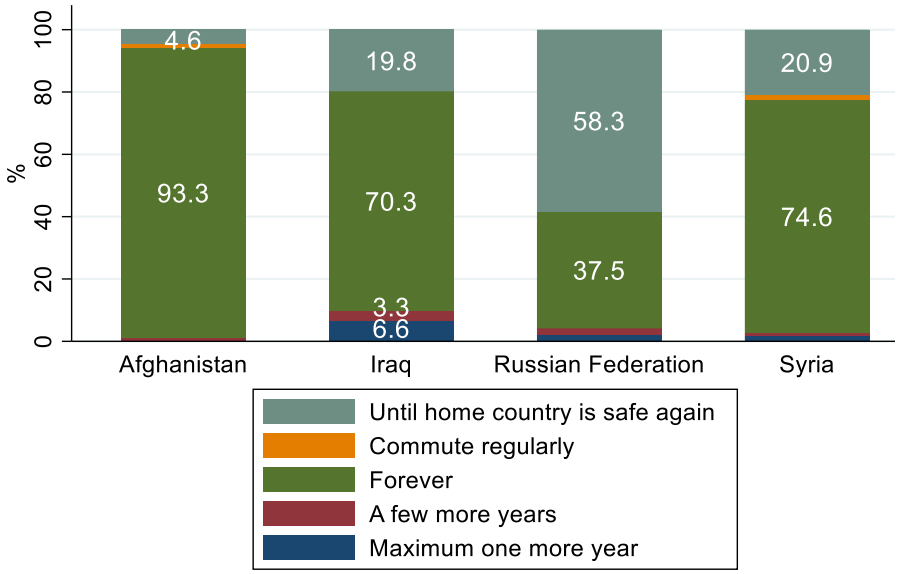


Source: FIMAS data

The question as to respondents' **wishes regarding how long they would like to stay in Austria** further confirmed these findings. Of those who provided an answer, **more than three quarters of respondents (78%) stated that they wished to stay in Austria for an indefinite period of time**, regardless of gender. Less than one fifth of respondents (17%) would like to stay at least until their country of origin is safe again. Of all the respondents, 22% said they "did not know", representing a very high proportion of missing values, which in this case is likely to be related to the uncertainty surrounding their country of origin or personal circumstances, making planning more difficult.

However, there were some group-specific differences regarding this question (Graph 63). Intentions to return were more common among Chechens/Russians (58%), as soon as the situation in their country of origin was safe again. Iraqis (20%) and Syrians (21%) reported having such an intention far less frequently. The intention to remain in Austria permanently was particularly high among Afghans, with 93% of respondents stating they wanted to stay in Austria forever.

Graph 63. Wish to stay in Austria by analytic group (n=888)



Source: FIMAS data

5 Multivariate analysis of the effect of education on labour market integration

In the multivariate model below, a logistic regression approach is used to examine whether persons with different educational backgrounds and work experience have a different likelihood of ever being gainfully employed in Austria. It also includes the factors described above which affect individual chances of labour market participation, such as health and care responsibilities. In doing so, it is assumed that the initial contact and experience with the formal labour market is crucial for further labour market integration in Austria.

The dependent variable therefore distinguishes between persons who have never been employed in Austria and those who had already been able to gain initial work experience in the country. Both groups also included persons who were not employed at the time of the interview: those who were registered as unemployed, those in education or training and those who were on parental leave or are otherwise not employed. A total of 33% of the individuals included in the sample had already held employment in Austria (see Graph 64, upper left pie share). Of these, 38% were currently employed, and a total of 7% had either an unofficial job, worked by helping out with a family business, were self-employed or worked as interns. The remaining 55% of respondents who had work experience in Austria were not employed at the time the survey was conducted.

Graph 64. Previous employment in Austria (dependent variable) by current employment status

Source: FIMAS data

Of the respondents, 67% had never worked in Austria (Graph 64, lower right pie share). Around 9% of those were still in school or university. 6% stated that they had care responsibilities or were homemakers, and 2% reported that they had health limitations. The vast majority of respondents who had never worked in Austria were unemployed (63%).

In the logistic regression, persons who had previously worked in Austria constituted the reference category. With regard to the effects of education and employment experience on labour market integration, the following hypotheses were examined:

1) It was assumed that qualifications acquired after immigrating to Austria had a higher value for labour market integration than qualifications acquired abroad (cf. e.g. Bock-Schappelwein/Huber, 2016; Koppenberg, 2016, p. 22).
2) Because the low marketability of foreign educational qualifications is, among others reasons, a result of limited comparability between foreign and Austrian educational qualifications, it was further assumed that the recognition of foreign educational qualifications had a positive effect on refugees' labour market integration.
3) At the same time, it was assumed that due to the generally low chances for labour market integration among persons with low formal levels of qualification, recognised refugees and beneficiaries of subsidiary protection who had acquired higher qualifications in their countries of origin also had better chances of accessing the labour market in Austria (Prettenthaler et al., 2017, p. 23).
4) It was also assumed that employment experience abroad had a positive effect on the employment integration in Austria. In this context, the effect of industry-specific knowledge was analysed.

This analysis is based on those respondents for whom information on their educational background and employment situation in Austria was available. Overall, this was the case for 76% of all respondents.

In addition to the influence of foreign educational qualifications and their recognition in Austria, participation in education and training in Austria, language skills and work experience abroad, the following models also take into account additional factors that might have impacted the employment situation in Austria: the place of residence (federal province), health restrictions and possible care responsibilities. Furthermore, the duration of stay, which impacts opportunities for attending or completing education or training in Austria, as well as language skills were integrated into the model.

Education in the country of origin was measured using the aggregated ISCED categories. Here, the analysis differentiated between low level qualifications (ISCED 0-2), intermediate level qualifications (ISCED 3-4) and higher level qualifications (ISCED 5-8). In addition, information regarding the recognition of qualifications was taken into account in the model. Whether a foreign qualification had already been recognised was recorded as well. Information on the country of education was included in the model in order to properly measure educational background prior to arrival in Austria. If there was no information on the primary country of education, or if the person indicated that they were unable to read or write, then the country of birth or citizenship information was used. Those who stated that Austria was their primary country of education were categorised under their country of origin (according to the information provided regarding citizenship). Whether respondents already had German language skills prior to entering Austria was also recorded. Because only very few had basic knowledge of German upon arrival, this variable was recorded dichotomously.

In addition to information on formal qualifications acquired abroad, information on professional experience and occupational fields abroad were also included in the model. A distinction was made between the categories of "employment experience", "employment experience relevant to area of study", "was in education/training" and "homemaker or unemployed". The latter were grouped together as only 17 persons stated that they were unemployed. Information on whether respondents were self-employed abroad was also included. Occupational groups were coded dichotomously. The occupational group "military personnel" was not included in the model because, according to the information provided by respondents, only a small number had served in the military in their country of origin. Because the questionnaire could cover two different occupations, respondents could be assigned to more than one occupational group as well.

Language courses and acquired language skills, course participation and formal education were counted under "education in Austria". A distinction was made between secondary school qualifications, ISCED category 3 (university entrance qualification, apprenticeship or vocational school) qualifications, and further training courses and academic degrees acquired in Austria. Computer courses and similar qualifications were not taken into account due to the small number of cases reported, as well as the fact

that they could not be grouped with the other categories. Furthermore, it was analysed whether the effects of integration measures designed to offer orientation and identify skills to promote labour market integration can be observed.

German language skills were classified under three categories: basic or no language proficiency (corresponds to A), independent user (corresponds to B) and proficient user (corresponds to C), taking into account communication skills in German (speaking and understanding). English language skills were assessed exclusively by respondents themselves. Communication skills (speaking and understanding) were included in the model, where the ability to speak English, in particular, was used as a reference point for assessing language skills. This implied that the English communication skills of those who believed they had an advanced understanding of English, but tended to assess their ability to speak English as average, were classified as average. Advanced knowledge and first language knowledge were in the category of "proficient use of language".

In addition, the following control variables were included in the analysis: duration of stay in Austria (in years), the federal state in which the person stayed in Austria, gender, age, information on care responsibilities, as well as any restrictions on activity that might have affected employment status. The interaction between gender and care responsibilities was also taken into account in the model.

A total of three models were analysed and compared with one another to illustrate the influence of formal education and work experience abroad or in Austria on labour market integration in Austria. The first model only included the control variables describing the situation in Austria. In the second model, information on education and training and employment experience prior to moving to Austria was included as well. Finally, the third model also took into account the qualifications obtained in Austria and language level at the time of the survey, as well as course attendance.

This approach made it possible to assess the concrete contribution of educational qualifications and employment experience acquired in Austria, as well as prior to moving to Austria, to the overall explanatory power of the model. In each model, the primary interpretation focused on the direction of influence of different skills and educational backgrounds on labour market participation in Austria (positive or negative influence).

The odds ratio (Exp(B)) of the independent variable indicates the change in relative probability of having been employed in Austria when the independent variable increases by one unit and all other variables in the model remained constant. If the odds ratio is one, then there is no effect of the independent variable on the employment chances. If the odds ratio is greater than one, the chance of having been employed in Austria is higher due to the influence of the independent variable. If the odds ratio is less than one, the chance of having been employed in Austria is lower due to the influence of the independent variable. The variables with the strongest effects are listed.

Nagelkerke's R-square[15] was 28% in the first model, which included only explanatory variables describing the current status of respondents in Austria and socio-economic data that potentially impacted labour market integration. Integrating information on foreign qualifications and employment experience abroad in the second model provided an explanation for another 10% of the differences between the two groups. Integrating information on qualifications and courses in Austria in the third model explained another 5% of the differences (see table 12).

Table 12. Explanatory value of the three models

Employment in Austria	-2 log likelihood	Nagelkerke's R-square
Model 1: Situation in Austria	1,174	0.28
Model 2: + Foreign qualifications	839	0.38
Model 3: + Austrian qualifications	627	0.43

Source: FIMAS data

[15] A pseudo-coefficient of determination for logistic regression models.

5.1 Findings: Factors explaining labour force integration of recognised refugees and beneficiaries of subsidiary protection in Austria

Model 1 shows that the duration of stay, in particular, had a positive effect on the likelihood of having been employed in Austria (Table 13). Thus, according to the first model, the relative chance of employment increased by a factor of 1.4 for every year of residence.

In addition, the federal province in which respondents lived influenced their chances of employment as well. After accounting for the other factors included in the model, those who did not live in metropolitan areas (Graz or Vienna) had a higher chance of being employed in Austria. In Tyrol, the relative chance compared to Vienna was almost three times higher (Tyrol Exp(B): 2.66).

Table 13. Model 1: Employment in Austria explained by situation in Austria (n=1,133)

Employment in Austria	logit	Exp(B)	p
Health problems	-0.1	0.91	0.51
Vienna (=ref)			0.00
Upper Austria	0.2	1.23	0.33
Salzburg	0.2	1.24	0.35
Styria (Graz)	-0.1	0.92	0.74
Tyrol	1.0	2.66	0.00
Female	-0.4	0.67	0.04
Care responsibilities	-1.0	0.38	0.37
Female*Care responsibilities	-0.5	0.63	0.70
Length of stay (years)	0.3	1.40	0.00
Age	0.0	0.98	0.01
Currently in education/training	-0.9	0.41	0.00
Constant	-1.4	0.25	0.00

Source: FIMAS data

On the other hand, care responsibilities (Exp(B): 0.38) had a negative effect on employment integration, as had undertaking education/training. The latter, however, potentially promised future labour market integration. Since it was women who were most likely to take on care responsibilities, and almost all the women in the sample had children, the relative chances of women achieving employment with and without care-related situations were similarly low.

The **second model** included additional information on the highest level of qualifications obtained prior to moving to Austria, as well as information on the country of education, employment experience, and the employment field. It also took into account whether or not the work abroad was appropriate to the training or education respondents had obtained, and whether their foreign qualifications were recognised in Austria.

Table 14. Model 2: Employment in Austria explained by including information on education and experience prior to arrival (n=905)

Employment in Austria	logit	Exp(B)	p
Care responsibilities	-0.56	0.57	0.62
Health problems	-0.08	0.92	0.66
Vienna (=ref)			0.08
Upper Austria	0.26	1.32	0.31
Salzburg	0.37	1.45	0.20
Styria	-0.22	0.80	0.50
Tyrol	0.59	1.80	0.04
Female	-0.21	0.81	0.44
Female*Care responsibilities	-0.32	0.72	0.80
Length of stay (years)	0.38	1.47	0
Age	-0.02	0.98	0.03
Currently in education/training	-0.89	0.41	0.02
Employment experience (=ref)			0.18
Employment experience relating to education/training	0.39	1.48	0.11
In education/training abroad	-0.68	0.51	0.19
Homemaker (female) or unemployed abroad	-0.33	0.72	0.53
Managers	-0.15	0.87	0.76
Professionals	0.02	1.02	0.95
Technicians and associate professionals	-0.14	0.87	0.67
Clerical support workers	0.17	1.18	0.71
Service and sales workers	-0.14	0.87	0.64
Agricultural, forestry and fisheries workers	-0.61	0.54	0.38
Craft and related trades workers	0.12	1.13	0.68
Plant and machine operators and assemblers	0.04	1.04	0.92
Elementary (low-skilled) workers	0.57	1.78	0.11
Self-employed abroad	0.46	1.59	0.05
Country of education Afghanistan (=ref)			0
Country of education Iraq	-1.60	0.20	0
Country of education Russia	-1.56	0.21	0.01
Country of education Syria	-1.35	0.26	0
Country of education other	-0.55	0.58	0.13
German proficiency upon arrival	0.41	1.51	0.23
Recognition of educational qualifications	0.93	2.53	0.01
low-level qualification (=ref)			0.04
Intermediate qualification	0.27	1.31	0.22
Higher-level qualification	-0.39	0.68	0.23
Constant	-0.66	0.52	0.20

Source: FIMAS data

Compared to the first model, the advantages enjoyed by recognised refugees and beneficiaries of subsidiary protection living in Tyrol compared to those in Vienna were reduced by including information on education and training and employment experience acquired abroad (0). However, the basic structure of the correlation between the dependent variable and the variables already included in the first model remains unchanged.

This model confirmed the theory that seeking asylum leads to a devaluation of degrees acquired abroad. Nevertheless, the effect education has on labour market integration in Austria is not negative. This means that, taking into account all other variables included in the model, the persons who had acquired intermediate qualifications abroad had a relatively higher chance of being employed in Austria (Exp(B): 1.3) compared to those with low qualifications. Those with higher-level, potentially more specialised qualifications, however, had a relatively lower chance of employment in Austria (Exp(B): 0.7) compared to those with lower qualifications. Recognition of foreign qualifications (Exp.(B): 2.5) had a particularly positive effect on employment in Austria. German language skills upon arrival also had a positive effect on obtaining employment.

The fact that skills and experience acquired in Austria were difficult to exploit, and that it was those with higher qualifications that faced particularly low chances on the Austrian labour market, was also confirmed by the fact that experience gained prior to arriving in Austria had highly varied effects on labour market integration. In general, the relative chance of having already been employed in Austria was higher among those who had already been employed abroad. Those who had already worked abroad in fields relating to their education or training, and who therefore had less fragmented employment histories, had a relatively higher chance of employment in Austria, in particular when compared to others who had previously been otherwise employed (Exp(B): 1.5). Experience in self-employment also had a positive effect on employment in Austria (Exp(B): 1.6). At the same time, it became clear that persons who had worked in low-skilled occupations prior to entering Austria – including labourers – had a relatively higher chance of gainful employment in Austria in relation to those who had worked in other sectors (Exp.(B): 1.8). By comparison, the relative chance of employment in Austria was low for former technicians and executives.

Interestingly, those who had listed Afghanistan as their primary country of education had a relatively higher chance of labour market integration than other members of the target groups (Syrian, Iraqi and Chechen/Russian nationals). Obviously, it was persons with low level qualifications who encountered lower barriers for entering the labour market in Austria compared to those with higher qualifications.

The **third model** also included information on respondents' training and education situation in Austria (Table 15). The main purpose of this model was to test whether education or training in Austria can be better utilised than foreign qualifications. In addition to qualifications acquired in Austria, this model included information on German and English language skills acquired since arriving in Austria. The odds ratios show that qualifications obtained in Austria, as well as qualifications already obtained prior to arriving in Austria above the compulsory education level, had a highly positive influence on labour market integration.

Table 15. Model 3: Employment in Austria explained by including qualifications in Austria (n=720)

Employment in Austria	logit	Exp(B)	p
Length of stay (years)	0.40	1.49	0.00
Vienna (=ref)			0.15
Upper Austria	0.32	1.38	0.35
Salzburg	0.45	1.57	0.21
Styria	0.02	1.02	0.96
Tyrol	0.84	2.32	0.02
Female	-0.30	0.74	0.36
Female*Care responsibilities	-0.11	0.90	0.94
Care	-0.09	0.92	0.95
Health problems	-0.31	0.73	0.18
Currently in education/training	-0.68	0.51	0.10
Age	-0.02	0.98	0.21
Employment experience (=ref)			0.23
Employment experience relating to education/training	0.48	1.61	0.11
In education/training abroad	-0.57	0.57	0.32
Homemaker (female) or unemployed abroad	-0.44	0.65	0.53
Managers	-0.08	0.92	0.89
Professionals	-0.30	0.74	0.49
Technicians and associate professionals	-0.27	0.76	0.48
Clerical support workers	0.27	1.31	0.59
Service and sales workers	-0.08	0.92	0.81
Agriculture, forestry and fisheries workers	0.75	2.11	0.31
Craft and related trades workers	0.44	1.55	0.21
Plant and machinery operators and assemblers	0.41	1.51	0.32
Elementary (low-skilled) workers	1.15	3.14	0.01
Self-employed abroad	0.36	1.43	0.22
Country of education Afghanistan (=ref)			0.00
Country of education Iraq	-1.95	0.14	0.00
Country of education Russia	-2.53	0.08	0.00
Country of education Syria	-1.66	0.19	0.00
Country of education other	-0.84	0.43	0.08
German proficiency upon arrival	0.45	1.57	0.32
Recognition of educational qualifications	0.82	2.26	0.03
low-level qualification (=ref)			0.28
Intermediate qualification	0.07	1.07	0.79
Higher-level qualification	-0.43	0.65	0.24
No qualification in Austria (=ref)			0.04
Secondary school qualification	-0.66	0.52	0.27
Education level ISCED 3	1.83	6.22	0.04
academic qualifications and post-graduate qualifications	1.60	4.95	0.07
Participation in values course	-0.19	0.83	0.48
Participation in competence check	0.05	1.05	0.87
German course	0.08	1.08	0.76
Basic/no English (=ref)			0.60
Independent use of English language	0.27	1.31	0.31
Proficient use of English language	0.14	1.15	0.67
Basic/no German (=ref)			0.18
Independent use of German language	0.41	1.51	0.07
Proficient use of German language	0.28	1.32	0.70
Constant	-1.01	0.37	0.15

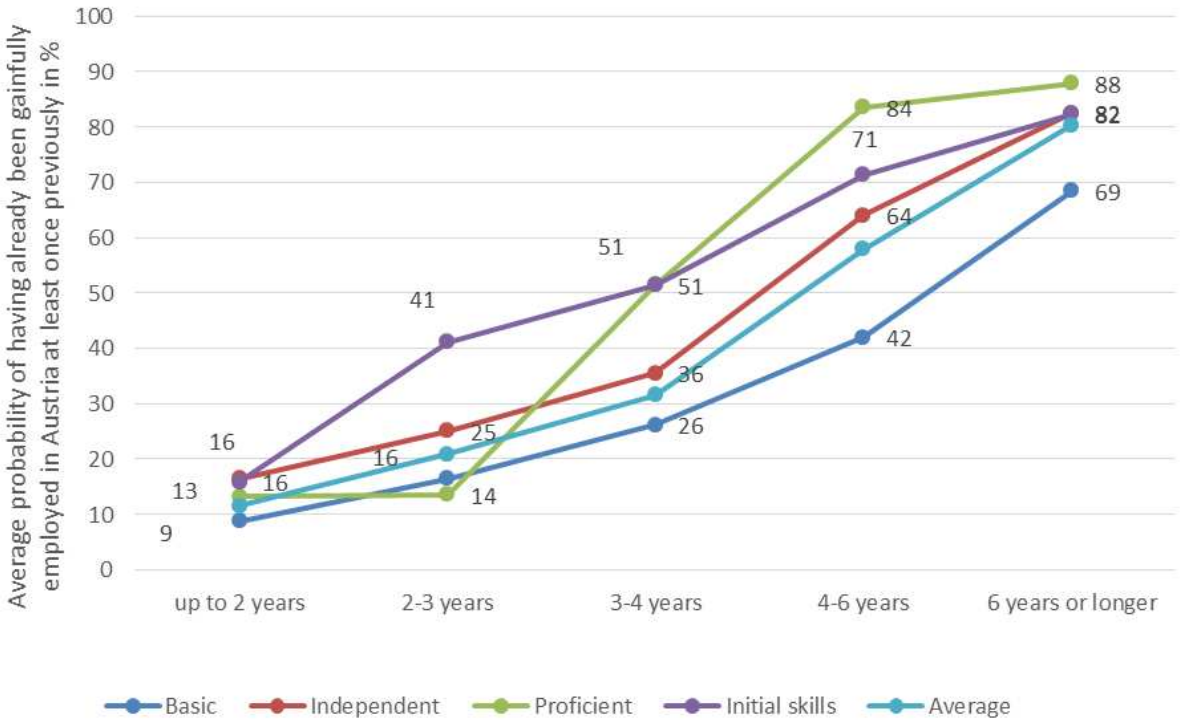
Source: FIMAS data

The relative chance of employment in Austria was six times higher for persons with an Austrian qualification equivalent to ISCED category 3 (university entrance qualification, apprenticeship or vocational school) (Exp.(B): 6.2) than it was for respondents who had not obtained any qualification in Austria. The same effect was identified for post-graduate and university degrees (Exp(B): 5.0). Participation in competence checks and values courses seems to have only limited impact on labour market integration. German language courses impact labour market integration by improving language skills. Because language skills are also crucial for educational participation, they are an important resource for successful integration at various levels. The negative influence of secondary school qualification on employment integration in Austria (Exp(B): 0.5) can be explained by the fact that this opens up opportunities for further education, which in the short term reduced labour market participation, but in the mid- to long term expands access to the labour market.

For persons living in Austria for up to two years, the average probability of having already worked in Austria was 11% (Graph 65), when controlling for all other influencing factors in the model. After living in Austria for six or more years, the average probability of having already worked in the country reached 80%, thereby reaching the level of the general population.

It can also be concluded that knowledge of **at least C-level German** is necessary for quick labour market participation. Among those who already had advanced German language skills prior to moving to Austria, the average probability of labour market participation was already 41% after only two to three years in Austria.

Graph 65. Average probability of having been employed in Austria (model 3) by language proficiency and duration of stay (n=721)

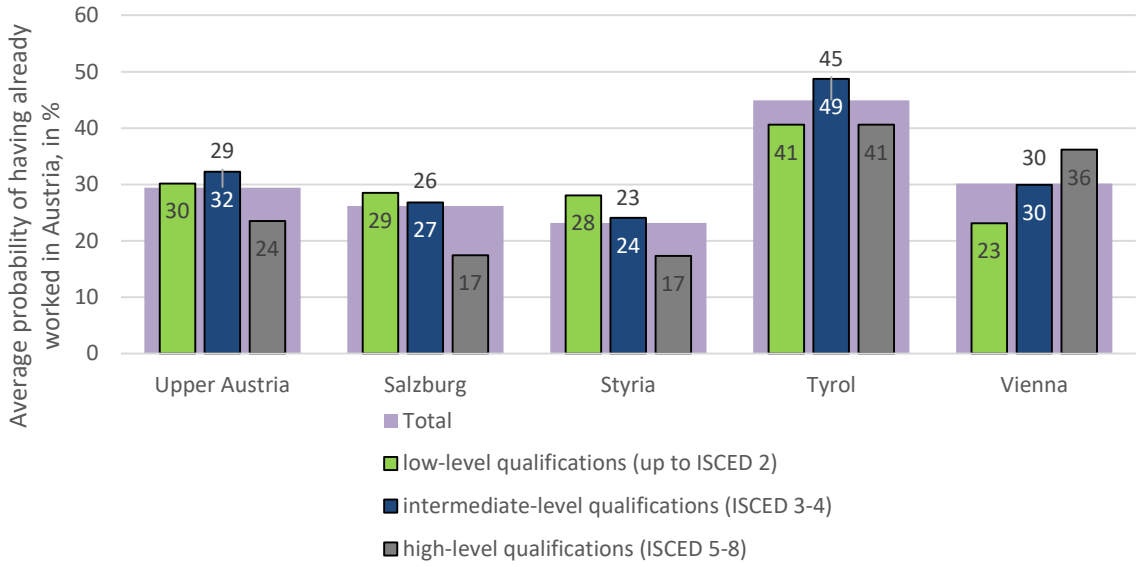


Source: FIMAS data

It is striking that, taking into account the effect of qualifications acquired in Austria, the negative effect of employment experience in high-skilled sectors was even more pronounced. This confirmed that respondents who had worked abroad in high-skilled sectors had a relatively lower chance of employment in Austria. Above all, the degree of labour market participation of those who were employed in academic professions abroad can be explained by their educational aspirations in Austria. However, taking into account qualifications completed in Austria, they still had a relatively lower chance of labour market participation, whereas the relative chance of gaining a foothold in the labour market in Austria for former low-skilled workers was three times higher than for all other groups.

Another reason for the comparatively lower chances of labour market participation for those with higher-level qualifications is the economic structure specific to each federal province, determining the types of opportunities for members of the target groups of the study. Except for Vienna, the average probability of labour market participation for those with intermediate or low level qualifications was higher than for those with higher level qualifications (Graph 66). This observation suggests differentiated opportunities for less qualified persons in each federal province's specific labour market. Placement policy in the federal states could also have an impact.

Graph 66. Average probability of having already worked in Austria, by qualification earned abroad and by federal province, model 3 (n=721)



Source: FIMAS data

In summary, it can be said that all four hypotheses formulated at the beginning of this section were confirmed in the respective models. Recognised refugees and beneficiaries of subsidiary protection who were at risk of comparatively low integration in the labour market were those who had no employment experience; who lived in urban centres (including Vienna); who were low-skilled, skilled or former executives who were unable to market their skills in Austria; and women.

Additional educational and vocational training measures as well as faster recognition of skills and qualifications are therefore particularly important for achieving greater participation in the labour market of recognised refugees and beneficiaries of subsidiary protection. For those who are highly skilled, in particular, this could open up new opportunities and allow them to put their skills to good use in line with their experience.

6 Summary

One of the main findings of the FIMAS study is that refugees in Austria face a **devaluation of educational qualifications** (also see Scheiber, 2007, for Upper Austria). When employed, the recognised refugees and beneficiaries of subsidiary protection in the sample worked in industries and occupations that did not correspond to their qualifications before relocating to Austria. Thus, it was those with higher qualifications (ISCED 5-8) who had particular difficulties in entering the Austrian labour market and finding a job that related to their qualifications. Only when **foreign degrees were recognised**, although only a very small number of FIMAS respondents had tried to get their qualifications recognised, or when respondents **had obtained Austrian degrees,** did the chances for successful labour market integration start to increase.

The FIMAS study also confirmed that it is not possible to view refugees and beneficiaries of subsidiary protection as a homogenous group in terms of their educational, professional and language backgrounds, and the resulting chances of integration in respective labour markets. The study found **major differences in educational qualifications across the different analytic groups**. The number of persons who were illiterate was highest among Afghans, as was the number of respondents considered low-skilled (two thirds). By contrast, almost half of Chechen/Russian and Iraqi respondents were considered highly-skilled. Syrians had the highest share of persons considered medium-skilled, compared to the three other groups. Female respondents across all nationalities had lower levels of education on the one hand, whilst being more likely to hold university degrees on the other.

Employment experience prior to relocating to Austria, particularly with regard to training, seemed to contribute positively to labour market integration in Austria. Notwithstanding this, it was primarily former agricultural and elementary workers who had a higher chance of finding gainful employment more quickly. Even though the Austrian labour market exhibits a shortage of skilled workers, specialists among the respondents with sector-specific training in fields where vacancies exist still had very little chance of finding a job. Transferring personal skills, overcoming the language barrier and aligning previously obtained qualifications with the formal and informal requirements of the Austrian labour market seem to constitute large obstacles for individuals. Consequently, the **mismatch** between qualifications and employment was high across the whole FIMAS sample – more than three quarters of respondents worked in an industry not (or only partially) related to their educational or training background.

This was also related to the high concentration of employed persons among the target group who worked in elementary occupations, particularly in the federal provinces. Only in the capital, Vienna, was there a wider distribution of jobs across different occupational groups, especially social work and technical occupations. The strong concentration in low-skilled jobs, the high rates of part-time employment, temporary employment and the all-in-all low salaries of those respondents who had found employment add up to a situation of **occupational segregation** in an environment of labour market segmentation. Respondents found employment in what is known as the **4D jobs (dull, demeaning, dirty and dangerous)**, i.e. low-status jobs that non-migrants typically avoid (such as kitchen assistants, cleaners, warehouse or security workers). Furthermore, recognised refugees and beneficiaries of subsidiary protection were particularly likely to work in **atypical employment relationships**, and were much more likely to do so than other migrants or Austrian citizens, especially when they took up work for the first time.

When employed, **female respondents** and **persons with higher qualifications** found jobs that better corresponded to their educational backgrounds. At the same time, women entered the labour market much later than men, and had an overall lower rate of employment. The clear differences in employment status can be attributed to women being more likely to assume responsibilities in housework and childcare.

Applying multivariate models showed that participation in **competence checks and values courses** had no immediate impact on labour market integration. The positive effects of these measures, which were recent at the time of the FIMAS study, might become apparent only in the medium term. **German language courses** indirectly impacted labour market integration by improving language skills.

Despite **Afghans often having lower levels of education**, it did not take them longer to integrate in the labour market and their statistical chances of finding employment were not any worse than for the other groups in the sample. Thus, they tended to work in elementary occupations, and were rarely employed in highly-paid jobs. In terms of overall labour market participation, the comparative disadvantage of lower educational qualifications and less professional experience was outweighed by the greater readiness to accept unattractive jobs.

Thus far, little was known about the labour market situation of the Chechen community in Austria. The findings of the FIMAS study revealed a comparatively **long path to the labour market for this particular group.** Although higher participation would be expected given the longer duration of residence, the time it took for Chechen respondents to find their first job was much longer than for the other groups – eight years on average. Like for the other groups in the sample, the comparatively poor labour market performance could be the result of educational and skills backgrounds. However, the high rate of university graduates (52%) in the sample was probably due to distortion in the sampling method. Although previous results of an asylum screening by the federal province of Salzburg showed results along this trend (Federal Province of Salzburg, 2017), the 2015 ICMPD survey on asylum seekers (EQUAS) had identified only a rate of 15% of university graduates (ICMPD, 2015).[16]

All told, **Iraqis** were not able to bring their **high levels of education** (40% are university graduates) to bear in terms of higher incomes or better qualified occupations, and were often still looking for jobs at the time of the survey. With 51% of Iraqi respondents reporting they completed post-secondary level education, they were the group with the highest proportion of highly educated respondents. **Syrians** represented the group which had been in Austria for the shortest amount of time and, as a consequence, were often still looking for jobs, undergoing training or education or learning German. However, among all respondents, they stood out as the group that found **their first jobs the fastest.**

With due methodological caution, the integration processes of past cohorts of recognised refugees and beneficiaries of subsidiary protection suggest that success in labour market integration is to be achieved mainly through language acquisition, education and training, recognition of qualifications, and the existence of social networks. If these factors come into play in a similar manner for the more recent cohorts, one can expect a duration of **six to nine years for the labour force participation rate of recent cohorts to approximate to that of the total population**, and **nine to ten years for the employment rate to reach 55%**. This scenario would also be expected based on OECD (2015) findings. Taking into account the systematic under-coverage of employed persons (about 15%) in the sample, the employment rate could even reach the level of the overall total population during this period. These processes could be accelerated even further, once the effects of the additional resources provided recently to the variety of existing integration measures, as well as the newly initiated measures and, in particular, the increase in places for German language courses, are seen.

Asked about past experience in Austria and their **hopes for the future, respondents gave highest importance to work and education**. Family and friends were considered most helpful in terms of building a life in Austria, but respondents also stressed the positive role of the Austrian state and its institutions, rights and freedoms, the welcoming culture and friendliness of Austrian society, and support from NGOs in helping them get settled. **Lacking German language proficiency was seen as the greatest barrier** to settling in and integrating, with six out of ten respondents mentioning this. Unemployment and the housing situation took second and third place, respectively. In terms of their hopes for the future, respondents most often mentioned **finding a job**, followed by **learning the**

[16] Despite small sub-samples in both studies, this very large deviation points to systematic differences.

language, with pursuing **training or education** taking third place. Intentions about returning to refugees' home countries varied by group. Chechens were the only group where more than half of respondents noted a wish to return once their country of origin was safe again. All respondents emphasised **education and participating in the labour market as their personal main priorities**, a priority which was partly impeded by the need to acquire language skills and the right educational qualifications; and for women additionally by childcare and household responsibilities.

In terms of education, vocational training and professional backgrounds, the majority of respondents assessed themselves as medium- and high-skilled, although there were stark **group-specific differences**. Thus, the debate must duly reflect group-specific differences based on country of origin as well as gender, education and training/profession and tailor integration measures accordingly.

Methodologically speaking, it has to be stressed that the results of the FIMAS study are largely descriptive in nature.

Whilst the survey covered the most important nationalities among the overall population of recognised refugees and beneficiaries of subsidiary protection, males and females, the most recent arrival cohorts, five federal provinces and larger and smaller cities, it is not possible to draw conclusions for the total population of all refugees in Austria on the basis of this sample. This is due to the fact that four federal provinces, smaller nationalities and earlier cohorts were not included in the sample. Also, the findings presented here are based on **unweighted data**, so results for the overall sample reflect the structure of the sample, for example by citizenship groups.

Taking into account these inherent limitations, the FIMAS project still made an important contribution to the Austrian and international debate on labour market integration of recognised refugees and beneficiaries of subsidiary protection, including by improving the **basis for model calculations** regarding the economic effects of related processes (Holler und Schuster, 2016; EcoAustria, 2017). The survey of recognised refugees and beneficiaries of subsidiary protection in Austria is currently continued in the form of a **panel survey**. Such **longitudinal analysis** will provide the basis for modelling processes on an individual level and for developing causal analyses. Also, the provision of long-term data on integration fills a highly topical research gap (UNHCR, 2015, p. 47) that could be bridged by the FIMAS surveys, at least for the Austrian case.

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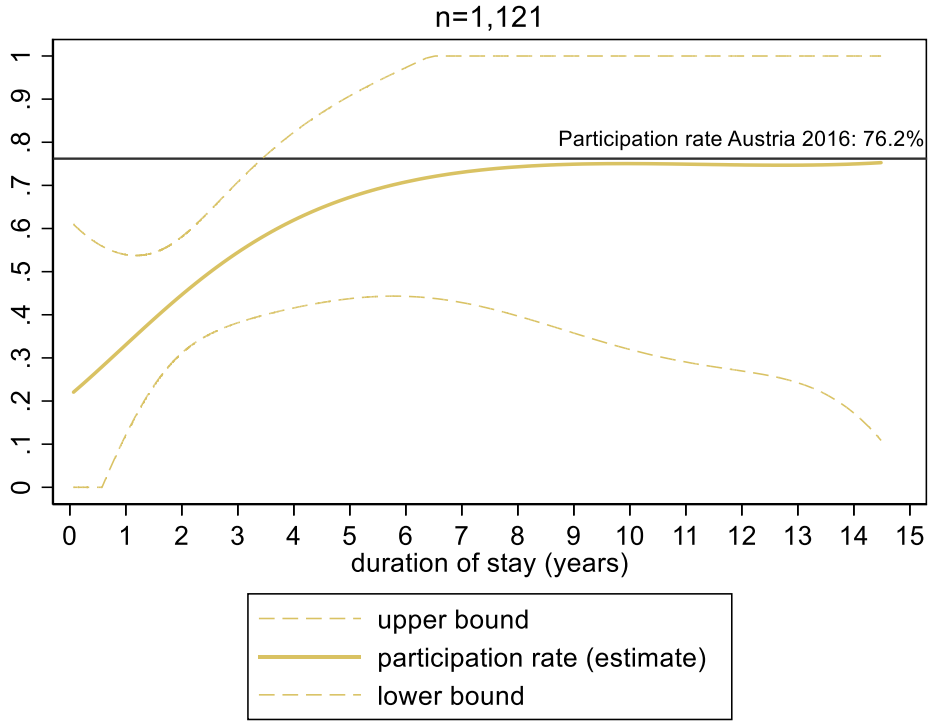
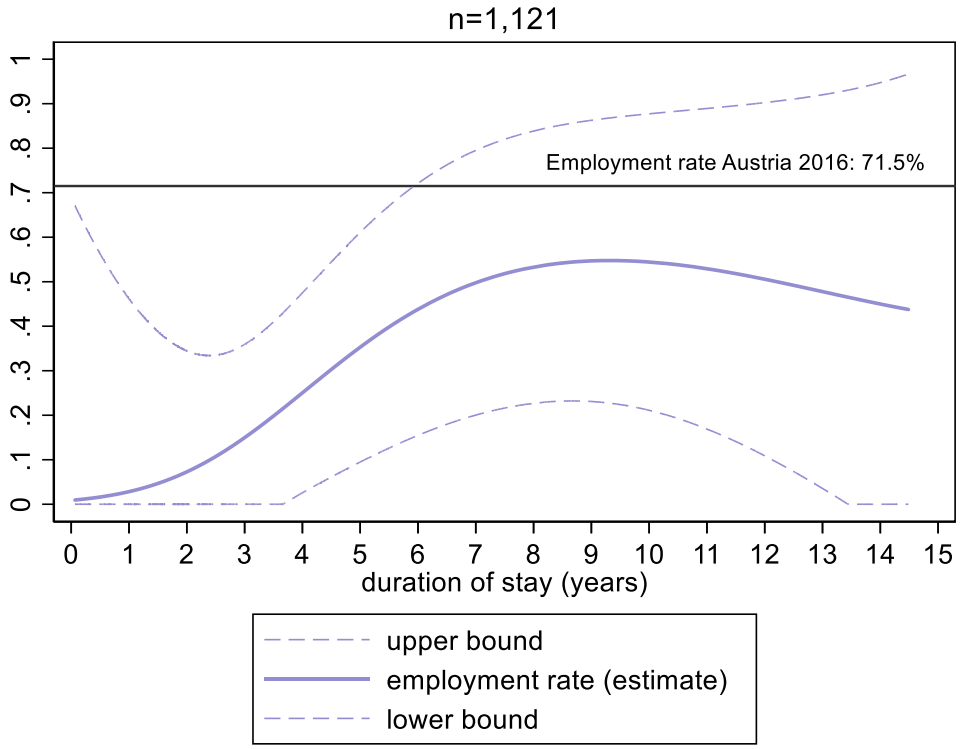
8 Appendix

Graph 67. Estimated trajectory of the labour force participation rate by duration of stay in Austria (age 18+)



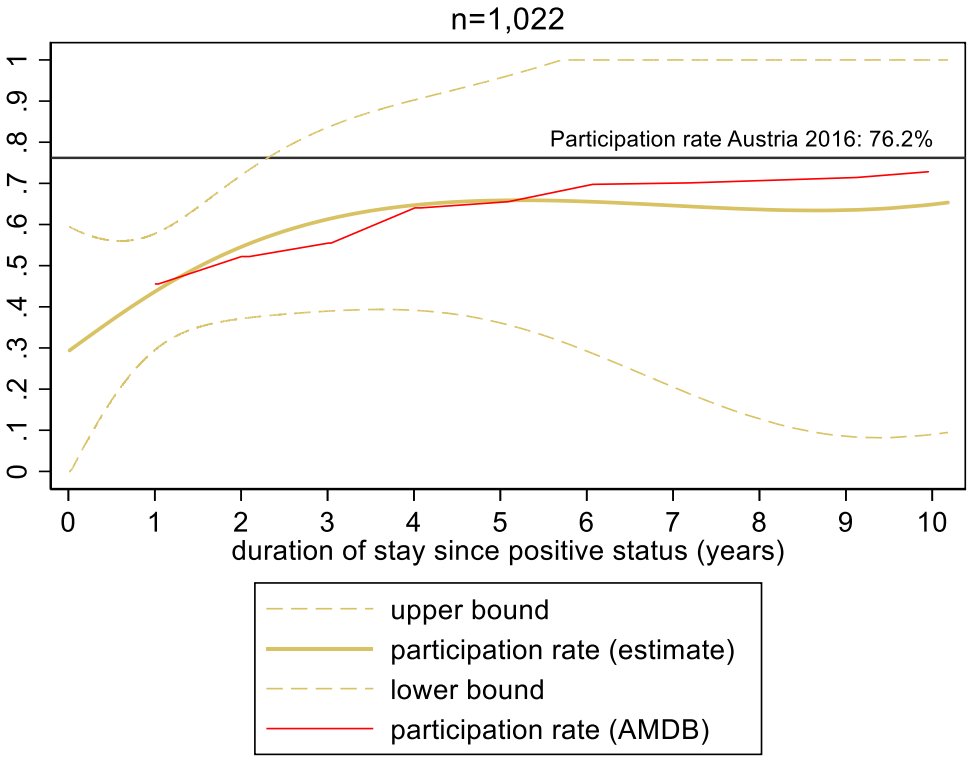
Logit model, including confidence intervals
Source: FIMAS data, excluding asylum seekers; Statistics Austria 2017d

Graph 68. Estimated trajectory of employment rate by duration of stay in Austria (age 18+)



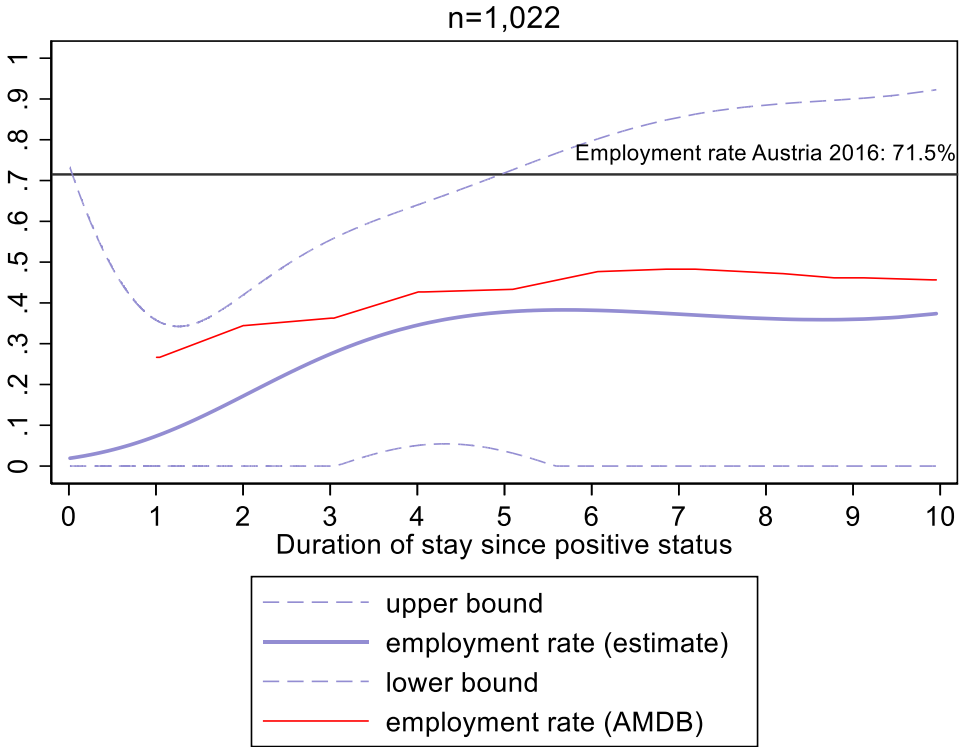
Logit model, including confidence intervals
Source: FIMAS data, excluding asylum seekers; Statistics Austria 2017d

Graph 69. Estimated trajectory of labour force participation rate by duration of stay in Austria since being granted asylum or protection (age 18+)



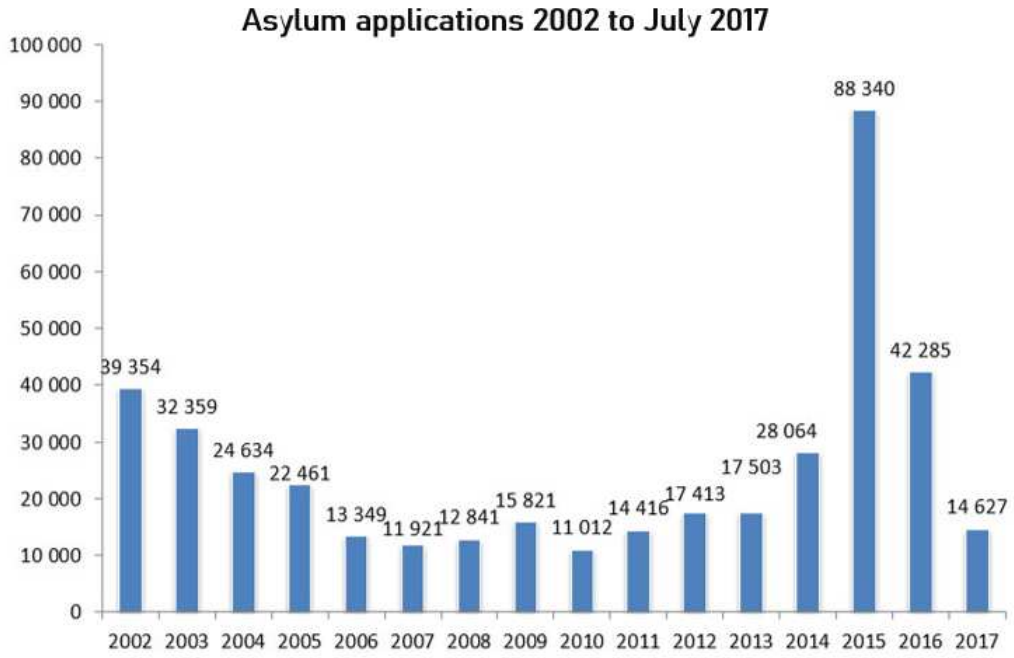
Logit model, including confidence intervals
Source: FIMAS data, excluding asylum seekers; Prettenthaler et al. 2017; Statistics Austria 2017d

Graph 70. Estimated trajectory of the employment rate by duration of stay in Austria since being granted asylum or protection (age 18+)



Logit model, including confidence intervals
Source: FIMAS data, excluding asylum seekers; Prettenthaler et al. 2017; Statistics Austria 2017d

Graph 71. Asylum applications in Austria, initial and subsequent applications, 2002 to July 2017



Source: Austrian Federal Ministry of the Interior, 2017

Table 16. Positive final asylum decisions, 2000-2008, by nationality, top 12 nationalities and stateless persons

Nationality	2000	2001	2002	2003	2004	2005	2006	2007	2008
Total	**1,002**	**1,152**	**1,073**	**2,084**	**5,136**	**4,528**	**4,063**	**5,197**	**3,753**
Russian Federation	21	18	36	788	2,905	2,427	2,090	2,636	1,557
Afghanistan	295	432	263	339	744	517	475	497	486
Turkey	18	49	68	73	94	70	113	195	246
Iraq	98	87	117	143	128	130	92	215	240
Iran	63	134	160	164	414	247	211	318	198
Somalia	10	7	15	24	12	58	103	191	118
Serbia[1)]	292	206	191	162	419	462	318	240	117
Armenia	20	6	10	26	35	39	100	135	105
Syria	1	11	9	18	21	53	47	86	68
Georgia	3	6	13	60	53	58	38	56	61
Azerbaijan	12	5	7	13	23	26	31	73	59
Stateless	5	13	6	24	8	32	41	50	55

Source: Statistics Austria (2017c). Austrian Federal Ministry of the Interior asylum statistics data. Created on 10 August 2017. 1) Up to and including 2005 including Montenegro; up to and including 2008 including Kosovo.

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